

Chapter 9

One- and Two-Sample Estimation Problems

9.1 Introduction

In previous chapters, we emphasized sampling properties of the sample mean and variance. We also emphasized displays of data in various forms. The purpose of these presentations is to build a foundation that allows us to draw conclusions about the population parameters from experimental data. For example, the Central Limit Theorem provides information about the distribution of the sample mean $\bar{X}$. The distribution involves the population mean μ . Thus, any conclusions concerning μ drawn from an observed sample average must depend on knowledge of this sampling distribution. Similar comments apply to S^2 and σ^2 . Clearly, any conclusions we draw about the variance of a normal distribution will likely involve the sampling distribution of S^2 .

In this chapter, we begin by formally outlining the purpose of statistical inference. We follow this by discussing the problem of **estimation of population parameters**. We confine our formal developments of specific estimation procedures to problems involving one and two samples.

9.2 Statistical Inference

In Chapter 1, we discussed the general philosophy of formal statistical inference. **Statistical inference** consists of those methods by which one makes inferences or generalizations about a population. The trend today is to distinguish between the **classical method** of estimating a population parameter, whereby inferences are based strictly on information obtained from a random sample selected from the population, and the **Bayesian method**, which utilizes prior subjective knowledge about the probability distribution of the unknown parameters in conjunction with the information provided by the sample data. Throughout most of this chapter, we shall use classical methods to estimate unknown population parameters such as the mean, the proportion, and the variance by computing statistics from random

samples and applying the theory of sampling distributions, much of which was covered in Chapter 8. Bayesian estimation will be discussed in Chapter 18.

Statistical inference may be divided into two major areas: **estimation** and **tests of hypotheses**. We treat these two areas separately, dealing with theory and applications of estimation in this chapter and hypothesis testing in Chapter 10. To distinguish clearly between the two areas, consider the following examples. A candidate for public office may wish to estimate the true proportion of voters favoring him by obtaining opinions from a random sample of 100 eligible voters. The fraction of voters in the sample favoring the candidate could be used as an estimate of the true proportion in the population of voters. A knowledge of the sampling distribution of a proportion enables one to establish the degree of accuracy of such an estimate. This problem falls in the area of estimation.

Now consider the case in which one is interested in finding out whether brand *A* floor wax is more scuff-resistant than brand *B* floor wax. He or she might hypothesize that brand *A* is better than brand *B* and, after proper testing, accept or reject this hypothesis. In this example, we do not attempt to estimate a parameter, but instead we try to arrive at a correct decision about a prestatd hypothesis. Once again we are dependent on sampling theory and the use of data to provide us with some measure of accuracy for our decision.

9.3 Classical Methods of Estimation

A **point estimate** of some population parameter θ is a single value $\hat{\theta}$ of a statistic $\hat{\Theta}$. For example, the value $\bar{x}$ of the statistic $\bar{X}$, computed from a sample of size n , is a point estimate of the population parameter μ . Similarly, $\hat{p} = x/n$ is a point estimate of the true proportion p for a binomial experiment.

An estimator is not expected to estimate the population parameter without error. We do not expect $\bar{X}$ to estimate μ exactly, but we certainly hope that it is not far off. For a particular sample, it is possible to obtain a closer estimate of μ by using the sample median $\tilde{X}$ as an estimator. Consider, for instance, a sample consisting of the values 2, 5, and 11 from a population whose mean is 4 but is supposedly unknown. We would estimate μ to be $\bar{x} = 6$, using the sample mean as our estimate, or $\tilde{x} = 5$, using the sample median as our estimate. In this case, the estimator $\tilde{X}$ produces an estimate closer to the true parameter than does the estimator $\bar{X}$. On the other hand, if our random sample contains the values 2, 6, and 7, then $\bar{x} = 5$ and $\tilde{x} = 6$, so $\bar{X}$ is the better estimator. Not knowing the true value of μ , we must decide in advance whether to use $\bar{X}$ or $\tilde{X}$ as our estimator.

Unbiased Estimator

What are the desirable properties of a “good” decision function that would influence us to choose one estimator rather than another? Let $\hat{\Theta}$ be an estimator whose value $\hat{\theta}$ is a point estimate of some unknown population parameter θ . Certainly, we would like the sampling distribution of $\hat{\Theta}$ to have a mean equal to the parameter estimated. An estimator possessing this property is said to be **unbiased**.

Definition 9.1: A statistic $\hat{\Theta}$ is said to be an **unbiased estimator** of the parameter θ if

$$\mu_{\hat{\Theta}} = E(\hat{\Theta}) = \theta.$$

Example 9.1: Show that S^2 is an unbiased estimator of the parameter σ^2 .

Solution: In Section 8.5 on page 244, we showed that

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

Now

$$\begin{aligned} E(S^2) &= E \left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \right] \\ &= \frac{1}{n-1} \left[\sum_{i=1}^n E(X_i - \mu)^2 - nE(\bar{X} - \mu)^2 \right] = \frac{1}{n-1} \left(\sum_{i=1}^n \sigma_{X_i}^2 - n\sigma_{\bar{X}}^2 \right). \end{aligned}$$

However,

$$\sigma_{X_i}^2 = \sigma^2, \text{ for } i = 1, 2, \dots, n, \text{ and } \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}.$$

Therefore,

$$E(S^2) = \frac{1}{n-1} \left(n\sigma^2 - n\frac{\sigma^2}{n} \right) = \sigma^2.$$

Although S^2 is an unbiased estimator of σ^2 , S , on the other hand, is usually a biased estimator of σ , with the bias becoming insignificant for large samples. This example illustrates **why we divide by $n - 1$** rather than n when the variance is estimated.

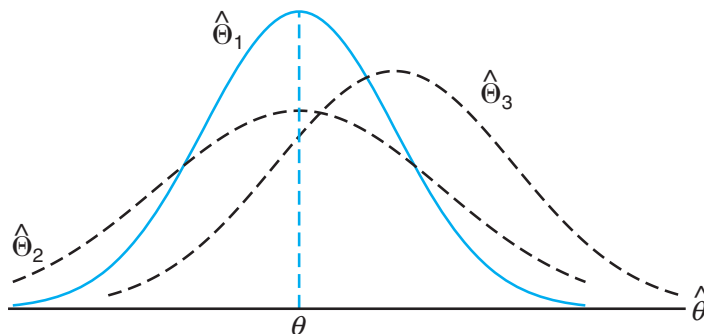
Variance of a Point Estimator

If $\hat{\Theta}_1$ and $\hat{\Theta}_2$ are two unbiased estimators of the same population parameter θ , we want to choose the estimator whose sampling distribution has the smaller variance. Hence, if $\sigma_{\hat{\Theta}_1}^2 < \sigma_{\hat{\Theta}_2}^2$, we say that $\hat{\Theta}_1$ is a **more efficient estimator** of θ than $\hat{\Theta}_2$.

Definition 9.2: If we consider all possible unbiased estimators of some parameter θ , the one with the smallest variance is called the **most efficient estimator** of θ .

Figure 9.1 illustrates the sampling distributions of three different estimators, $\hat{\Theta}_1$, $\hat{\Theta}_2$, and $\hat{\Theta}_3$, all estimating θ . It is clear that only $\hat{\Theta}_1$ and $\hat{\Theta}_2$ are unbiased, since their distributions are centered at θ . The estimator $\hat{\Theta}_1$ has a smaller variance than $\hat{\Theta}_2$ and is therefore more efficient. Hence, our choice for an estimator of θ , among the three considered, would be $\hat{\Theta}_1$.

For normal populations, one can show that both $\bar{X}$ and $\tilde{X}$ are unbiased estimators of the population mean μ , but the variance of $\bar{X}$ is smaller than the variance

Figure 9.1: Sampling distributions of different estimators of θ .

of $\tilde{X}$. Thus, both estimates $\bar{x}$ and $\tilde{x}$ will, on average, equal the population mean μ , but $\bar{x}$ is likely to be closer to μ for a given sample, and thus $\bar{X}$ is more efficient than $\tilde{X}$.

Interval Estimation

Even the most efficient unbiased estimator is unlikely to estimate the population parameter exactly. It is true that estimation accuracy increases with large samples, but there is still no reason we should expect a **point estimate** from a given sample to be exactly equal to the population parameter it is supposed to estimate. There are many situations in which it is preferable to determine an interval within which we would expect to find the value of the parameter. Such an interval is called an **interval estimate**.

An interval estimate of a population parameter θ is an interval of the form $\hat{\theta}_L < \theta < \hat{\theta}_U$, where $\hat{\theta}_L$ and $\hat{\theta}_U$ depend on the value of the statistic $\hat{\Theta}$ for a particular sample and also on the sampling distribution of $\hat{\Theta}$. For example, a random sample of SAT verbal scores for students in the entering freshman class might produce an interval from 530 to 550, within which we expect to find the true average of all SAT verbal scores for the freshman class. The values of the endpoints, 530 and 550, will depend on the computed sample mean $\bar{x}$ and the sampling distribution of $\bar{X}$. As the sample size increases, we know that $\sigma_{\bar{X}}^2 = \sigma^2/n$ decreases, and consequently our estimate is likely to be closer to the parameter μ , resulting in a shorter interval. Thus, the interval estimate indicates, by its length, the accuracy of the point estimate. An engineer will gain some insight into the population proportion defective by taking a sample and computing the *sample proportion defective*. But an interval estimate might be more informative.

Interpretation of Interval Estimates

Since different samples will generally yield different values of $\hat{\Theta}$ and, therefore, different values for $\hat{\theta}_L$ and $\hat{\theta}_U$, these endpoints of the interval are values of corresponding random variables $\hat{\Theta}_L$ and $\hat{\Theta}_U$. From the sampling distribution of $\hat{\Theta}$ we shall be able to determine $\hat{\Theta}_L$ and $\hat{\Theta}_U$ such that $P(\hat{\Theta}_L < \theta < \hat{\Theta}_U)$ is equal to any

positive fractional value we care to specify. If, for instance, we find $\hat{\Theta}_L$ and $\hat{\Theta}_U$ such that

$$P(\hat{\Theta}_L < \theta < \hat{\Theta}_U) = 1 - \alpha,$$

for $0 < \alpha < 1$, then we have a probability of $1 - \alpha$ of selecting a random sample that will produce an interval containing θ . The interval $\hat{\theta}_L < \theta < \hat{\theta}_U$, computed from the selected sample, is called a $100(1 - \alpha)\%$ **confidence interval**, the fraction $1 - \alpha$ is called the **confidence coefficient** or the **degree of confidence**, and the endpoints, $\hat{\theta}_L$ and $\hat{\theta}_U$, are called the lower and upper **confidence limits**. Thus, when $\alpha = 0.05$, we have a 95% confidence interval, and when $\alpha = 0.01$, we obtain a wider 99% confidence interval. The wider the confidence interval is, the more confident we can be that the interval contains the unknown parameter. Of course, it is better to be 95% confident that the average life of a certain television transistor is between 6 and 7 years than to be 99% confident that it is between 3 and 10 years. Ideally, we prefer a short interval with a high degree of confidence. Sometimes, restrictions on the size of our sample prevent us from achieving short intervals without sacrificing some degree of confidence.

In the sections that follow, we pursue the notions of point and interval estimation, with each section presenting a different special case. The reader should notice that while point and interval estimation represent different approaches to gaining information regarding a parameter, they are related in the sense that confidence interval estimators are based on point estimators. In the following section, for example, we will see that $\bar{X}$ is a very reasonable point estimator of μ . As a result, the important confidence interval estimator of μ depends on knowledge of the sampling distribution of $\bar{X}$.

We begin the following section with the simplest case of a confidence interval. The scenario is simple and yet unrealistic. We are interested in estimating a population mean μ and yet σ is known. Clearly, if μ is unknown, it is quite unlikely that σ is known. Any historical results that produced enough information to allow the assumption that σ is known would likely have produced similar information about μ . Despite this argument, we begin with this case because the concepts and indeed the resulting mechanics associated with confidence interval estimation remain the same for the more realistic situations presented later in Section 9.4 and beyond.

9.4 Single Sample: Estimating the Mean

The sampling distribution of $\bar{X}$ is centered at μ , and in most applications the variance is smaller than that of any other estimators of μ . Thus, the sample mean $\bar{x}$ will be used as a point estimate for the population mean μ . Recall that $\sigma_{\bar{X}}^2 = \sigma^2/n$, so a large sample will yield a value of $\bar{X}$ that comes from a sampling distribution with a small variance. Hence, $\bar{x}$ is likely to be a very accurate estimate of μ when n is large.

Let us now consider the interval estimate of μ . If our sample is selected from a normal population or, failing this, if n is sufficiently large, we can establish a confidence interval for μ by considering the sampling distribution of $\bar{X}$.

According to the Central Limit Theorem, we can expect the sampling distribution of $\bar{X}$ to be approximately normally distributed with mean $\mu_{\bar{X}} = \mu$ and

standard deviation $\sigma_{\bar{X}} = \sigma/\sqrt{n}$. Writing $z_{\alpha/2}$ for the z -value above which we find an area of $\alpha/2$ under the normal curve, we can see from Figure 9.2 that

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha,$$

where

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

Hence,

$$P\left(-z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = 1 - \alpha.$$

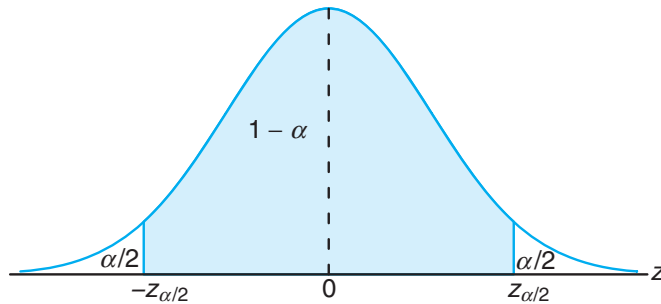


Figure 9.2: $P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$.

Multiplying each term in the inequality by $\sigma/\sqrt{n}$ and then subtracting $\bar{X}$ from each term and multiplying by -1 (reversing the sense of the inequalities), we obtain

$$P\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

A random sample of size n is selected from a population whose variance σ^2 is known, and the mean $\bar{x}$ is computed to give the $100(1 - \alpha)\%$ confidence interval below. It is important to emphasize that we have invoked the Central Limit Theorem above. As a result, it is important to note the conditions for applications that follow.

Confidence
Interval on μ , σ^2
Known

If $\bar{x}$ is the mean of a random sample of size n from a population with known variance σ^2 , a $100(1 - \alpha)\%$ confidence interval for μ is given by

$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}},$$

where $z_{\alpha/2}$ is the z -value leaving an area of $\alpha/2$ to the right.

For small samples selected from nonnormal populations, we cannot expect our degree of confidence to be accurate. However, for samples of size $n \geq 30$, with

the shape of the distributions not too skewed, sampling theory guarantees good results.

Clearly, the values of the random variables $\hat{\Theta}_L$ and $\hat{\Theta}_U$, defined in Section 9.3, are the confidence limits

$$\hat{\theta}_L = \bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \text{and} \quad \hat{\theta}_U = \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

Different samples will yield different values of $\bar{x}$ and therefore produce different interval estimates of the parameter μ , as shown in Figure 9.3. The dot at the center of each interval indicates the position of the point estimate $\bar{x}$ for that random sample. Note that all of these intervals are of the same width, since their widths depend only on the choice of $z_{\alpha/2}$ once $\bar{x}$ is determined. The larger the value we choose for $z_{\alpha/2}$, the wider we make all the intervals and the more confident we can be that the particular sample selected will produce an interval that contains the unknown parameter μ . In general, for a selection of $z_{\alpha/2}$, $100(1 - \alpha)\%$ of the intervals will cover μ .

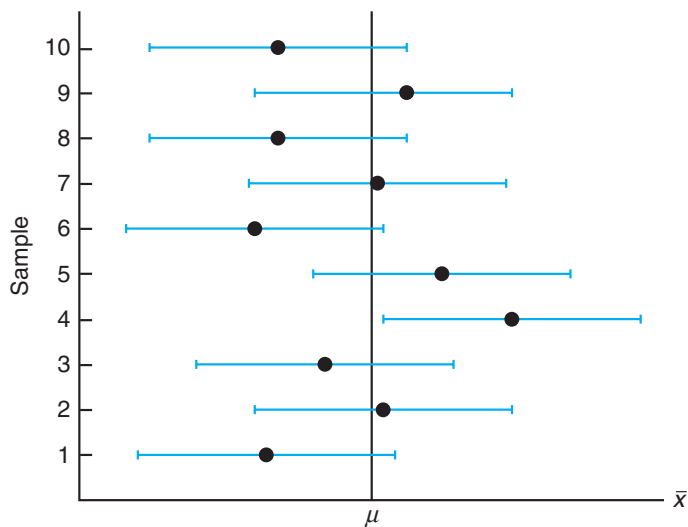


Figure 9.3: Interval estimates of μ for different samples.

Example 9.2: The average zinc concentration recovered from a sample of measurements taken in 36 different locations in a river is found to be 2.6 grams per milliliter. Find the 95% and 99% confidence intervals for the mean zinc concentration in the river. Assume that the population standard deviation is 0.3 gram per milliliter.

Solution: The point estimate of μ is $\bar{x} = 2.6$. The z -value leaving an area of 0.025 to the right, and therefore an area of 0.975 to the left, is $z_{0.025} = 1.96$ (Table A.3). Hence, the 95% confidence interval is

$$2.6 - (1.96) \left(\frac{0.3}{\sqrt{36}} \right) < \mu < 2.6 + (1.96) \left(\frac{0.3}{\sqrt{36}} \right),$$

which reduces to $2.50 < \mu < 2.70$. To find a 99% confidence interval, we find the z -value leaving an area of 0.005 to the right and 0.995 to the left. From Table A.3 again, $z_{0.005} = 2.575$, and the 99% confidence interval is

$$2.6 - (2.575) \left(\frac{0.3}{\sqrt{36}} \right) < \mu < 2.6 + (2.575) \left(\frac{0.3}{\sqrt{36}} \right),$$

or simply

$$2.47 < \mu < 2.73.$$

We now see that a longer interval is required to estimate μ with a higher degree of confidence. ▮

The $100(1 - \alpha)\%$ confidence interval provides an estimate of the accuracy of our point estimate. If μ is actually the center value of the interval, then $\bar{x}$ estimates μ without error. Most of the time, however, $\bar{x}$ will not be exactly equal to μ and the point estimate will be in error. The size of this error will be the absolute value of the difference between μ and $\bar{x}$, and we can be $100(1 - \alpha)\%$ confident that this difference will not exceed $z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$. We can readily see this if we draw a diagram of a hypothetical confidence interval, as in Figure 9.4.

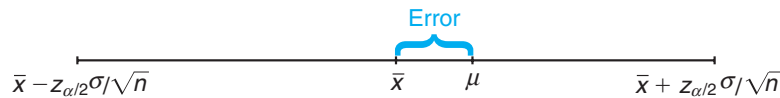


Figure 9.4: Error in estimating μ by $\bar{x}$.

Theorem 9.1: If $\bar{x}$ is used as an estimate of μ , we can be $100(1 - \alpha)\%$ confident that the error will not exceed $z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$.

In Example 9.2, we are 95% confident that the sample mean $\bar{x} = 2.6$ differs from the true mean μ by an amount less than $(1.96)(0.3)/\sqrt{36} = 0.1$ and 99% confident that the difference is less than $(2.575)(0.3)/\sqrt{36} = 0.13$.

Frequently, we wish to know how large a sample is necessary to ensure that the error in estimating μ will be less than a specified amount e . By Theorem 9.1, we must choose n such that $z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = e$. Solving this equation gives the following formula for n .

Theorem 9.2: If $\bar{x}$ is used as an estimate of μ , we can be $100(1 - \alpha)\%$ confident that the error will not exceed a specified amount e when the sample size is

$$n = \left(\frac{z_{\alpha/2} \sigma}{e} \right)^2.$$

When solving for the sample size, n , we round all fractional values up to the next whole number. By adhering to this principle, we can be sure that our degree of confidence never falls below $100(1 - \alpha)\%$.

Strictly speaking, the formula in Theorem 9.2 is applicable only if we know the variance of the population from which we select our sample. Lacking this information, we could take a preliminary sample of size $n \geq 30$ to provide an estimate of σ . Then, using s as an approximation for σ in Theorem 9.2, we could determine approximately how many observations are needed to provide the desired degree of accuracy.

Example 9.3: How large a sample is required if we want to be 95% confident that our estimate of μ in Example 9.2 is off by less than 0.05?

Solution: The population standard deviation is $\sigma = 0.3$. Then, by Theorem 9.2,

$$n = \left[\frac{(1.96)(0.3)}{0.05} \right]^2 = 138.3.$$

Therefore, we can be 95% confident that a random sample of size 139 will provide an estimate $\bar{x}$ differing from μ by an amount less than 0.05. ■

One-Sided Confidence Bounds

The confidence intervals and resulting confidence bounds discussed thus far are *two-sided* (i.e., both upper and lower bounds are given). However, there are many applications in which only one bound is sought. For example, if the measurement of interest is tensile strength, the engineer receives better information from a lower bound only. This bound communicates the worst-case scenario. On the other hand, if the measurement is something for which a relatively large value of μ is not profitable or desirable, then an upper confidence bound is of interest. An example would be a case in which inferences need to be made concerning the mean mercury composition in a river. An upper bound is very informative in this case.

One-sided confidence bounds are developed in the same fashion as two-sided intervals. However, the source is a one-sided probability statement that makes use of the Central Limit Theorem:

$$P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_\alpha\right) = 1 - \alpha.$$

One can then manipulate the probability statement much as before and obtain

$$P(\mu > \bar{X} - z_\alpha \sigma / \sqrt{n}) = 1 - \alpha.$$

Similar manipulation of $P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > -z_\alpha\right) = 1 - \alpha$ gives

$$P(\mu < \bar{X} + z_\alpha \sigma / \sqrt{n}) = 1 - \alpha.$$

As a result, the upper and lower one-sided bounds follow.

One-Sided
Confidence
Bounds on μ , σ^2
Known

If $\bar{X}$ is the mean of a random sample of size n from a population with variance σ^2 , the one-sided $100(1 - \alpha)\%$ confidence bounds for μ are given by

upper one-sided bound:	$\bar{x} + z_\alpha \sigma / \sqrt{n};$
lower one-sided bound:	$\bar{x} - z_\alpha \sigma / \sqrt{n}.$

Example 9.4: In a psychological testing experiment, 25 subjects are selected randomly and their reaction time, in seconds, to a particular stimulus is measured. Past experience suggests that the variance in reaction times to these types of stimuli is 4 sec^2 and that the distribution of reaction times is approximately normal. The average time for the subjects is 6.2 seconds. Give an upper 95% bound for the mean reaction time.

Solution: The upper 95% bound is given by

$$\begin{aligned}\bar{x} + z_{\alpha}\sigma/\sqrt{n} &= 6.2 + (1.645)\sqrt{4/25} = 6.2 + 0.658 \\ &= 6.858 \text{ seconds.}\end{aligned}$$

Hence, we are 95% confident that the mean reaction time is less than 6.858 seconds. J

The Case of σ Unknown

Frequently, we must attempt to estimate the mean of a population when the variance is unknown. The reader should recall learning in Chapter 8 that if we have a random sample from a *normal distribution*, then the random variable

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a Student t -distribution with $n - 1$ degrees of freedom. Here S is the sample standard deviation. In this situation, with σ unknown, T can be used to construct a confidence interval on μ . The procedure is the same as that with σ known except that σ is replaced by S and the standard normal distribution is replaced by the t -distribution. Referring to Figure 9.5, we can assert that

$$P(-t_{\alpha/2} < T < t_{\alpha/2}) = 1 - \alpha,$$

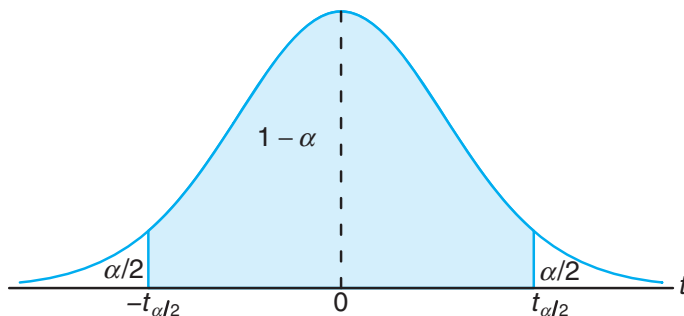
where $t_{\alpha/2}$ is the t -value with $n - 1$ degrees of freedom, above which we find an area of $\alpha/2$. Because of symmetry, an equal area of $\alpha/2$ will fall to the left of $-t_{\alpha/2}$. Substituting for T , we write

$$P\left(-t_{\alpha/2} < \frac{\bar{X} - \mu}{S/\sqrt{n}} < t_{\alpha/2}\right) = 1 - \alpha.$$

Multiplying each term in the inequality by $S/\sqrt{n}$, and then subtracting $\bar{X}$ from each term and multiplying by -1 , we obtain

$$P\left(\bar{X} - t_{\alpha/2} \frac{S}{\sqrt{n}} < \mu < \bar{X} + t_{\alpha/2} \frac{S}{\sqrt{n}}\right) = 1 - \alpha.$$

For a particular random sample of size n , the mean $\bar{x}$ and standard deviation s are computed and the following $100(1 - \alpha)\%$ confidence interval for μ is obtained.

Figure 9.5: $P(-t_{\alpha/2} < T < t_{\alpha/2}) = 1 - \alpha$.

**Confidence
Interval on μ , σ^2
Unknown**

If $\bar{x}$ and s are the mean and standard deviation of a random sample from a normal population with unknown variance σ^2 , a $100(1 - \alpha)\%$ confidence interval for μ is

$$\bar{x} - t_{\alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2} \frac{s}{\sqrt{n}},$$

where $t_{\alpha/2}$ is the t -value with $v = n - 1$ degrees of freedom, leaving an area of $\alpha/2$ to the right.

We have made a distinction between the cases of σ known and σ unknown in computing confidence interval estimates. We should emphasize that for σ known we exploited the Central Limit Theorem, whereas for σ unknown we made use of the sampling distribution of the random variable T . However, the use of the t -distribution is based on the premise that the sampling is from a normal distribution. As long as the distribution is approximately bell shaped, confidence intervals can be computed when σ^2 is unknown by using the t -distribution and we may expect very good results.

Computed one-sided confidence bounds for μ with σ unknown are as the reader would expect, namely

$$\bar{x} + t_{\alpha} \frac{s}{\sqrt{n}} \quad \text{and} \quad \bar{x} - t_{\alpha} \frac{s}{\sqrt{n}}.$$

They are the upper and lower $100(1 - \alpha)\%$ bounds, respectively. Here t_{α} is the t -value having an area of α to the right.

Example 9.5: The contents of seven similar containers of sulfuric acid are 9.8, 10.2, 10.4, 9.8, 10.0, 10.2, and 9.6 liters. Find a 95% confidence interval for the mean contents of all such containers, assuming an approximately normal distribution.

Solution: The sample mean and standard deviation for the given data are

$$\bar{x} = 10.0 \quad \text{and} \quad s = 0.283.$$

Using Table A.4, we find $t_{0.025} = 2.447$ for $v = 6$ degrees of freedom. Hence, the

95% confidence interval for μ is

$$10.0 - (2.447) \left(\frac{0.283}{\sqrt{7}} \right) < \mu < 10.0 + (2.447) \left(\frac{0.283}{\sqrt{7}} \right),$$

which reduces to $9.74 < \mu < 10.26$. ┐

Concept of a Large-Sample Confidence Interval

Often statisticians recommend that even when normality cannot be assumed, σ is unknown, and $n \geq 30$, s can replace σ and the confidence interval

$$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$

may be used. This is often referred to as a *large-sample confidence interval*. The justification lies only in the presumption that with a sample as large as 30 and the population distribution not too skewed, s will be very close to the true σ and thus the Central Limit Theorem prevails. It should be emphasized that this is only an approximation and the quality of the result becomes better as the sample size grows larger.

Example 9.6: Scholastic Aptitude Test (SAT) mathematics scores of a random sample of 500 high school seniors in the state of Texas are collected, and the sample mean and standard deviation are found to be 501 and 112, respectively. Find a 99% confidence interval on the mean SAT mathematics score for seniors in the state of Texas.

Solution: Since the sample size is large, it is reasonable to use the normal approximation. Using Table A.3, we find $z_{0.005} = 2.575$. Hence, a 99% confidence interval for μ is

$$501 \pm (2.575) \left(\frac{112}{\sqrt{500}} \right) = 501 \pm 12.9,$$

which yields $488.1 < \mu < 513.9$. ┐

9.5 Standard Error of a Point Estimate

We have made a rather sharp distinction between the goal of a point estimate and that of a confidence interval estimate. The former supplies a single number extracted from a set of experimental data, and the latter provides an interval that is reasonable for the parameter, *given the experimental data*; that is, $100(1 - \alpha)\%$ of such computed intervals “cover” the parameter.

These two approaches to estimation are related to each other. The common thread is the sampling distribution of the point estimator. Consider, for example, the estimator $\bar{X}$ of μ with σ known. We indicated earlier that a measure of the quality of an unbiased estimator is its variance. The variance of $\bar{X}$ is

$$\sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}.$$

Thus, the standard deviation of $\bar{X}$, or *standard error* of $\bar{X}$, is $\sigma/\sqrt{n}$. Simply put, the standard error of an estimator is its standard deviation. For $\bar{X}$, the computed confidence limit

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \text{ is written as } \bar{x} \pm z_{\alpha/2} \text{ s.e.}(\bar{x}),$$

where “s.e.” is the “standard error.” The important point is that the width of the confidence interval on μ is dependent on the quality of the point estimator through its standard error. In the case where σ is unknown and sampling is from a normal distribution, s replaces σ and the *estimated standard error* $s/\sqrt{n}$ is involved. Thus, the confidence limits on μ are

Confidence
Limits on μ , σ^2
Unknown

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} = \bar{x} \pm t_{\alpha/2} \text{ s.e.}(\bar{x})$$

Again, the confidence interval is *no better* (in terms of width) *than the quality of the point estimate*, in this case through its estimated standard error. Computer packages often refer to estimated standard errors simply as “standard errors.”

As we move to more complex confidence intervals, there is a prevailing notion that widths of confidence intervals become shorter as the quality of the corresponding point estimate becomes better, although it is not always quite as simple as we have illustrated here. It can be argued that a confidence interval is merely an augmentation of the point estimate to take into account the precision of the point estimate.

9.6 Prediction Intervals

The point and interval estimations of the mean in Sections 9.4 and 9.5 provide good information about the unknown parameter μ of a normal distribution or a nonnormal distribution from which a large sample is drawn. Sometimes, other than the population mean, the experimenter may also be interested in predicting the possible **value of a future observation**. For instance, in quality control, the experimenter may need to use the observed data to predict a new observation. A process that produces a metal part may be evaluated on the basis of whether the part meets specifications on tensile strength. On certain occasions, a customer may be interested in purchasing a **single part**. In this case, a confidence interval on the mean tensile strength does not capture the required information. The customer requires a statement regarding the uncertainty of a **single observation**. This type of requirement is nicely fulfilled by the construction of a **prediction interval**.

It is quite simple to obtain a prediction interval for the situations we have considered so far. Assume that the random sample comes from a normal population with unknown mean μ and known variance σ^2 . A natural point estimator of a new observation is $\bar{X}$. It is known, from Section 8.4, that the variance of $\bar{X}$ is σ^2/n . However, to predict a new observation, not only do we need to account for the variation due to estimating the mean, but also we should account for the **variation of a future observation**. From the assumption, we know that the variance of the random error in a new observation is σ^2 . The development of a

prediction interval is best illustrated by beginning with a normal random variable $x_0 - \bar{x}$, where x_0 is the new observation and $\bar{x}$ comes from the sample. Since x_0 and $\bar{x}$ are independent, we know that

$$z = \frac{x_0 - \bar{x}}{\sqrt{\sigma^2 + \sigma^2/n}} = \frac{x_0 - \bar{x}}{\sigma\sqrt{1 + 1/n}}$$

is $n(z; 0, 1)$. As a result, if we use the probability statement

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$$

with the z -statistic above and place x_0 in the center of the probability statement, we have the following event occurring with probability $1 - \alpha$:

$$\bar{x} - z_{\alpha/2}\sigma\sqrt{1 + 1/n} < x_0 < \bar{x} + z_{\alpha/2}\sigma\sqrt{1 + 1/n}.$$

As a result, computation of the prediction interval is formalized as follows.

Prediction Interval of a Future Observation, σ^2 Known	For a normal distribution of measurements with unknown mean μ and known variance σ^2 , a $100(1 - \alpha)\%$ prediction interval of a future observation x_0 is $\bar{x} - z_{\alpha/2}\sigma\sqrt{1 + 1/n} < x_0 < \bar{x} + z_{\alpha/2}\sigma\sqrt{1 + 1/n},$ where $z_{\alpha/2}$ is the z -value leaving an area of $\alpha/2$ to the right.
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Example 9.7: Due to the decrease in interest rates, the First Citizens Bank received a lot of mortgage applications. A recent sample of 50 mortgage loans resulted in an average loan amount of \$257,300. Assume a population standard deviation of \$25,000. For the next customer who fills out a mortgage application, find a 95% prediction interval for the loan amount.

Solution: The point prediction of the next customer's loan amount is $\bar{x} = \$257,300$. The z -value here is $z_{0.025} = 1.96$. Hence, a 95% prediction interval for the future loan amount is

$$257,300 - (1.96)(25,000)\sqrt{1 + 1/50} < x_0 < 257,300 + (1.96)(25,000)\sqrt{1 + 1/50},$$

which gives the interval (\$207,812.43, \$306,787.57). ┐

The prediction interval provides a good estimate of the location of a future observation, which is quite different from the estimate of the sample mean value. It should be noted that the variation of this prediction is the sum of the variation due to an estimation of the mean and the variation of a single observation. However, as in the past, we first consider the case with known variance. It is also important to deal with the prediction interval of a future observation in the situation where the variance is unknown. Indeed a Student t -distribution may be used in this case, as described in the following result. The normal distribution is merely replaced by the t -distribution.

Prediction
Interval of a
Future
Observation, σ^2
Unknown

For a normal distribution of measurements with unknown mean μ and unknown variance σ^2 , a $100(1 - \alpha)\%$ **prediction interval** of a future observation x_0 is

$$\bar{x} - t_{\alpha/2}s\sqrt{1 + 1/n} < x_0 < \bar{x} + t_{\alpha/2}s\sqrt{1 + 1/n},$$

where $t_{\alpha/2}$ is the t -value with $v = n - 1$ degrees of freedom, leaving an area of $\alpha/2$ to the right.

One-sided prediction intervals can also be constructed. Upper prediction bounds apply in cases where focus must be placed on future large observations. Concern over future small observations calls for the use of lower prediction bounds. The upper bound is given by

$$\bar{x} + t_{\alpha}s\sqrt{1 + 1/n}$$

and the lower bound by

$$\bar{x} - t_{\alpha}s\sqrt{1 + 1/n}.$$

Example 9.8: A meat inspector has randomly selected 30 packs of 95% lean beef. The sample resulted in a mean of 96.2% with a sample standard deviation of 0.8%. Find a 99% prediction interval for the leanness of a new pack. Assume normality.

Solution: For $v = 29$ degrees of freedom, $t_{0.005} = 2.756$. Hence, a 99% prediction interval for a new observation x_0 is

$$96.2 - (2.756)(0.8)\sqrt{1 + \frac{1}{30}} < x_0 < 96.2 + (2.756)(0.8)\sqrt{1 + \frac{1}{30}},$$

which reduces to (93.96, 98.44). ┐

Use of Prediction Limits for Outlier Detection

To this point in the text very little attention has been paid to the concept of **outliers**, or aberrant observations. The majority of scientific investigators are keenly sensitive to the existence of outlying observations or so-called faulty or “bad data.” We deal with the concept of outlier detection extensively in Chapter 12. However, it is certainly of interest here since there is an important relationship between outlier detection and prediction intervals.

It is convenient for our purposes to view an outlying observation as one that comes from a population with a mean that is different from the mean that governs the rest of the sample of size n being studied. The prediction interval produces a bound that “covers” a future single observation with probability $1 - \alpha$ if it comes from the population from which the sample was drawn. As a result, a methodology for outlier detection involves the rule that **an observation is an outlier if it falls outside the prediction interval computed without including the questionable observation in the sample**. As a result, for the prediction interval of Example 9.8, if a new pack of beef is measured and its leanness is outside the interval (93.96, 98.44), that observation can be viewed as an outlier.

9.7 Tolerance Limits

As discussed in Section 9.6, the scientist or engineer may be less interested in estimating parameters than in gaining a notion about where an individual *observation* or measurement might fall. Such situations call for the use of prediction intervals. However, there is yet a third type of interval that is of interest in many applications. Once again, suppose that interest centers around the manufacturing of a component part and specifications exist on a dimension of that part. In addition, there is little concern about the mean of the dimension. But unlike in the scenario in Section 9.6, one may be less interested in a single observation and more interested in where the majority of the population falls. If process specifications are important, the manager of the process is concerned about long-range performance, **not the next observation**. One must attempt to determine bounds that, in some probabilistic sense, “cover” values in the population (i.e., the measured values of the dimension).

One method of establishing the desired bounds is to determine a confidence interval on a *fixed proportion* of the measurements. This is best motivated by visualizing a situation in which we are doing random sampling from a normal distribution with known mean μ and variance σ^2 . Clearly, a bound that covers the middle 95% of the population of observations is

$$\mu \pm 1.96\sigma.$$

This is called a **tolerance interval**, and indeed its coverage of 95% of measured observations is exact. However, in practice, μ and σ are seldom known; thus, the user must apply

$$\bar{x} \pm ks.$$

Now, of course, the interval is a random variable, and hence the *coverage* of a proportion of the population by the interval is not exact. As a result, a $100(1-\gamma)\%$ confidence interval must be used since $\bar{x} \pm ks$ cannot be expected to cover any specified proportion all the time. As a result, we have the following definition.

Tolerance Limits	For a normal distribution of measurements with unknown mean μ and unknown standard deviation σ , tolerance limits are given by $\bar{x} \pm ks$, where k is determined such that one can assert with $100(1-\gamma)\%$ confidence that the given limits contain at least the proportion $1-\alpha$ of the measurements.
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Table A.7 gives values of k for $1-\alpha = 0.90, 0.95, 0.99$; $\gamma = 0.05, 0.01$; and selected values of n from 2 to 300.

Example 9.9: Consider Example 9.8. With the information given, find a tolerance interval that gives two-sided 95% bounds on 90% of the distribution of packages of 95% lean beef. Assume the data came from an approximately normal distribution.

Solution: Recall from Example 9.8 that $n = 30$, the sample mean is 96.2%, and the sample standard deviation is 0.8%. From Table A.7, $k = 2.14$. Using

$$\bar{x} \pm ks = 96.2 \pm (2.14)(0.8),$$

we find that the lower and upper bounds are 94.5 and 97.9.

We are 95% confident that the above range covers the central 90% of the distribution of 95% lean beef packages. 

Distinction among Confidence Intervals, Prediction Intervals, and Tolerance Intervals

It is important to reemphasize the difference among the three types of intervals discussed and illustrated in the preceding sections. The computations are straightforward, but interpretation can be confusing. In real-life applications, these intervals are not interchangeable because their interpretations are quite distinct.

In the case of confidence intervals, one is attentive only to the **population mean**. For example, Exercise 9.13 on page 283 deals with an engineering process that produces shearing pins. A specification will be set on Rockwell hardness, below which a customer will not accept any pins. Here, a population parameter must take a backseat. It is important that the engineer know where the *majority of the values of Rockwell hardness are going to be*. Thus, tolerance limits should be used. Surely, when tolerance limits on any process output are tighter than process specifications, that is good news for the process manager.

It is true that the tolerance limit interpretation is somewhat related to the confidence interval. The $100(1-\alpha)\%$ tolerance interval on, say, the proportion 0.95 can be viewed as a confidence interval **on the middle 95%** of the corresponding normal distribution. One-sided tolerance limits are also relevant. In the case of the Rockwell hardness problem, it is desirable to have a lower bound of the form $\bar{x} - ks$ such that there is 99% confidence that at least 99% of Rockwell hardness values will exceed the computed value.

Prediction intervals are applicable when it is important to determine a bound on a **single value**. The mean is not the issue here, nor is the location of the majority of the population. Rather, the location of a single new observation is required.

Case Study 9.1: **Machine Quality:** A machine produces metal pieces that are cylindrical in shape. A sample of these pieces is taken and the diameters are found to be 1.01, 0.97, 1.03, 1.04, 0.99, 0.98, 0.99, 1.01, and 1.03 centimeters. Use these data to calculate three interval types and draw interpretations that illustrate the distinction between them in the context of the system. For all computations, assume an approximately normal distribution. The sample mean and standard deviation for the given data are $\bar{x} = 1.0056$ and $s = 0.0246$.

- Find a 99% confidence interval on the mean diameter.
- Compute a 99% prediction interval on a measured diameter of a single metal piece taken from the machine.
- Find the 99% tolerance limits that will contain 95% of the metal pieces produced by this machine.

Solution: (a) The 99% confidence interval for the mean diameter is given by

$$\bar{x} \pm t_{0.005} s / \sqrt{n} = 1.0056 \pm (3.355)(0.0246/3) = 1.0056 \pm 0.0275.$$

Thus, the 99% confidence bounds are 0.9781 and 1.0331.

- (b) The 99% prediction interval for a future observation is given by

$$\bar{x} \pm t_{0.005} s \sqrt{1 + 1/n} = 1.0056 \pm (3.355)(0.0246) \sqrt{1 + 1/9},$$

with the bounds being 0.9186 and 1.0926.

- (c) From Table A.7, for $n = 9$, $1 - \gamma = 0.99$, and $1 - \alpha = 0.95$, we find $k = 4.550$ for two-sided limits. Hence, the 99% tolerance limits are given by

$$\bar{x} + ks = 1.0056 \pm (4.550)(0.0246),$$

with the bounds being 0.8937 and 1.1175. We are 99% confident that the tolerance interval from 0.8937 to 1.1175 will contain the central 95% of the distribution of diameters produced.

This case study illustrates that the three types of limits can give appreciably different results even though they are all 99% bounds. In the case of the confidence interval on the mean, 99% of such intervals cover the population mean diameter. Thus, we say that we are 99% confident that the mean diameter produced by the process is between 0.9781 and 1.0331 centimeters. Emphasis is placed on the mean, with less concern about a single reading or the general nature of the distribution of diameters in the population. In the case of the prediction limits, the bounds 0.9186 and 1.0926 are based on the distribution of a single “new” metal piece taken from the process, and again 99% of such limits will cover the diameter of a new measured piece. On the other hand, the tolerance limits, as suggested in the previous section, give the engineer a sense of where the “majority,” say the central 95%, of the diameters of measured pieces in the population reside. The 99% tolerance limits, 0.8937 and 1.1175, are numerically quite different from the other two bounds. If these bounds appear alarmingly wide to the engineer, it reflects negatively on process quality. On the other hand, if the bounds represent a desirable result, the engineer may conclude that a majority (95% in here) of the diameters are in a desirable range. Again, a confidence interval interpretation may be used: namely, 99% of such calculated bounds will cover the middle 95% of the population of diameters. ■

Exercises

9.1 A UCLA researcher claims that the life span of mice can be extended by as much as 25% when the calories in their diet are reduced by approximately 40% from the time they are weaned. The restricted diet is enriched to normal levels by vitamins and protein. Assuming that it is known from previous studies that $\sigma = 5.8$ months, how many mice should be included in our sample if we wish to be 99% confident that the mean life span of the sample will be within 2 months of the population mean for all mice subjected to this reduced diet?

9.2 An electrical firm manufactures light bulbs that have a length of life that is approximately normally distributed with a standard deviation of 40 hours. If a sample of 30 bulbs has an average life of 780 hours, find a 96% confidence interval for the population mean of all bulbs produced by this firm.

9.3 Many cardiac patients wear an implanted pacemaker to control their heartbeat. A plastic connector module mounts on the top of the pacemaker. Assuming a standard deviation of 0.0015 inch and an approximately normal distribution, find a 95% confidence

interval for the mean of the depths of all connector modules made by a certain manufacturing company. A random sample of 75 modules has an average depth of 0.310 inch.

9.4 The heights of a random sample of 50 college students showed a mean of 174.5 centimeters and a standard deviation of 6.9 centimeters.

- Construct a 98% confidence interval for the mean height of all college students.
- What can we assert with 98% confidence about the possible size of our error if we estimate the mean height of all college students to be 174.5 centimeters?

9.5 A random sample of 100 automobile owners in the state of Virginia shows that an automobile is driven on average 23,500 kilometers per year with a standard deviation of 3900 kilometers. Assume the distribution of measurements to be approximately normal.

- Construct a 99% confidence interval for the average number of kilometers an automobile is driven annually in Virginia.
- What can we assert with 99% confidence about the possible size of our error if we estimate the average number of kilometers driven by car owners in Virginia to be 23,500 kilometers per year?

9.6 How large a sample is needed in Exercise 9.2 if we wish to be 96% confident that our sample mean will be within 10 hours of the true mean?

9.7 How large a sample is needed in Exercise 9.3 if we wish to be 95% confident that our sample mean will be within 0.0005 inch of the true mean?

9.8 An efficiency expert wishes to determine the average time that it takes to drill three holes in a certain metal clamp. How large a sample will she need to be 95% confident that her sample mean will be within 15 seconds of the true mean? Assume that it is known from previous studies that $\sigma = 40$ seconds.

9.9 Regular consumption of presweetened cereals contributes to tooth decay, heart disease, and other degenerative diseases, according to studies conducted by Dr. W. H. Bowen of the National Institute of Health and Dr. J. Yudben, Professor of Nutrition and Dietetics at the University of London. In a random sample consisting of 20 similar single servings of Alpha-Bits, the average sugar content was 11.3 grams with a standard deviation of 2.45 grams. Assuming that the sugar contents are normally distributed, construct a 95% confidence interval for the mean sugar content for single servings of Alpha-Bits.

9.10 A random sample of 12 graduates of a certain secretarial school typed an average of 79.3 words per minute with a standard deviation of 7.8 words per minute. Assuming a normal distribution for the number of words typed per minute, find a 95% confidence interval for the average number of words typed by all graduates of this school.

9.11 A machine produces metal pieces that are cylindrical in shape. A sample of pieces is taken, and the diameters are found to be 1.01, 0.97, 1.03, 1.04, 0.99, 0.98, 0.99, 1.01, and 1.03 centimeters. Find a 99% confidence interval for the mean diameter of pieces from this machine, assuming an approximately normal distribution.

9.12 A random sample of 10 chocolate energy bars of a certain brand has, on average, 230 calories per bar, with a standard deviation of 15 calories. Construct a 99% confidence interval for the true mean calorie content of this brand of energy bar. Assume that the distribution of the calorie content is approximately normal.

9.13 A random sample of 12 shearing pins is taken in a study of the Rockwell hardness of the pin head. Measurements on the Rockwell hardness are made for each of the 12, yielding an average value of 48.50 with a sample standard deviation of 1.5. Assuming the measurements to be normally distributed, construct a 90% confidence interval for the mean Rockwell hardness.

9.14 The following measurements were recorded for the drying time, in hours, of a certain brand of latex paint:

3.4	2.5	4.8	2.9	3.6
2.8	3.3	5.6	3.7	2.8
4.4	4.0	5.2	3.0	4.8

Assuming that the measurements represent a random sample from a normal population, find a 95% prediction interval for the drying time for the next trial of the paint.

9.15 Referring to Exercise 9.5, construct a 99% prediction interval for the kilometers traveled annually by an automobile owner in Virginia.

9.16 Consider Exercise 9.10. Compute the 95% prediction interval for the next observed number of words per minute typed by a graduate of the secretarial school.

9.17 Consider Exercise 9.9. Compute a 95% prediction interval for the sugar content of the next single serving of Alpha-Bits.

9.18 Referring to Exercise 9.13, construct a 95% tolerance interval containing 90% of the measurements.

9.19 A random sample of 25 tablets of buffered aspirin contains, on average, 325.05 mg of aspirin per tablet, with a standard deviation of 0.5 mg. Find the 95% tolerance limits that will contain 90% of the tablet contents for this brand of buffered aspirin. Assume that the aspirin content is normally distributed.

9.20 Consider the situation of Exercise 9.11. Estimation of the mean diameter, while important, is not nearly as important as trying to pin down the location of the majority of the distribution of diameters. Find the 95% tolerance limits that contain 95% of the diameters.

9.21 In a study conducted by the Department of Zoology at Virginia Tech, fifteen samples of water were collected from a certain station in the James River in order to gain some insight regarding the amount of orthophosphorus in the river. The concentration of the chemical is measured in milligrams per liter. Let us suppose that the mean at the station is not as important as the upper extreme of the distribution of the concentration of the chemical at the station. Concern centers around whether the concentration at the extreme is too large. Readings for the fifteen water samples gave a sample mean of 3.84 milligrams per liter and a sample standard deviation of 3.07 milligrams per liter. Assume that the readings are a random sample from a normal distribution. Calculate a prediction interval (upper 95% prediction limit) and a tolerance limit (95% upper tolerance limit that exceeds 95% of the population of values). Interpret both; that is, tell what each communicates about the upper extreme of the distribution of orthophosphorus at the sampling station.

9.22 A type of thread is being studied for its tensile strength properties. Fifty pieces were tested under similar conditions, and the results showed an average tensile strength of 78.3 kilograms and a standard deviation of 5.6 kilograms. Assuming a normal distribution of tensile strengths, give a lower 95% prediction limit on a single observed tensile strength value. In addition, give a lower 95% tolerance limit that is exceeded by 99% of the tensile strength values.

9.23 Refer to Exercise 9.22. Why are the quantities requested in the exercise likely to be more important to the manufacturer of the thread than, say, a confidence interval on the mean tensile strength?

9.24 Refer to Exercise 9.22 again. Suppose that specifications by a buyer of the thread are that the tensile strength of the material must be at least 62 kilograms. The manufacturer is satisfied if at most 5% of the manufactured pieces have tensile strength less than 62 kilograms. Is there cause for concern? Use a one-sided 99% tolerance limit that is exceeded by 95% of the tensile strength values.

9.25 Consider the drying time measurements in Exercise 9.14. Suppose the 15 observations in the data set are supplemented by a 16th value of 6.9 hours. In the context of the original 15 observations, is the 16th value an outlier? Show work.

9.26 Consider the data in Exercise 9.13. Suppose the manufacturer of the shearing pins insists that the Rockwell hardness of the product be less than or equal to 44.0 only 5% of the time. What is your reaction? Use a tolerance limit calculation as the basis for your judgment.

9.27 Consider the situation of Case Study 9.1 on page 281 with a larger sample of metal pieces. The diameters are as follows: 1.01, 0.97, 1.03, 1.04, 0.99, 0.98, 1.01, 1.03, 0.99, 1.00, 1.00, 0.99, 0.98, 1.01, 1.02, 0.99 centimeters. Once again the normality assumption may be made. Do the following and compare your results to those of the case study. Discuss how they are different and why.

- Compute a 99% confidence interval on the mean diameter.
- Compute a 99% prediction interval on the next diameter to be measured.
- Compute a 99% tolerance interval for coverage of the central 95% of the distribution of diameters.

9.28 In Section 9.3, we emphasized the notion of “most efficient estimator” by comparing the variance of two unbiased estimators $\hat{\Theta}_1$ and $\hat{\Theta}_2$. However, this does not take into account bias in case one or both estimators are not unbiased. Consider the quantity

$$MSE = E(\hat{\Theta} - \theta),$$

where MSE denotes **mean squared error**. The MSE is often used to compare two estimators $\hat{\Theta}_1$ and $\hat{\Theta}_2$ of θ when either or both is unbiased because (i) it is intuitively reasonable and (ii) it accounts for bias. Show that MSE can be written

$$\begin{aligned} MSE &= E[\hat{\Theta} - E(\hat{\Theta})]^2 + [E(\hat{\Theta} - \theta)]^2 \\ &= \text{Var}(\hat{\Theta}) + [\text{Bias}(\hat{\Theta})]^2. \end{aligned}$$

9.29 Let us define $S'^2 = \sum_{i=1}^n (X_i - \bar{X})^2/n$. Show that

$$E(S'^2) = [(n-1)/n]\sigma^2,$$

and hence S'^2 is a biased estimator for σ^2 .

9.30 Consider S'^2 , the estimator of σ^2 , from Exercise 9.29. Analysts often use S'^2 rather than dividing $\sum_{i=1}^n (X_i - \bar{X})^2$ by $n-1$, the degrees of freedom in the sample.

- (a) What is the bias of S'^2 ?
 (b) Show that the bias of S'^2 approaches zero as $n \rightarrow \infty$.

9.31 If X is a binomial random variable, show that

(a) $\hat{P} = X/n$ is an unbiased estimator of p ;

(b) $P' = \frac{X + \sqrt{n}/2}{n + \sqrt{n}}$ is a biased estimator of p .

9.32 Show that the estimator P' of Exercise 9.31(b) becomes unbiased as $n \rightarrow \infty$.

9.33 Compare S^2 and S'^2 (see Exercise 9.29), the

two estimators of σ^2 , to determine which is more efficient. Assume these estimators are found using $X_1, X_2, \dots, X_n$, independent random variables from $n(x; \mu, \sigma)$. Which estimator is more efficient considering only the variance of the estimators? [Hint: Make use of Theorem 8.4 and the fact that the variance of χ_v^2 is $2v$, from Section 6.7.]

9.34 Consider Exercise 9.33. Use the MSE discussed in Exercise 9.28 to determine which estimator is more efficient. Write out

$$\frac{MSE(S^2)}{MSE(S'^2)}.$$

9.8 Two Samples: Estimating the Difference between Two Means

If we have two populations with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2 , respectively, a point estimator of the difference between μ_1 and μ_2 is given by the statistic $\bar{X}_1 - \bar{X}_2$. Therefore, to obtain a point estimate of $\mu_1 - \mu_2$, we shall select two independent random samples, one from each population, of sizes n_1 and n_2 , and compute $\bar{x}_1 - \bar{x}_2$, the difference of the sample means. Clearly, we must consider the sampling distribution of $\bar{X}_1 - \bar{X}_2$.

According to Theorem 8.3, we can expect the sampling distribution of $\bar{X}_1 - \bar{X}_2$ to be approximately normally distributed with mean $\mu_{\bar{X}_1 - \bar{X}_2} = \mu_1 - \mu_2$ and standard deviation $\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$. Therefore, we can assert with a probability of $1 - \alpha$ that the standard normal variable

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

will fall between $-z_{\alpha/2}$ and $z_{\alpha/2}$. Referring once again to Figure 9.2, we write

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha.$$

Substituting for Z , we state equivalently that

$$P\left(-z_{\alpha/2} < \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} < z_{\alpha/2}\right) = 1 - \alpha,$$

which leads to the following $100(1 - \alpha)\%$ confidence interval for $\mu_1 - \mu_2$.

Confidence
Interval for
 $\mu_1 - \mu_2$, σ_1^2 and
 σ_2^2 Known

If $\bar{x}_1$ and $\bar{x}_2$ are means of independent random samples of sizes n_1 and n_2 from populations with known variances σ_1^2 and σ_2^2 , respectively, a $100(1 - \alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is given by

$$(\bar{x}_1 - \bar{x}_2) - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}},$$

where $z_{\alpha/2}$ is the z -value leaving an area of $\alpha/2$ to the right.

The degree of confidence is exact when samples are selected from normal populations. For nonnormal populations, the Central Limit Theorem allows for a good approximation for reasonable size samples.

The Experimental Conditions and the Experimental Unit

For the case of confidence interval estimation on the difference between two means, we need to consider the experimental conditions in the data-taking process. It is assumed that we have two independent random samples from distributions with means μ_1 and μ_2 , respectively. It is important that experimental conditions emulate this ideal described by these assumptions as closely as possible. Quite often, the experimenter should plan the strategy of the experiment accordingly. For almost any study of this type, there is a so-called *experimental unit*, which is that part of the experiment that produces experimental error and is responsible for the population variance we refer to as σ^2 . In a drug study, the experimental unit is the patient or subject. In an agricultural experiment, it may be a plot of ground. In a chemical experiment, it may be a quantity of raw materials. It is important that differences between the experimental units have minimal impact on the results. The experimenter will have a degree of insurance that experimental units will not bias results if the conditions that define the two populations are *randomly assigned* to the experimental units. We shall again focus on randomization in future chapters that deal with hypothesis testing.

Example 9.10: A study was conducted in which two types of engines, A and B , were compared. Gas mileage, in miles per gallon, was measured. Fifty experiments were conducted using engine type A and 75 experiments were done with engine type B . The gasoline used and other conditions were held constant. The average gas mileage was 36 miles per gallon for engine A and 42 miles per gallon for engine B . Find a 96% confidence interval on $\mu_B - \mu_A$, where μ_A and μ_B are population mean gas mileages for engines A and B , respectively. Assume that the population standard deviations are 6 and 8 for engines A and B , respectively.

Solution: The point estimate of $\mu_B - \mu_A$ is $\bar{x}_B - \bar{x}_A = 42 - 36 = 6$. Using $\alpha = 0.04$, we find $z_{0.02} = 2.05$ from Table A.3. Hence, with substitution in the formula above, the 96% confidence interval is

$$6 - 2.05\sqrt{\frac{64}{75} + \frac{36}{50}} < \mu_B - \mu_A < 6 + 2.05\sqrt{\frac{64}{75} + \frac{36}{50}},$$

or simply $3.43 < \mu_B - \mu_A < 8.57$. └

This procedure for estimating the difference between two means is applicable if σ_1^2 and σ_2^2 are known. If the variances are not known and the two distributions involved are approximately normal, the t -distribution becomes involved, as in the case of a single sample. If one is not willing to assume normality, large samples (say greater than 30) will allow the use of s_1 and s_2 in place of σ_1 and σ_2 , respectively, with the rationale that $s_1 \approx \sigma_1$ and $s_2 \approx \sigma_2$. Again, of course, the confidence interval is an approximate one.

Variances Unknown but Equal

Consider the case where σ_1^2 and σ_2^2 are unknown. If $\sigma_1^2 = \sigma_2^2 = \sigma^2$, we obtain a standard normal variable of the form

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma^2[(1/n_1) + (1/n_2)]}}.$$

According to Theorem 8.4, the two random variables

$$\frac{(n_1 - 1)S_1^2}{\sigma^2} \quad \text{and} \quad \frac{(n_2 - 1)S_2^2}{\sigma^2}$$

have chi-squared distributions with $n_1 - 1$ and $n_2 - 1$ degrees of freedom, respectively. Furthermore, they are independent chi-squared variables, since the random samples were selected independently. Consequently, their sum

$$V = \frac{(n_1 - 1)S_1^2}{\sigma^2} + \frac{(n_2 - 1)S_2^2}{\sigma^2} = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{\sigma^2}$$

has a chi-squared distribution with $v = n_1 + n_2 - 2$ degrees of freedom.

Since the preceding expressions for Z and V can be shown to be independent, it follows from Theorem 8.5 that the statistic

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma^2[(1/n_1) + (1/n_2)]}} \bigg/ \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{\sigma^2(n_1 + n_2 - 2)}}$$

has the t -distribution with $v = n_1 + n_2 - 2$ degrees of freedom.

A point estimate of the unknown common variance σ^2 can be obtained by pooling the sample variances. Denoting the pooled estimator by S_p^2 , we have the following.

Pooled Estimate
of Variance

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.$$

Substituting S_p^2 in the T statistic, we obtain the less cumbersome form

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{(1/n_1) + (1/n_2)}}.$$

Using the T statistic, we have

$$P(-t_{\alpha/2} < T < t_{\alpha/2}) = 1 - \alpha,$$

where $t_{\alpha/2}$ is the t -value with $n_1 + n_2 - 2$ degrees of freedom, above which we find an area of $\alpha/2$. Substituting for T in the inequality, we write

$$P \left[-t_{\alpha/2} < \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{(1/n_1) + (1/n_2)}} < t_{\alpha/2} \right] = 1 - \alpha.$$

After the usual mathematical manipulations, the difference of the sample means $\bar{x}_1 - \bar{x}_2$ and the pooled variance are computed and then the following $100(1 - \alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is obtained.

The value of s_p^2 is easily seen to be a weighted average of the two sample variances s_1^2 and s_2^2 , where the weights are the degrees of freedom.

Confidence
Interval for
 $\mu_1 - \mu_2$, $\sigma_1^2 = \sigma_2^2$
but Both
Unknown

If $\bar{x}_1$ and $\bar{x}_2$ are the means of independent random samples of sizes n_1 and n_2 , respectively, from approximately normal populations with unknown but equal variances, a $100(1 - \alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is given by

$$(\bar{x}_1 - \bar{x}_2) - t_{\alpha/2}s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{\alpha/2}s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}},$$

where s_p is the pooled estimate of the population standard deviation and $t_{\alpha/2}$ is the t -value with $v = n_1 + n_2 - 2$ degrees of freedom, leaving an area of $\alpha/2$ to the right.

Example 9.11: The article “Macroinvertebrate Community Structure as an Indicator of Acid Mine Pollution,” published in the *Journal of Environmental Pollution*, reports on an investigation undertaken in Cane Creek, Alabama, to determine the relationship between selected physiochemical parameters and different measures of macroinvertebrate community structure. One facet of the investigation was an evaluation of the effectiveness of a numerical species diversity index to indicate aquatic degradation due to acid mine drainage. Conceptually, a high index of macroinvertebrate species diversity should indicate an unstressed aquatic system, while a low diversity index should indicate a stressed aquatic system.

Two independent sampling stations were chosen for this study, one located downstream from the acid mine discharge point and the other located upstream. For 12 monthly samples collected at the downstream station, the species diversity index had a mean value $\bar{x}_1 = 3.11$ and a standard deviation $s_1 = 0.771$, while 10 monthly samples collected at the upstream station had a mean index value $\bar{x}_2 = 2.04$ and a standard deviation $s_2 = 0.448$. Find a 90% confidence interval for the difference between the population means for the two locations, assuming that the populations are approximately normally distributed with equal variances.

Solution: Let μ_1 and μ_2 represent the population means, respectively, for the species diversity indices at the downstream and upstream stations. We wish to find a 90% confidence interval for $\mu_1 - \mu_2$. Our point estimate of $\mu_1 - \mu_2$ is

$$\bar{x}_1 - \bar{x}_2 = 3.11 - 2.04 = 1.07.$$

The pooled estimate, s_p^2 , of the common variance, σ^2 , is

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{(11)(0.771^2) + (9)(0.448^2)}{12 + 10 - 2} = 0.417.$$

Taking the square root, we obtain $s_p = 0.646$. Using $\alpha = 0.1$, we find in Table A.4 that $t_{0.05} = 1.725$ for $v = n_1 + n_2 - 2 = 20$ degrees of freedom. Therefore, the 90% confidence interval for $\mu_1 - \mu_2$ is

$$1.07 - (1.725)(0.646)\sqrt{\frac{1}{12} + \frac{1}{10}} < \mu_1 - \mu_2 < 1.07 + (1.725)(0.646)\sqrt{\frac{1}{12} + \frac{1}{10}},$$

which simplifies to $0.593 < \mu_1 - \mu_2 < 1.547$. ■

Interpretation of the Confidence Interval

For the case of a single parameter, the confidence interval simply provides error bounds on the parameter. Values contained in the interval should be viewed as reasonable values given the experimental data. In the case of a difference between two means, the interpretation can be extended to one of comparing the two means. For example, if we have high confidence that a difference $\mu_1 - \mu_2$ is positive, we would certainly infer that $\mu_1 > \mu_2$ with little risk of being in error. For example, in Example 9.11, we are 90% confident that the interval from 0.593 to 1.547 contains the difference of the population means for values of the species diversity index at the two stations. The fact that both confidence limits are positive indicates that, on the average, the index for the station located downstream from the discharge point is greater than the index for the station located upstream.

Equal Sample Sizes

The procedure for constructing confidence intervals for $\mu_1 - \mu_2$ with $\sigma_1 = \sigma_2 = \sigma$ unknown requires the assumption that the populations are normal. Slight departures from either the equal variance or the normality assumption do not seriously alter the degree of confidence for our interval. (A procedure is presented in Chapter 10 for testing the equality of two unknown population variances based on the information provided by the sample variances.) If the population variances are considerably different, we still obtain reasonable results when the populations are normal, provided that $n_1 = n_2$. Therefore, in planning an experiment, one should make every effort to equalize the size of the samples.

Unknown and Unequal Variances

Let us now consider the problem of finding an interval estimate of $\mu_1 - \mu_2$ when the unknown population variances are not likely to be equal. The statistic most often used in this case is

$$T' = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{(S_1^2/n_1) + (S_2^2/n_2)}},$$

which has approximately a t -distribution with v degrees of freedom, where

$$v = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{[(s_1^2/n_1)^2/(n_1 - 1)] + [(s_2^2/n_2)^2/(n_2 - 1)]}.$$

Since v is seldom an integer, we *round it down* to the nearest whole number. The above estimate of the degrees of freedom is called the Satterthwaite approximation (Satterthwaite, 1946, in the Bibliography).

Using the statistic T' , we write

$$P(-t_{\alpha/2} < T' < t_{\alpha/2}) \approx 1 - \alpha,$$

where $t_{\alpha/2}$ is the value of the t -distribution with v degrees of freedom, above which we find an area of $\alpha/2$. Substituting for T' in the inequality and following the same steps as before, we state the final result.

Confidence
Interval for
 $\mu_1 - \mu_2$, $\sigma_1^2 \neq \sigma_2^2$
and Both
Unknown

If $\bar{x}_1$ and s_1^2 and $\bar{x}_2$ and s_2^2 are the means and variances of independent random samples of sizes n_1 and n_2 , respectively, from approximately normal populations with unknown and unequal variances, an approximate $100(1 - \alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is given by

$$(\bar{x}_1 - \bar{x}_2) - t_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}},$$

where $t_{\alpha/2}$ is the t -value with

$$v = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{[(s_1^2/n_1)^2/(n_1 - 1)] + [(s_2^2/n_2)^2/(n_2 - 1)]}$$

degrees of freedom, leaving an area of $\alpha/2$ to the right.

Note that the expression for v above involves random variables, and thus v is an *estimate* of the degrees of freedom. In applications, this estimate will not result in a whole number, and thus the analyst must round down to the nearest integer to achieve the desired confidence.

Before we illustrate the above confidence interval with an example, we should point out that all the confidence intervals on $\mu_1 - \mu_2$ are of the same general form as those on a single mean; namely, they can be written as

$$\text{point estimate} \pm t_{\alpha/2} \widehat{\text{s.e.}}(\text{point estimate})$$

or

$$\text{point estimate} \pm z_{\alpha/2} \text{s.e.}(\text{point estimate}).$$

For example, in the case where $\sigma_1 = \sigma_2 = \sigma$, the estimated standard error of $\bar{x}_1 - \bar{x}_2$ is $s_p \sqrt{1/n_1 + 1/n_2}$. For the case where $\sigma_1^2 \neq \sigma_2^2$,

$$\widehat{\text{s.e.}}(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}.$$

Example 9.12: A study was conducted by the Department of Zoology at the Virginia Tech to estimate the difference in the amounts of the chemical orthophosphorus measured at two different stations on the James River. Orthophosphorus was measured in milligrams per liter. Fifteen samples were collected from station 1, and 12 samples were obtained from station 2. The 15 samples from station 1 had an average orthophosphorus content of 3.84 milligrams per liter and a standard deviation of 3.07 milligrams per liter, while the 12 samples from station 2 had an average content of 1.49 milligrams per liter and a standard deviation of 0.80 milligram per liter. Find a 95% confidence interval for the difference in the true average orthophosphorus contents at these two stations, assuming that the observations came from normal populations with different variances.

Solution: For station 1, we have $\bar{x}_1 = 3.84$, $s_1 = 3.07$, and $n_1 = 15$. For station 2, $\bar{x}_2 = 1.49$, $s_2 = 0.80$, and $n_2 = 12$. We wish to find a 95% confidence interval for $\mu_1 - \mu_2$.

Since the population variances are assumed to be unequal, we can only find an approximate 95% confidence interval based on the t -distribution with v degrees of freedom, where

$$v = \frac{(3.07^2/15 + 0.80^2/12)^2}{[(3.07^2/15)^2/14] + [(0.80^2/12)^2/11]} = 16.3 \approx 16.$$

Our point estimate of $\mu_1 - \mu_2$ is

$$\bar{x}_1 - \bar{x}_2 = 3.84 - 1.49 = 2.35.$$

Using $\alpha = 0.05$, we find in Table A.4 that $t_{0.025} = 2.120$ for $v = 16$ degrees of freedom. Therefore, the 95% confidence interval for $\mu_1 - \mu_2$ is

$$2.35 - 2.120\sqrt{\frac{3.07^2}{15} + \frac{0.80^2}{12}} < \mu_1 - \mu_2 < 2.35 + 2.120\sqrt{\frac{3.07^2}{15} + \frac{0.80^2}{12}},$$

which simplifies to $0.60 < \mu_1 - \mu_2 < 4.10$. Hence, we are 95% confident that the interval from 0.60 to 4.10 milligrams per liter contains the difference of the true average orthophosphorus contents for these two locations. 

When two population variances are unknown, the assumption of equal variances or unequal variances may be precarious. In Section 10.10, a procedure will be introduced that will aid in discriminating between the equal variance and the unequal variance situation.

9.9 Paired Observations

At this point, we shall consider estimation procedures for the difference of two means when the samples are not independent and the variances of the two populations are not necessarily equal. The situation considered here deals with a very special experimental condition, namely that of *paired observations*. Unlike in the situation described earlier, the conditions of the two populations are not assigned randomly to experimental units. Rather, each homogeneous experimental unit receives both population conditions; as a result, each experimental unit has a pair of observations, one for each population. For example, if we run a test on a new diet using 15 individuals, the weights before and after going on the diet form the information for our two samples. The two populations are “before” and “after,” and the experimental unit is the individual. Obviously, the observations in a pair have something in common. To determine if the diet is effective, we consider the differences $d_1, d_2, \dots, d_n$ in the paired observations. These differences are the values of a random sample $D_1, D_2, \dots, D_n$ from a population of differences that we shall assume to be normally distributed with mean $\mu_D = \mu_1 - \mu_2$ and variance σ_D^2 . We estimate σ_D^2 by s_d^2 , the variance of the differences that constitute our sample. The point estimator of μ_D is given by $\bar{D}$.

When Should Pairing Be Done?

Pairing observations in an experiment is a strategy that can be employed in many fields of application. The reader will be exposed to this concept in material related

to hypothesis testing in Chapter 10 and experimental design issues in Chapters 13 and 15. Selecting experimental units that are relatively homogeneous (within the units) and allowing each unit to experience both population conditions reduces the effective experimental error variance (in this case, σ_D^2). The reader may visualize the i th pair difference as

$$D_i = X_{1i} - X_{2i}.$$

Since the two observations are taken on the sample experimental unit, they are not independent and, in fact,

$$\text{Var}(D_i) = \text{Var}(X_{1i} - X_{2i}) = \sigma_1^2 + \sigma_2^2 - 2 \text{Cov}(X_{1i}, X_{2i}).$$

Now, intuitively, we expect that σ_D^2 should be reduced because of the similarity in nature of the “errors” of the two observations within a given experimental unit, and this comes through in the expression above. One certainly expects that if the unit is homogeneous, the covariance is positive. As a result, the gain in quality of the confidence interval over that obtained without pairing will be greatest when there is homogeneity within units and large differences as one goes from unit to unit. One should keep in mind that the performance of the confidence interval will depend on the standard error of $\bar{D}$, which is, of course, $\sigma_D/\sqrt{n}$, where n is the number of pairs. As we indicated earlier, the intent of pairing is to reduce σ_D .

Tradeoff between Reducing Variance and Losing Degrees of Freedom

Comparing the confidence intervals obtained with and without pairing makes apparent that there is a tradeoff involved. Although pairing should indeed reduce variance and hence reduce the standard error of the point estimate, the degrees of freedom are reduced by reducing the problem to a one-sample problem. As a result, the $t_{\alpha/2}$ point attached to the standard error is adjusted accordingly. Thus, pairing may be counterproductive. This would certainly be the case if one experienced only a modest reduction in variance (through σ_D^2) by pairing.

Another illustration of pairing involves choosing n pairs of subjects, with each pair having a similar characteristic such as IQ, age, or breed, and then selecting one member of each pair at random to yield a value of X_1 , leaving the other member to provide the value of X_2 . In this case, X_1 and X_2 might represent the grades obtained by two individuals of equal IQ when one of the individuals is assigned at random to a class using the conventional lecture approach while the other individual is assigned to a class using programmed materials.

A $100(1 - \alpha)\%$ confidence interval for μ_D can be established by writing

$$P(-t_{\alpha/2} < T < t_{\alpha/2}) = 1 - \alpha,$$

where $T = \frac{\bar{D} - \mu_D}{S_d/\sqrt{n}}$ and $t_{\alpha/2}$, as before, is a value of the t -distribution with $n - 1$ degrees of freedom.

It is now a routine procedure to replace T by its definition in the inequality above and carry out the mathematical steps that lead to the following $100(1 - \alpha)\%$ confidence interval for $\mu_1 - \mu_2 = \mu_D$.

Confidence Interval for $\mu_D = \mu_1 - \mu_2$ for Paired Observations If $\bar{d}$ and s_d are the mean and standard deviation, respectively, of the normally distributed differences of n random pairs of measurements, a $100(1 - \alpha)\%$ confidence interval for $\mu_D = \mu_1 - \mu_2$ is

$$\bar{d} - t_{\alpha/2} \frac{s_d}{\sqrt{n}} < \mu_D < \bar{d} + t_{\alpha/2} \frac{s_d}{\sqrt{n}},$$

where $t_{\alpha/2}$ is the t -value with $v = n - 1$ degrees of freedom, leaving an area of $\alpha/2$ to the right.

Example 9.13: A study published in *Chemosphere* reported the levels of the dioxin TCDD of 20 Massachusetts Vietnam veterans who were possibly exposed to Agent Orange. The TCDD levels in plasma and in fat tissue are listed in Table 9.1.

Find a 95% confidence interval for $\mu_1 - \mu_2$, where μ_1 and μ_2 represent the true mean TCDD levels in plasma and in fat tissue, respectively. Assume the distribution of the differences to be approximately normal.

Table 9.1: Data for Example 9.13

Veteran	TCDD Levels in Plasma	TCDD Levels in Fat Tissue	d_i	Veteran	TCDD Levels in Plasma	TCDD Levels in Fat Tissue	d_i
1	2.5	4.9	-2.4	11	6.9	7.0	-0.1
2	3.1	5.9	-2.8	12	3.3	2.9	0.4
3	2.1	4.4	-2.3	13	4.6	4.6	0.0
4	3.5	6.9	-3.4	14	1.6	1.4	0.2
5	3.1	7.0	-3.9	15	7.2	7.7	-0.5
6	1.8	4.2	-2.4	16	1.8	1.1	0.7
7	6.0	10.0	-4.0	17	20.0	11.0	9.0
8	3.0	5.5	-2.5	18	2.0	2.5	-0.5
9	36.0	41.0	-5.0	19	2.5	2.3	0.2
10	4.7	4.4	0.3	20	4.1	2.5	1.6

Source: Schecter, A. et al. "Partitioning of 2,3,7,8-chlorinated dibenzo-*p*-dioxins and dibenzofurans between adipose tissue and plasma lipid of 20 Massachusetts Vietnam veterans," *Chemosphere*, Vol. 20, Nos. 7-9, 1990, pp. 954-955 (Tables I and II).

Solution: We wish to find a 95% confidence interval for $\mu_1 - \mu_2$. Since the observations are paired, $\mu_1 - \mu_2 = \mu_D$. The point estimate of μ_D is $\bar{d} = -0.87$. The standard deviation, s_d , of the sample differences is

$$s_d = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (d_i - \bar{d})^2} = \sqrt{\frac{168.4220}{19}} = 2.9773.$$

Using $\alpha = 0.05$, we find in Table A.4 that $t_{0.025} = 2.093$ for $v = n - 1 = 19$ degrees of freedom. Therefore, the 95% confidence interval is

$$-0.8700 - (2.093) \left(\frac{2.9773}{\sqrt{20}} \right) < \mu_D < -0.8700 + (2.093) \left(\frac{2.9773}{\sqrt{20}} \right),$$

or simply $-2.2634 < \mu_D < 0.5234$, from which we can conclude that there is no significant difference between the mean TCDD level in plasma and the mean TCDD level in fat tissue. J

Exercises

9.35 A random sample of size $n_1 = 25$, taken from a normal population with a standard deviation $\sigma_1 = 5$, has a mean $\bar{x}_1 = 80$. A second random sample of size $n_2 = 36$, taken from a different normal population with a standard deviation $\sigma_2 = 3$, has a mean $\bar{x}_2 = 75$. Find a 94% confidence interval for $\mu_1 - \mu_2$.

9.36 Two kinds of thread are being compared for strength. Fifty pieces of each type of thread are tested under similar conditions. Brand A has an average tensile strength of 78.3 kilograms with a standard deviation of 5.6 kilograms, while brand B has an average tensile strength of 87.2 kilograms with a standard deviation of 6.3 kilograms. Construct a 95% confidence interval for the difference of the population means.

9.37 A study was conducted to determine if a certain treatment has any effect on the amount of metal removed in a pickling operation. A random sample of 100 pieces was immersed in a bath for 24 hours without the treatment, yielding an average of 12.2 millimeters of metal removed and a sample standard deviation of 1.1 millimeters. A second sample of 200 pieces was exposed to the treatment, followed by the 24-hour immersion in the bath, resulting in an average removal of 9.1 millimeters of metal with a sample standard deviation of 0.9 millimeter. Compute a 98% confidence interval estimate for the difference between the population means. Does the treatment appear to reduce the mean amount of metal removed?

9.38 Two catalysts in a batch chemical process, are being compared for their effect on the output of the process reaction. A sample of 12 batches was prepared using catalyst 1, and a sample of 10 batches was prepared using catalyst 2. The 12 batches for which catalyst 1 was used in the reaction gave an average yield of 85 with a sample standard deviation of 4, and the 10 batches for which catalyst 2 was used gave an average yield of 81 and a sample standard deviation of 5. Find a 90% confidence interval for the difference between the population means, assuming that the populations are approximately normally distributed with equal variances.

9.39 Students may choose between a 3-semester-hour physics course without labs and a 4-semester-hour course with labs. The final written examination is the same for each section. If 12 students in the section with

labs made an average grade of 84 with a standard deviation of 4, and 18 students in the section without labs made an average grade of 77 with a standard deviation of 6, find a 99% confidence interval for the difference between the average grades for the two courses. Assume the populations to be approximately normally distributed with equal variances.

9.40 In a study conducted at Virginia Tech on the development of ectomycorrhizal, a symbiotic relationship between the roots of trees and a fungus, in which minerals are transferred from the fungus to the trees and sugars from the trees to the fungus, 20 northern red oak seedlings exposed to the fungus *Pisolithus tinctorius* were grown in a greenhouse. All seedlings were planted in the same type of soil and received the same amount of sunshine and water. Half received no nitrogen at planting time, to serve as a control, and the other half received 368 ppm of nitrogen in the form NaNO_3 . The stem weights, in grams, at the end of 140 days were recorded as follows:

No Nitrogen	Nitrogen
0.32	0.26
0.53	0.43
0.28	0.47
0.37	0.49
0.47	0.52
0.43	0.75
0.36	0.79
0.42	0.86
0.38	0.62
0.43	0.46

Construct a 95% confidence interval for the difference in the mean stem weight between seedlings that receive no nitrogen and those that receive 368 ppm of nitrogen. Assume the populations to be normally distributed with equal variances.

9.41 The following data represent the length of time, in days, to recovery for patients randomly treated with one of two medications to clear up severe bladder infections:

Medication 1	Medication 2
$n_1 = 14$	$n_2 = 16$
$\bar{x}_1 = 17$	$\bar{x}_2 = 19$
$s_1^2 = 1.5$	$s_2^2 = 1.8$

Find a 99% confidence interval for the difference $\mu_2 - \mu_1$

in the mean recovery times for the two medications, assuming normal populations with equal variances.

9.42 An experiment reported in *Popular Science* compared fuel economies for two types of similarly equipped diesel mini-trucks. Let us suppose that 12 Volkswagen and 10 Toyota trucks were tested in 90-kilometer-per-hour steady-paced trials. If the 12 Volkswagen trucks averaged 16 kilometers per liter with a standard deviation of 1.0 kilometer per liter and the 10 Toyota trucks averaged 11 kilometers per liter with a standard deviation of 0.8 kilometer per liter, construct a 90% confidence interval for the difference between the average kilometers per liter for these two mini-trucks. Assume that the distances per liter for the truck models are approximately normally distributed with equal variances.

9.43 A taxi company is trying to decide whether to purchase brand *A* or brand *B* tires for its fleet of taxis. To estimate the difference in the two brands, an experiment is conducted using 12 of each brand. The tires are run until they wear out. The results are

Brand *A*: $\bar{x}_1 = 36,300$ kilometers,
 $s_1 = 5000$ kilometers.
 Brand *B*: $\bar{x}_2 = 38,100$ kilometers,
 $s_2 = 6100$ kilometers.

Compute a 95% confidence interval for $\mu_A - \mu_B$ assuming the populations to be approximately normally distributed. You may not assume that the variances are equal.

9.44 Referring to Exercise 9.43, find a 99% confidence interval for $\mu_1 - \mu_2$ if tires of the two brands are assigned at random to the left and right rear wheels of 8 taxis and the following distances, in kilometers, are recorded:

Taxi	Brand <i>A</i>	Brand <i>B</i>
1	34,400	36,700
2	45,500	46,800
3	36,700	37,700
4	32,000	31,100
5	48,400	47,800
6	32,800	36,400
7	38,100	38,900
8	30,100	31,500

Assume that the differences of the distances are approximately normally distributed.

9.45 The federal government awarded grants to the agricultural departments of 9 universities to test the yield capabilities of two new varieties of wheat. Each variety was planted on a plot of equal area at each university, and the yields, in kilograms per plot, were recorded as follows:

Variety	University								
	1	2	3	4	5	6	7	8	9
1	38	23	35	41	44	29	37	31	38
2	45	25	31	38	50	33	36	40	43

Find a 95% confidence interval for the mean difference between the yields of the two varieties, assuming the differences of yields to be approximately normally distributed. Explain why pairing is necessary in this problem.

9.46 The following data represent the running times of films produced by two motion-picture companies.

Company	Time (minutes)							
I	103	94	110	87	98			
II	97	82	123	92	175	88	118	

Compute a 90% confidence interval for the difference between the average running times of films produced by the two companies. Assume that the running-time differences are approximately normally distributed with unequal variances.

9.47 *Fortune* magazine (March 1997) reported the total returns to investors for the 10 years prior to 1996 and also for 1996 for 431 companies. The total returns for 10 of the companies are listed below. Find a 95% confidence interval for the mean change in percent return to investors.

Company	Total Return to Investors	
	1986–96	1996
Coca-Cola	29.8%	43.3%
Mirage Resorts	27.9%	25.4%
Merck	22.1%	24.0%
Microsoft	44.5%	88.3%
Johnson & Johnson	22.2%	18.1%
Intel	43.8%	131.2%
Pfizer	21.7%	34.0%
Procter & Gamble	21.9%	32.1%
Berkshire Hathaway	28.3%	6.2%
S&P 500	11.8%	20.3%

9.48 An automotive company is considering two types of batteries for its automobile. Sample information on battery life is collected for 20 batteries of type *A* and 20 batteries of type *B*. The summary statistics are $\bar{x}_A = 32.91$, $\bar{x}_B = 30.47$, $s_A = 1.57$, and $s_B = 1.74$. Assume the data on each battery are normally distributed and assume $\sigma_A = \sigma_B$.

- Find a 95% confidence interval on $\mu_A - \mu_B$.
- Draw a conclusion from (a) that provides insight into whether *A* or *B* should be adopted.

9.49 Two different brands of latex paint are being considered for use. Fifteen specimens of each type of

paint were selected, and the drying times, in hours, were as follows:

Paint A					Paint B				
3.5	2.7	3.9	4.2	3.6	4.7	3.9	4.5	5.5	4.0
2.7	3.3	5.2	4.2	2.9	5.3	4.3	6.0	5.2	3.7
4.4	5.2	4.0	4.1	3.4	5.5	6.2	5.1	5.4	4.8

Assume the drying time is normally distributed with $\sigma_A = \sigma_B$. Find a 95% confidence interval on $\mu_B - \mu_A$, where μ_A and μ_B are the mean drying times.

9.50 Two levels (low and high) of insulin doses are given to two groups of diabetic rats to check the insulin-binding capacity, yielding the following data:

Low dose:	$n_1 = 8$	$\bar{x}_1 = 1.98$	$s_1 = 0.51$
High dose:	$n_2 = 13$	$\bar{x}_2 = 1.30$	$s_2 = 0.35$

Assume that the variances are equal. Give a 95% confidence interval for the difference in the true average insulin-binding capacity between the two samples.

9.10 Single Sample: Estimating a Proportion

A point estimator of the proportion p in a binomial experiment is given by the statistic $\hat{P} = X/n$, where X represents the number of successes in n trials. Therefore, the sample proportion $\hat{p} = x/n$ will be used as the point estimate of the parameter p .

If the unknown proportion p is not expected to be too close to 0 or 1, we can establish a confidence interval for p by considering the sampling distribution of $\hat{P}$. Designating a failure in each binomial trial by the value 0 and a success by the value 1, the number of successes, x , can be interpreted as the sum of n values consisting only of 0 and 1s, and $\hat{p}$ is just the sample mean of these n values. Hence, by the Central Limit Theorem, for n sufficiently large, $\hat{P}$ is approximately normally distributed with mean

$$\mu_{\hat{P}} = E(\hat{P}) = E\left(\frac{X}{n}\right) = \frac{np}{n} = p$$

and variance

$$\sigma_{\hat{P}}^2 = \sigma_{X/n}^2 = \frac{\sigma_X^2}{n^2} = \frac{npq}{n^2} = \frac{pq}{n}.$$

Therefore, we can assert that

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha, \text{ with } Z = \frac{\hat{P} - p}{\sqrt{pq/n}},$$

and $z_{\alpha/2}$ is the value above which we find an area of $\alpha/2$ under the standard normal curve. Substituting for Z , we write

$$P\left(-z_{\alpha/2} < \frac{\hat{P} - p}{\sqrt{pq/n}} < z_{\alpha/2}\right) = 1 - \alpha.$$

When n is large, very little error is introduced by substituting the point estimate $\hat{p} = x/n$ for the p under the radical sign. Then we can write

$$P\left(\hat{P} - z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}} < p < \hat{P} + z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}\right) \approx 1 - \alpha.$$

On the other hand, by solving for p in the quadratic inequality above,

$$-z_{\alpha/2} < \frac{\hat{P} - p}{\sqrt{pq/n}} < z_{\alpha/2},$$

we obtain another form of the confidence interval for p with limits

$$\frac{\hat{p} + \frac{z_{\alpha/2}^2}{2n}}{1 + \frac{z_{\alpha/2}^2}{n}} \pm \frac{z_{\alpha/2}}{1 + \frac{z_{\alpha/2}^2}{n}} \sqrt{\frac{\hat{p}\hat{q}}{n} + \frac{z_{\alpha/2}^2}{4n^2}}.$$

For a random sample of size n , the sample proportion $\hat{p} = x/n$ is computed, and the following approximate $100(1 - \alpha)\%$ confidence intervals for p can be obtained.

Large-Sample Confidence Intervals for p	If $\hat{p}$ is the proportion of successes in a random sample of size n and $\hat{q} = 1 - \hat{p}$, an approximate $100(1 - \alpha)\%$ confidence interval, for the binomial parameter p is given by (method 1)
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$$\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} < p < \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

or by (method 2)

$$\frac{\hat{p} + \frac{z_{\alpha/2}^2}{2n}}{1 + \frac{z_{\alpha/2}^2}{n}} - \frac{z_{\alpha/2}}{1 + \frac{z_{\alpha/2}^2}{n}} \sqrt{\frac{\hat{p}\hat{q}}{n} + \frac{z_{\alpha/2}^2}{4n^2}} < p < \frac{\hat{p} + \frac{z_{\alpha/2}^2}{2n}}{1 + \frac{z_{\alpha/2}^2}{n}} + \frac{z_{\alpha/2}}{1 + \frac{z_{\alpha/2}^2}{n}} \sqrt{\frac{\hat{p}\hat{q}}{n} + \frac{z_{\alpha/2}^2}{4n^2}},$$

where $z_{\alpha/2}$ is the z -value leaving an area of $\alpha/2$ to the right.

When n is small and the unknown proportion p is believed to be close to 0 or to 1, the confidence-interval procedure established here is unreliable and, therefore, should not be used. To be on the safe side, one should require both $n\hat{p}$ and $n\hat{q}$ to be greater than or equal to 5. The methods for finding a confidence interval for the binomial parameter p are also applicable when the binomial distribution is being used to approximate the hypergeometric distribution, that is, when n is small relative to N , as illustrated by Example 9.14.

Note that although method 2 yields more accurate results, it is more complicated to calculate, and the gain in accuracy that it provides diminishes when the sample size is large enough. Hence, method 1 is commonly used in practice.

Example 9.14: In a random sample of $n = 500$ families owning television sets in the city of Hamilton, Canada, it is found that $x = 340$ subscribe to HBO. Find a 95% confidence interval for the actual proportion of families with television sets in this city that subscribe to HBO.

Solution: The point estimate of p is $\hat{p} = 340/500 = 0.68$. Using Table A.3, we find that $z_{0.025} = 1.96$. Therefore, using method 1, the 95% confidence interval for p is

$$0.68 - 1.96 \sqrt{\frac{(0.68)(0.32)}{500}} < p < 0.68 + 1.96 \sqrt{\frac{(0.68)(0.32)}{500}},$$

which simplifies to $0.6391 < p < 0.7209$.

If we use method 2, we can obtain

$$\frac{0.68 + \frac{1.96^2}{(2)(500)}}{1 + \frac{1.96^2}{500}} \pm \frac{1.96}{1 + \frac{1.96^2}{500}} \sqrt{\frac{(0.68)(0.32)}{500} + \frac{1.96^2}{(4)(500^2)}} = 0.6786 \pm 0.0408,$$

which simplifies to $0.6378 < p < 0.7194$. Apparently, when n is large (500 here), both methods yield very similar results. ┐

If p is the center value of a $100(1 - \alpha)\%$ confidence interval, then $\hat{p}$ estimates p without error. Most of the time, however, $\hat{p}$ will not be exactly equal to p and the point estimate will be in error. The size of this error will be the positive difference that separates p and $\hat{p}$, and we can be $100(1 - \alpha)\%$ confident that this difference will not exceed $z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n}$. We can readily see this if we draw a diagram of a typical confidence interval, as in Figure 9.6. Here we use method 1 to estimate the error.

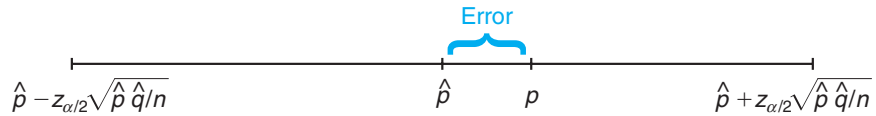


Figure 9.6: Error in estimating p by $\hat{p}$.

Theorem 9.3: If $\hat{p}$ is used as an estimate of p , we can be $100(1 - \alpha)\%$ confident that the error will not exceed $z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n}$.

In Example 9.14, we are 95% confident that the sample proportion $\hat{p} = 0.68$ differs from the true proportion p by an amount not exceeding 0.04.

Choice of Sample Size

Let us now determine how large a sample is necessary to ensure that the error in estimating p will be less than a specified amount e . By Theorem 9.3, we must choose n such that $z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n} = e$.

Theorem 9.4: If $\hat{p}$ is used as an estimate of p , we can be $100(1 - \alpha)\%$ confident that the error will be less than a specified amount e when the sample size is approximately

$$n = \frac{z_{\alpha/2}^2 \hat{p}\hat{q}}{e^2}.$$

Theorem 9.4 is somewhat misleading in that we must use $\hat{p}$ to determine the sample size n , but $\hat{p}$ is computed from the sample. If a crude estimate of p can be made without taking a sample, this value can be used to determine n . Lacking such an estimate, we could take a preliminary sample of size $n \geq 30$ to provide an estimate of p . Using Theorem 9.4, we could determine approximately how many observations are needed to provide the desired degree of accuracy. Note that fractional values of n are rounded up to the next whole number.

Example 9.15: How large a sample is required if we want to be 95% confident that our estimate of p in Example 9.14 is within 0.02 of the true value?

Solution: Let us treat the 500 families as a preliminary sample, providing an estimate $\hat{p} = 0.68$. Then, by Theorem 9.4,

$$n = \frac{(1.96)^2(0.68)(0.32)}{(0.02)^2} = 2089.8 \approx 2090.$$

Therefore, if we base our estimate of p on a random sample of size 2090, we can be 95% confident that our sample proportion will not differ from the true proportion by more than 0.02. ■

Occasionally, it will be impractical to obtain an estimate of p to be used for determining the sample size for a specified degree of confidence. If this happens, an upper bound for n is established by noting that $\hat{p}\hat{q} = \hat{p}(1 - \hat{p})$, which must be at most $1/4$, since $\hat{p}$ must lie between 0 and 1. This fact may be verified by completing the square. Hence

$$\hat{p}(1 - \hat{p}) = -(\hat{p}^2 - \hat{p}) = \frac{1}{4} - \left(\hat{p}^2 - \hat{p} + \frac{1}{4}\right) = \frac{1}{4} - \left(\hat{p} - \frac{1}{2}\right)^2,$$

which is always less than $1/4$ except when $\hat{p} = 1/2$, and then $\hat{p}\hat{q} = 1/4$. Therefore, if we substitute $\hat{p} = 1/2$ into the formula for n in Theorem 9.4 when, in fact, p actually differs from $1/2$, n will turn out to be larger than necessary for the specified degree of confidence; as a result, our degree of confidence will increase.

Theorem 9.5: If $\hat{p}$ is used as an estimate of p , we can be **at least** $100(1 - \alpha)\%$ confident that the error will not exceed a specified amount e when the sample size is

$$n = \frac{z_{\alpha/2}^2}{4e^2}.$$

Example 9.16: How large a sample is required if we want to be at least 95% confident that our estimate of p in Example 9.14 is within 0.02 of the true value?

Solution: Unlike in Example 9.15, we shall now assume that no preliminary sample has been taken to provide an estimate of p . Consequently, we can be at least 95% confident that our sample proportion will not differ from the true proportion by more than 0.02 if we choose a sample of size

$$n = \frac{(1.96)^2}{(4)(0.02)^2} = 2401.$$

Comparing the results of Examples 9.15 and 9.16, we see that information concerning p , provided by a preliminary sample or from experience, enables us to choose a smaller sample while maintaining our required degree of accuracy. ■

9.11 Two Samples: Estimating the Difference between Two Proportions

Consider the problem where we wish to estimate the difference between two binomial parameters p_1 and p_2 . For example, p_1 might be the proportion of smokers with lung cancer and p_2 the proportion of nonsmokers with lung cancer, and the problem is to estimate the difference between these two proportions. First, we select independent random samples of sizes n_1 and n_2 from the two binomial populations with means n_1p_1 and n_2p_2 and variances $n_1p_1q_1$ and $n_2p_2q_2$, respectively; then we determine the numbers x_1 and x_2 of people in each sample with lung cancer and form the proportions $\hat{p}_1 = x_1/n$ and $\hat{p}_2 = x_2/n$. A point estimator of the difference between the two proportions, $p_1 - p_2$, is given by the statistic $\hat{P}_1 - \hat{P}_2$. Therefore, the difference of the sample proportions, $\hat{p}_1 - \hat{p}_2$, will be used as the point estimate of $p_1 - p_2$.

A confidence interval for $p_1 - p_2$ can be established by considering the sampling distribution of $\hat{P}_1 - \hat{P}_2$. From Section 9.10 we know that $\hat{P}_1$ and $\hat{P}_2$ are each approximately normally distributed, with means p_1 and p_2 and variances p_1q_1/n_1 and p_2q_2/n_2 , respectively. Choosing independent samples from the two populations ensures that the variables $\hat{P}_1$ and $\hat{P}_2$ will be independent, and then by the reproductive property of the normal distribution established in Theorem 7.11, we conclude that $\hat{P}_1 - \hat{P}_2$ is approximately normally distributed with mean

$$\mu_{\hat{P}_1 - \hat{P}_2} = p_1 - p_2$$

and variance

$$\sigma_{\hat{P}_1 - \hat{P}_2}^2 = \frac{p_1q_1}{n_1} + \frac{p_2q_2}{n_2}.$$

Therefore, we can assert that

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha,$$

where

$$Z = \frac{(\hat{P}_1 - \hat{P}_2) - (p_1 - p_2)}{\sqrt{p_1q_1/n_1 + p_2q_2/n_2}}$$

and $z_{\alpha/2}$ is the value above which we find an area of $\alpha/2$ under the standard normal curve. Substituting for Z , we write

$$P \left[-z_{\alpha/2} < \frac{(\hat{P}_1 - \hat{P}_2) - (p_1 - p_2)}{\sqrt{p_1q_1/n_1 + p_2q_2/n_2}} < z_{\alpha/2} \right] = 1 - \alpha.$$

After performing the usual mathematical manipulations, we replace p_1 , p_2 , q_1 , and q_2 under the radical sign by their estimates $\hat{p}_1 = x_1/n_1$, $\hat{p}_2 = x_2/n_2$, $\hat{q}_1 = 1 - \hat{p}_1$, and $\hat{q}_2 = 1 - \hat{p}_2$, provided that $n_1\hat{p}_1$, $n_1\hat{q}_1$, $n_2\hat{p}_2$, and $n_2\hat{q}_2$ are all greater than or equal to 5, and the following approximate $100(1 - \alpha)\%$ confidence interval for $p_1 - p_2$ is obtained.

Large-Sample Confidence Interval for $p_1 - p_2$ If $\hat{p}_1$ and $\hat{p}_2$ are the proportions of successes in random samples of sizes n_1 and n_2 , respectively, $\hat{q}_1 = 1 - \hat{p}_1$, and $\hat{q}_2 = 1 - \hat{p}_2$, an approximate $100(1 - \alpha)\%$ confidence interval for the difference of two binomial parameters, $p_1 - p_2$, is given by

$$(\hat{p}_1 - \hat{p}_2) - z_{\alpha/2} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}} < p_1 - p_2 < (\hat{p}_1 - \hat{p}_2) + z_{\alpha/2} \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}},$$

where $z_{\alpha/2}$ is the z -value leaving an area of $\alpha/2$ to the right.

Example 9.17: A certain change in a process for manufacturing component parts is being considered. Samples are taken under both the existing and the new process so as to determine if the new process results in an improvement. If 75 of 1500 items from the existing process are found to be defective and 80 of 2000 items from the new process are found to be defective, find a 90% confidence interval for the true difference in the proportion of defectives between the existing and the new process.

Solution: Let p_1 and p_2 be the true proportions of defectives for the existing and new processes, respectively. Hence, $\hat{p}_1 = 75/1500 = 0.05$ and $\hat{p}_2 = 80/2000 = 0.04$, and the point estimate of $p_1 - p_2$ is

$$\hat{p}_1 - \hat{p}_2 = 0.05 - 0.04 = 0.01.$$

Using Table A.3, we find $z_{0.05} = 1.645$. Therefore, substituting into the formula, with

$$1.645 \sqrt{\frac{(0.05)(0.95)}{1500} + \frac{(0.04)(0.96)}{2000}} = 0.0117,$$

we find the 90% confidence interval to be $-0.0017 < p_1 - p_2 < 0.0217$. Since the interval contains the value 0, there is no reason to believe that the new process produces a significant decrease in the proportion of defectives over the existing method. 

Up to this point, all confidence intervals presented were of the form

$$\text{point estimate} \pm K \text{ s.e.}(\text{point estimate}),$$

where K is a constant (either t or normal percent point). This form is valid when the parameter is a mean, a difference between means, a proportion, or a difference between proportions, due to the symmetry of the t - and Z -distributions. However, it does not extend to variances and ratios of variances, which will be discussed in Sections 9.12 and 9.13.

Exercises

In this set of exercises, for estimation concerning one proportion, use only method 1 to obtain the confidence intervals, unless instructed otherwise.

9.51 In a random sample of 1000 homes in a certain city, it is found that 228 are heated by oil. Find 99% confidence intervals for the proportion of homes in this city that are heated by oil using both methods presented on page 297.

9.52 Compute 95% confidence intervals, using both methods on page 297, for the proportion of defective items in a process when it is found that a sample of size 100 yields 8 defectives.

9.53 (a) A random sample of 200 voters in a town is selected, and 114 are found to support an annexation suit. Find the 96% confidence interval for the fraction of the voting population favoring the suit.

(b) What can we assert with 96% confidence about the possible size of our error if we estimate the fraction of voters favoring the annexation suit to be 0.57?

9.54 A manufacturer of MP3 players conducts a set of comprehensive tests on the electrical functions of its product. All MP3 players must pass all tests prior to being sold. Of a random sample of 500 MP3 players, 15 failed one or more tests. Find a 90% confidence interval for the proportion of MP3 players from the population that pass all tests.

9.55 A new rocket-launching system is being considered for deployment of small, short-range rockets. The existing system has $p = 0.8$ as the probability of a successful launch. A sample of 40 experimental launches is made with the new system, and 34 are successful.

(a) Construct a 95% confidence interval for p .

(b) Would you conclude that the new system is better?

9.56 A geneticist is interested in the proportion of African males who have a certain minor blood disorder. In a random sample of 100 African males, 24 are found to be afflicted.

(a) Compute a 99% confidence interval for the proportion of African males who have this blood disorder.

(b) What can we assert with 99% confidence about the possible size of our error if we estimate the proportion of African males with this blood disorder to be 0.24?

9.57 (a) According to a report in the *Roanoke Times* & *World-News*, approximately $2/3$ of 1600 adults

polled by telephone said they think the space shuttle program is a good investment for the country. Find a 95% confidence interval for the proportion of American adults who think the space shuttle program is a good investment for the country.

(b) What can we assert with 95% confidence about the possible size of our error if we estimate the proportion of American adults who think the space shuttle program is a good investment to be $2/3$?

9.58 In the newspaper article referred to in Exercise 9.57, 32% of the 1600 adults polled said the U.S. space program should emphasize scientific exploration. How large a sample of adults is needed for the poll if one wishes to be 95% confident that the estimated percentage will be within 2% of the true percentage?

9.59 How large a sample is needed if we wish to be 96% confident that our sample proportion in Exercise 9.53 will be within 0.02 of the true fraction of the voting population?

9.60 How large a sample is needed if we wish to be 99% confident that our sample proportion in Exercise 9.51 will be within 0.05 of the true proportion of homes in the city that are heated by oil?

9.61 How large a sample is needed in Exercise 9.52 if we wish to be 98% confident that our sample proportion will be within 0.05 of the true proportion defective?

9.62 A conjecture by a faculty member in the microbiology department at Washington University School of Dental Medicine in St. Louis, Missouri, states that a couple of cups of either green or oolong tea each day will provide sufficient fluoride to protect your teeth from decay. How large a sample is needed to estimate the percentage of citizens in a certain town who favor having their water fluoridated if one wishes to be at least 99% confident that the estimate is within 1% of the true percentage?

9.63 A study is to be made to estimate the percentage of citizens in a town who favor having their water fluoridated. How large a sample is needed if one wishes to be at least 95% confident that the estimate is within 1% of the true percentage?

9.64 A study is to be made to estimate the proportion of residents of a certain city and its suburbs who favor the construction of a nuclear power plant near the city. How large a sample is needed if one wishes to be at least 95% confident that the estimate is within 0.04 of the true proportion of residents who favor the construction of the nuclear power plant?

9.65 A certain geneticist is interested in the proportion of males and females in the population who have a minor blood disorder. In a random sample of 1000 males, 250 are found to be afflicted, whereas 275 of 1000 females tested appear to have the disorder. Compute a 95% confidence interval for the difference between the proportions of males and females who have the blood disorder.

9.66 Ten engineering schools in the United States were surveyed. The sample contained 250 electrical engineers, 80 being women; 175 chemical engineers, 40 being women. Compute a 90% confidence interval for the difference between the proportions of women in these two fields of engineering. Is there a significant difference between the two proportions?

9.67 A clinical trial was conducted to determine if a certain type of inoculation has an effect on the incidence of a certain disease. A sample of 1000 rats was kept in a controlled environment for a period of 1 year, and 500 of the rats were given the inoculation. In the group not inoculated, there were 120 incidences of the disease, while 98 of the rats in the inoculated group contracted it. If p_1 is the probability of incidence of the disease in uninoculated rats and p_2 the probability of incidence in inoculated rats, compute a 90% confidence interval for $p_1 - p_2$.

9.68 In the study *Germination and Emergence of Broccoli*, conducted by the Department of Horticulture at Virginia Tech, a researcher found that at 5°C, 10 broccoli seeds out of 20 germinated, while at 15°C, 15 out of 20 germinated. Compute a 95% confidence interval for the difference between the proportions of germination at the two different temperatures and decide if there is a significant difference.

9.69 A survey of 1000 students found that 274 chose professional baseball team *A* as their favorite team. In a similar survey involving 760 students, 240 of them chose team *A* as their favorite. Compute a 95% confidence interval for the difference between the proportions of students favoring team *A* in the two surveys. Is there a significant difference?

9.70 According to *USA Today* (March 17, 1997), women made up 33.7% of the editorial staff at local TV stations in the United States in 1990 and 36.2% in 1994. Assume 20 new employees were hired as editorial staff.

- Estimate the number that would have been women in 1990 and 1994, respectively.
- Compute a 95% confidence interval to see if there is evidence that the proportion of women hired as editorial staff was higher in 1994 than in 1990.

9.12 Single Sample: Estimating the Variance

If a sample of size n is drawn from a normal population with variance σ^2 and the sample variance s^2 is computed, we obtain a value of the statistic S^2 . This computed sample variance is used as a point estimate of σ^2 . Hence, the statistic S^2 is called an estimator of σ^2 .

An interval estimate of σ^2 can be established by using the statistic

$$X^2 = \frac{(n-1)S^2}{\sigma^2}.$$

According to Theorem 8.4, the statistic X^2 has a chi-squared distribution with $n-1$ degrees of freedom when samples are chosen from a normal population. We may write (see Figure 9.7)

$$P(\chi_{1-\alpha/2}^2 < X^2 < \chi_{\alpha/2}^2) = 1 - \alpha,$$

where $\chi_{1-\alpha/2}^2$ and $\chi_{\alpha/2}^2$ are values of the chi-squared distribution with $n-1$ degrees of freedom, leaving areas of $1-\alpha/2$ and $\alpha/2$, respectively, to the right. Substituting for X^2 , we write

$$P\left[\chi_{1-\alpha/2}^2 < \frac{(n-1)S^2}{\sigma^2} < \chi_{\alpha/2}^2\right] = 1 - \alpha.$$

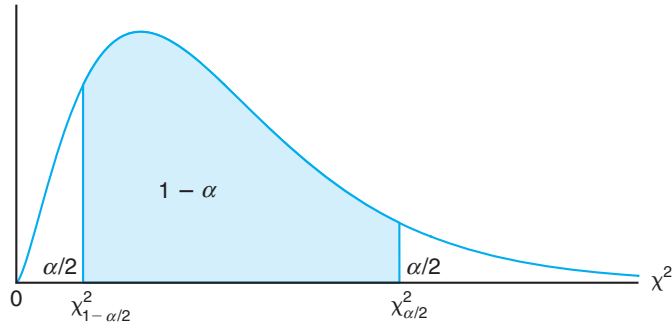


Figure 9.7: $P(\chi^2_{1-\alpha/2} < X^2 < \chi^2_{\alpha/2}) = 1 - \alpha$.

Dividing each term in the inequality by $(n-1)S^2$ and then inverting each term (thereby changing the sense of the inequalities), we obtain

$$P\left[\frac{(n-1)S^2}{\chi^2_{\alpha/2}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}}\right] = 1 - \alpha.$$

For a random sample of size n from a normal population, the sample variance s^2 is computed, and the following $100(1-\alpha)\%$ confidence interval for σ^2 is obtained.

Confidence Interval for σ^2	If s^2 is the variance of a random sample of size n from a normal population, a $100(1-\alpha)\%$ confidence interval for σ^2 is
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$$\frac{(n-1)s^2}{\chi^2_{\alpha/2}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}},$$

where $\chi^2_{\alpha/2}$ and $\chi^2_{1-\alpha/2}$ are χ^2 -values with $v = n-1$ degrees of freedom, leaving areas of $\alpha/2$ and $1-\alpha/2$, respectively, to the right.

An approximate $100(1-\alpha)\%$ confidence interval for σ is obtained by taking the square root of each endpoint of the interval for σ^2 .

Example 9.18: The following are the weights, in decagrams, of 10 packages of grass seed distributed by a certain company: 46.4, 46.1, 45.8, 47.0, 46.1, 45.9, 45.8, 46.9, 45.2, and 46.0. Find a 95% confidence interval for the variance of the weights of all such packages of grass seed distributed by this company, assuming a normal population.

Solution: First we find

$$\begin{aligned} s^2 &= \frac{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i\right)^2}{n(n-1)} \\ &= \frac{(10)(21,273.12) - (461.2)^2}{(10)(9)} = 0.286. \end{aligned}$$

To obtain a 95% confidence interval, we choose $\alpha = 0.05$. Then, using Table A.5 with $v = 9$ degrees of freedom, we find $\chi_{0.025}^2 = 19.023$ and $\chi_{0.975}^2 = 2.700$. Therefore, the 95% confidence interval for σ^2 is

$$\frac{(9)(0.286)}{19.023} < \sigma^2 < \frac{(9)(0.286)}{2.700},$$

or simply $0.135 < \sigma^2 < 0.953$. ┐

9.13 Two Samples: Estimating the Ratio of Two Variances

A point estimate of the ratio of two population variances σ_1^2/σ_2^2 is given by the ratio s_1^2/s_2^2 of the sample variances. Hence, the statistic S_1^2/S_2^2 is called an estimator of σ_1^2/σ_2^2 .

If σ_1^2 and σ_2^2 are the variances of normal populations, we can establish an interval estimate of σ_1^2/σ_2^2 by using the statistic

$$F = \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2}.$$

According to Theorem 8.8, the random variable F has an F -distribution with $v_1 = n_1 - 1$ and $v_2 = n_2 - 1$ degrees of freedom. Therefore, we may write (see Figure 9.8)

$$P[f_{1-\alpha/2}(v_1, v_2) < F < f_{\alpha/2}(v_1, v_2)] = 1 - \alpha,$$

where $f_{1-\alpha/2}(v_1, v_2)$ and $f_{\alpha/2}(v_1, v_2)$ are the values of the F -distribution with v_1 and v_2 degrees of freedom, leaving areas of $1 - \alpha/2$ and $\alpha/2$, respectively, to the right.

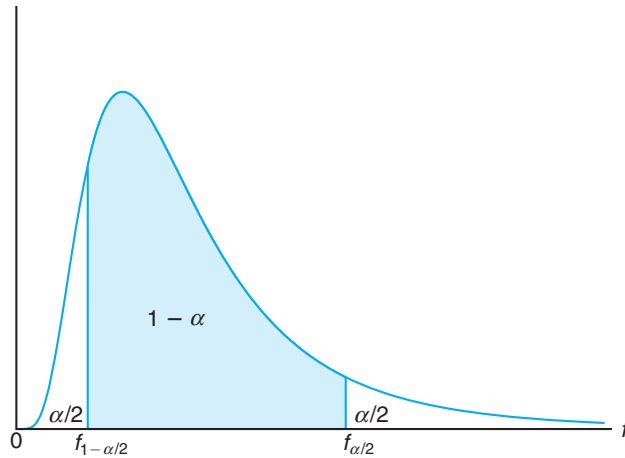


Figure 9.8: $P[f_{1-\alpha/2}(v_1, v_2) < F < f_{\alpha/2}(v_1, v_2)] = 1 - \alpha$.

Substituting for F , we write

$$P \left[f_{1-\alpha/2}(v_1, v_2) < \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2} < f_{\alpha/2}(v_1, v_2) \right] = 1 - \alpha.$$

Multiplying each term in the inequality by S_2^2/S_1^2 and then inverting each term, we obtain

$$P \left[\frac{S_1^2}{S_2^2} \frac{1}{f_{\alpha/2}(v_1, v_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{S_1^2}{S_2^2} \frac{1}{f_{1-\alpha/2}(v_1, v_2)} \right] = 1 - \alpha.$$

The results of Theorem 8.7 enable us to replace the quantity $f_{1-\alpha/2}(v_1, v_2)$ by $1/f_{\alpha/2}(v_2, v_1)$. Therefore,

$$P \left[\frac{S_1^2}{S_2^2} \frac{1}{f_{\alpha/2}(v_1, v_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{S_1^2}{S_2^2} f_{\alpha/2}(v_2, v_1) \right] = 1 - \alpha.$$

For any two independent random samples of sizes n_1 and n_2 selected from two normal populations, the ratio of the sample variances s_1^2/s_2^2 is computed, and the following $100(1 - \alpha)\%$ confidence interval for σ_1^2/σ_2^2 is obtained.

**Confidence
Interval for σ_1^2/σ_2^2**

If s_1^2 and s_2^2 are the variances of independent samples of sizes n_1 and n_2 , respectively, from normal populations, then a $100(1 - \alpha)\%$ confidence interval for σ_1^2/σ_2^2 is

$$\frac{s_1^2}{s_2^2} \frac{1}{f_{\alpha/2}(v_1, v_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2}{s_2^2} f_{\alpha/2}(v_2, v_1),$$

where $f_{\alpha/2}(v_1, v_2)$ is an f -value with $v_1 = n_1 - 1$ and $v_2 = n_2 - 1$ degrees of freedom, leaving an area of $\alpha/2$ to the right, and $f_{\alpha/2}(v_2, v_1)$ is a similar f -value with $v_2 = n_2 - 1$ and $v_1 = n_1 - 1$ degrees of freedom.

As in Section 9.12, an approximate $100(1 - \alpha)\%$ confidence interval for σ_1/σ_2 is obtained by taking the square root of each endpoint of the interval for σ_1^2/σ_2^2 .

Example 9.19: A confidence interval for the difference in the mean orthophosphorus contents, measured in milligrams per liter, at two stations on the James River was constructed in Example 9.12 on page 290 by assuming the normal population variance to be unequal. Justify this assumption by constructing 98% confidence intervals for σ_1^2/σ_2^2 and for σ_1/σ_2 , where σ_1^2 and σ_2^2 are the variances of the populations of orthophosphorus contents at station 1 and station 2, respectively.

Solution: From Example 9.12, we have $n_1 = 15$, $n_2 = 12$, $s_1 = 3.07$, and $s_2 = 0.80$. For a 98% confidence interval, $\alpha = 0.02$. Interpolating in Table A.6, we find $f_{0.01}(14, 11) \approx 4.30$ and $f_{0.01}(11, 14) \approx 3.87$. Therefore, the 98% confidence interval for σ_1^2/σ_2^2 is

$$\left(\frac{3.07^2}{0.80^2} \right) \left(\frac{1}{4.30} \right) < \frac{\sigma_1^2}{\sigma_2^2} < \left(\frac{3.07^2}{0.80^2} \right) (3.87),$$

which simplifies to $3.425 < \frac{\sigma_1^2}{\sigma_2^2} < 56.991$. Taking square roots of the confidence limits, we find that a 98% confidence interval for σ_1/σ_2 is

$$1.851 < \frac{\sigma_1}{\sigma_2} < 7.549.$$

Since this interval does not allow for the possibility of σ_1/σ_2 being equal to 1, we were correct in assuming that $\sigma_1 \neq \sigma_2$ or $\sigma_1^2 \neq \sigma_2^2$ in Example 9.12. ■

Exercises

9.71 A manufacturer of car batteries claims that the batteries will last, on average, 3 years with a variance of 1 year. If 5 of these batteries have lifetimes of 1.9, 2.4, 3.0, 3.5, and 4.2 years, construct a 95% confidence interval for σ^2 and decide if the manufacturer's claim that $\sigma^2 = 1$ is valid. Assume the population of battery lives to be approximately normally distributed.

9.72 A random sample of 20 students yielded a mean of $\bar{x} = 72$ and a variance of $s^2 = 16$ for scores on a college placement test in mathematics. Assuming the scores to be normally distributed, construct a 98% confidence interval for σ^2 .

9.73 Construct a 95% confidence interval for σ^2 in Exercise 9.9 on page 283.

9.74 Construct a 99% confidence interval for σ^2 in Exercise 9.11 on page 283.

9.75 Construct a 99% confidence interval for σ in Exercise 9.12 on page 283.

9.76 Construct a 90% confidence interval for σ in Exercise 9.13 on page 283.

9.77 Construct a 98% confidence interval for σ_1/σ_2 in Exercise 9.42 on page 295, where σ_1 and σ_2 are, respectively, the standard deviations for the distances traveled per liter of fuel by the Volkswagen and Toyota mini-trucks.

9.78 Construct a 90% confidence interval for σ_1^2/σ_2^2 in Exercise 9.43 on page 295. Were we justified in assuming that $\sigma_1^2 \neq \sigma_2^2$ when we constructed the confidence interval for $\mu_1 - \mu_2$?

9.79 Construct a 90% confidence interval for σ_1^2/σ_2^2 in Exercise 9.46 on page 295. Should we have assumed $\sigma_1^2 = \sigma_2^2$ in constructing our confidence interval for $\mu_I - \mu_{II}$?

9.80 Construct a 95% confidence interval for σ_A^2/σ_B^2 in Exercise 9.49 on page 295. Should the equal-variance assumption be used?

9.14 Maximum Likelihood Estimation (Optional)

Often the estimators of parameters have been those that appeal to intuition. The estimator $\bar{X}$ certainly seems reasonable as an estimator of a population mean μ . The virtue of S^2 as an estimator of σ^2 is underscored through the discussion of unbiasedness in Section 9.3. The estimator for a binomial parameter p is merely a sample proportion, which, of course, is an *average* and appeals to common sense. But there are many situations in which it is not at all obvious what the proper estimator should be. As a result, there is much to be learned by the student of statistics concerning different philosophies that produce different methods of estimation. In this section, we deal with the **method of maximum likelihood**.

Maximum likelihood estimation is one of the most important approaches to estimation in all of statistical inference. We will not give a thorough development of the method. Rather, we will attempt to communicate the philosophy of maximum likelihood and illustrate with examples that relate to other estimation problems discussed in this chapter.

The Likelihood Function

As the name implies, the method of maximum likelihood is that for which the *likelihood function* is maximized. The likelihood function is best illustrated through the use of an example with a discrete distribution and a single parameter. Denote by $X_1, X_2, \dots, X_n$ the independent random variables taken from a discrete probability distribution represented by $f(\mathbf{x}, \theta)$, where θ is a single parameter of the distribution. Now

$$\begin{aligned} L(x_1, x_2, \dots, x_n; \theta) &= f(x_1, x_2, \dots, x_n; \theta) \\ &= f(x_1, \theta) f(x_2, \theta) \cdots f(x_n, \theta) \end{aligned}$$

is the *joint distribution of the random variables*, often referred to as the likelihood function. Note that the variable of the likelihood function is θ , not $\mathbf{x}$. Denote by $x_1, x_2, \dots, x_n$ the observed values in a sample. In the case of a discrete random variable, the interpretation is very clear. The quantity $L(x_1, x_2, \dots, x_n; \theta)$, the *likelihood of the sample*, is the following joint probability:

$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n \mid \theta),$$

which is the probability of obtaining the sample values $x_1, x_2, \dots, x_n$. For the discrete case, the maximum likelihood estimator is one that results in a maximum value for this joint probability or maximizes the likelihood of the sample.

Consider a fictitious example where three items from an assembly line are inspected. The items are ruled either defective or nondefective, and thus the Bernoulli process applies. Testing the three items results in two nondefective items followed by a defective item. It is of interest to estimate p , the proportion nondefective in the process. The likelihood of the sample for this illustration is given by

$$p \cdot p \cdot q = p^2 q = p^2 - p^3,$$

where $q = 1 - p$. Maximum likelihood estimation would give an estimate of p for which the likelihood is maximized. It is clear that if we differentiate the likelihood with respect to p , set the derivative to zero, and solve, we obtain the value

$$\hat{p} = \frac{2}{3}.$$

Now, of course, in this situation $\hat{p} = 2/3$ is the sample proportion defective and is thus a reasonable estimator of the probability of a defective. The reader should attempt to understand that the philosophy of maximum likelihood estimation evolves from the notion that the reasonable estimator of a parameter based on sample information is *that parameter value that produces the largest probability of obtaining the sample*. This is, indeed, the interpretation for the discrete case, since the likelihood is the probability of jointly observing the values in the sample.

Now, while the interpretation of the likelihood function as a joint probability is confined to the discrete case, the notion of maximum likelihood extends to the estimation of parameters of a continuous distribution. We now present a formal definition of maximum likelihood estimation.

Definition 9.3:

Given independent observations $x_1, x_2, \dots, x_n$ from a probability density function (continuous case) or probability mass function (discrete case) $f(\mathbf{x}; \theta)$, the maximum likelihood estimator $\hat{\theta}$ is that which maximizes the likelihood function

$$L(x_1, x_2, \dots, x_n; \theta) = f(\mathbf{x}; \theta) = f(x_1, \theta)f(x_2, \theta) \cdots f(x_n, \theta).$$

Quite often it is convenient to work with the natural log of the likelihood function in finding the maximum of that function. Consider the following example dealing with the parameter μ of a Poisson distribution.

Example 9.20: Consider a Poisson distribution with probability mass function

$$f(x|\mu) = \frac{e^{-\mu} \mu^x}{x!}, \quad x = 0, 1, 2, \dots$$

Suppose that a random sample $x_1, x_2, \dots, x_n$ is taken from the distribution. What is the maximum likelihood estimate of μ ?

Solution: The likelihood function is

$$L(x_1, x_2, \dots, x_n; \mu) = \prod_{i=1}^n f(x_i|\mu) = \frac{e^{-n\mu} \mu^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}.$$

Now consider

$$\begin{aligned} \ln L(x_1, x_2, \dots, x_n; \mu) &= -n\mu + \sum_{i=1}^n x_i \ln \mu - \ln \prod_{i=1}^n x_i! \\ \frac{\partial \ln L(x_1, x_2, \dots, x_n; \mu)}{\partial \mu} &= -n + \sum_{i=1}^n \frac{x_i}{\mu}. \end{aligned}$$

Solving for $\hat{\mu}$, the maximum likelihood estimator, involves setting the derivative to zero and solving for the parameter. Thus,

$$\hat{\mu} = \sum_{i=1}^n \frac{x_i}{n} = \bar{x}.$$

The second derivative of the log-likelihood function is negative, which implies that the solution above indeed is a maximum. Since μ is the mean of the Poisson distribution (Chapter 5), the sample average would certainly seem like a reasonable estimator. └

The following example shows the use of the method of maximum likelihood for finding estimates of two parameters. We simply find the values of the parameters that maximize (jointly) the likelihood function.

Example 9.21: Consider a random sample $x_1, x_2, \dots, x_n$ from a normal distribution $N(\mu, \sigma)$. Find the maximum likelihood estimators for μ and σ^2 .

Solution: The likelihood function for the normal distribution is

$$L(x_1, x_2, \dots, x_n; \mu, \sigma^2) = \frac{1}{(2\pi)^{n/2}(\sigma^2)^{n/2}} \exp \left[-\frac{1}{2} \sum_{i=1}^n \left(\frac{x_i - \mu}{\sigma} \right)^2 \right].$$

Taking logarithms gives us

$$\ln L(x_1, x_2, \dots, x_n; \mu, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma^2 - \frac{1}{2} \sum_{i=1}^n \left(\frac{x_i - \mu}{\sigma} \right)^2.$$

Hence,

$$\frac{\partial \ln L}{\partial \mu} = \sum_{i=1}^n \left(\frac{x_i - \mu}{\sigma^2} \right)$$

and

$$\frac{\partial \ln L}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Setting both derivatives equal to 0, we obtain

$$\sum_{i=1}^n x_i - n\mu = 0 \quad \text{and} \quad n\sigma^2 = \sum_{i=1}^n (x_i - \mu)^2.$$

Thus, the maximum likelihood estimator of μ is given by

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x},$$

which is a pleasing result since $\bar{x}$ has played such an important role in this chapter as a point estimate of μ . On the other hand, the maximum likelihood estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Checking the second-order partial derivative matrix confirms that the solution results in a maximum of the likelihood function. ▮

It is interesting to note the distinction between the maximum likelihood estimator of σ^2 and the unbiased estimator S^2 developed earlier in this chapter. The numerators are identical, of course, and the denominator is the degrees of freedom $n-1$ for the unbiased estimator and n for the maximum likelihood estimator. Maximum likelihood estimators do not necessarily enjoy the property of unbiasedness. However, they do have very important asymptotic properties.

Example 9.22: Suppose 10 rats are used in a biomedical study where they are injected with cancer cells and then given a cancer drug that is designed to increase their survival rate. The survival times, in months, are 14, 17, 27, 18, 12, 8, 22, 13, 19, and 12. Assume

that the exponential distribution applies. Give a maximum likelihood estimate of the mean survival time.

Solution: From Chapter 6, we know that the probability density function for the exponential random variable X is

$$f(x, \beta) = \begin{cases} \frac{1}{\beta} e^{-x/\beta}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Thus, the log-likelihood function for the data, given $n = 10$, is

$$\ln L(x_1, x_2, \dots, x_{10}; \beta) = -10 \ln \beta - \frac{1}{\beta} \sum_{i=1}^{10} x_i.$$

Setting

$$\frac{\partial \ln L}{\partial \beta} = -\frac{10}{\beta} + \frac{1}{\beta^2} \sum_{i=1}^{10} x_i = 0$$

implies that

$$\hat{\beta} = \frac{1}{10} \sum_{i=1}^{10} x_i = \bar{x} = 16.2.$$

Evaluating the second derivative of the log-likelihood function at the value $\hat{\beta}$ above yields a negative value. As a result, the estimator of the parameter β , the population mean, is the sample average $\bar{x}$. ■

The following example shows the maximum likelihood estimator for a distribution that does not appear in previous chapters.

Example 9.23: It is known that a sample consisting of the values 12, 11.2, 13.5, 12.3, 13.8, and 11.9 comes from a population with the density function

$$f(x; \theta) = \begin{cases} \frac{\theta}{x^{\theta+1}}, & x > 1, \\ 0, & \text{elsewhere,} \end{cases}$$

where $\theta > 0$. Find the maximum likelihood estimate of θ .

Solution: The likelihood function of n observations from this population can be written as

$$L(x_1, x_2, \dots, x_{10}; \theta) = \prod_{i=1}^n \frac{\theta}{x_i^{\theta+1}} = \frac{\theta^n}{(\prod_{i=1}^n x_i)^{\theta+1}},$$

which implies that

$$\ln L(x_1, x_2, \dots, x_{10}; \theta) = n \ln(\theta) - (\theta + 1) \sum_{i=1}^n \ln(x_i).$$

Setting $0 = \frac{\partial \ln L}{\partial \theta} = \frac{n}{\theta} - \sum_{i=1}^n \ln(x_i)$ results in

$$\begin{aligned}\hat{\theta} &= \frac{n}{\sum_{i=1}^n \ln(x_i)} \\ &= \frac{6}{\ln(12) + \ln(11.2) + \ln(13.5) + \ln(12.3) + \ln(13.8) + \ln(11.9)} = 0.3970.\end{aligned}$$

Since the second derivative of L is $-n/\theta^2$, which is always negative, the likelihood function does achieve its maximum value at $\hat{\theta}$. ▮

Additional Comments Concerning Maximum Likelihood Estimation

A thorough discussion of the properties of maximum likelihood estimation is beyond the scope of this book and is usually a major topic of a course in the theory of statistical inference. The method of maximum likelihood allows the analyst to make use of knowledge of the distribution in determining an appropriate estimator. *The method of maximum likelihood cannot be applied without knowledge of the underlying distribution.* We learned in Example 9.21 that the maximum likelihood estimator is not necessarily unbiased. The maximum likelihood estimator is unbiased *asymptotically* or *in the limit*; that is, the amount of bias approaches zero as the sample size becomes large. Earlier in this chapter the notion of efficiency was discussed, efficiency being linked to the variance property of an estimator. Maximum likelihood estimators possess desirable variance properties in the limit. The reader should consult Lehmann and D'Abrera (1998) for details.

Exercises

9.81 Suppose that there are n trials $x_1, x_2, \dots, x_n$ from a Bernoulli process with parameter p , the probability of a success. That is, the probability of r successes is given by $\binom{n}{r} p^r (1-p)^{n-r}$. Work out the maximum likelihood estimator for the parameter p .

9.82 Consider the lognormal distribution with the density function given in Section 6.9. Suppose we have a random sample $x_1, x_2, \dots, x_n$ from a lognormal distribution.

- (a) Write out the likelihood function.
- (b) Develop the maximum likelihood estimators of μ and σ^2 .

9.83 Consider a random sample of $x_1, \dots, x_n$ coming from the gamma distribution discussed in Section 6.6. Suppose the parameter α is known, say 5, and determine the maximum likelihood estimation for parameter β .

9.84 Consider a random sample of $x_1, x_2, \dots, x_n$ ob-

servations from a Weibull distribution with parameters α and β and density function

$$f(x) = \begin{cases} \alpha \beta x^{\beta-1} e^{-\alpha x^\beta}, & x > 0, \\ 0, & \text{elsewhere,} \end{cases}$$

for $\alpha, \beta > 0$.

- (a) Write out the likelihood function.
- (b) Write out the equations that, when solved, give the maximum likelihood estimators of α and β .

9.85 Consider a random sample of $x_1, \dots, x_n$ from a uniform distribution $U(0, \theta)$ with unknown parameter θ , where $\theta > 0$. Determine the maximum likelihood estimator of θ .

9.86 Consider the independent observations $x_1, x_2, \dots, x_n$ from the gamma distribution discussed in Section 6.6.

- (a) Write out the likelihood function.

- (b) Write out a set of equations that, when solved, give the maximum likelihood estimators of α and β .

9.87 Consider a hypothetical experiment where a man with a fungus uses an antifungal drug and is cured. Consider this, then, a sample of one from a Bernoulli distribution with probability function

$$f(x) = p^x q^{1-x}, \quad x = 0, 1,$$

Review Exercises

9.89 Consider two estimators of σ^2 for a sample $x_1, x_2, \dots, x_n$, which is drawn from a normal distribution with mean μ and variance σ^2 . The estimators are the unbiased estimator $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ and

the maximum likelihood estimator $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$.

Discuss the variance properties of these two estimators.

9.90 According to the *Roanoke Times*, McDonald's sold 42.1% of the market share of hamburgers. A random sample of 75 burgers sold resulted in 28 of them being from McDonald's. Use material in Section 9.10 to determine if this information supports the claim in the *Roanoke Times*.

9.91 It is claimed that a new diet will reduce a person's weight by 4.5 kilograms on average in a period of 2 weeks. The weights of 7 women who followed this diet were recorded before and after the 2-week period.

Woman	Weight Before	Weight After
1	58.5	60.0
2	60.3	54.9
3	61.7	58.1
4	69.0	62.1
5	64.0	58.5
6	62.6	59.9
7	56.7	54.4

Test the claim about the diet by computing a 95% confidence interval for the mean difference in weights. Assume the differences of weights to be approximately normally distributed.

9.92 A study was undertaken at Virginia Tech to determine if fire can be used as a viable management tool to increase the amount of forage available to deer during the critical months in late winter and early spring. Calcium is a required element for plants and animals. The amount taken up and stored in plants is closely correlated to the amount present in the soil. It was hypothesized that a fire may change the calcium levels

where p is the probability of a success (cure) and $q = 1 - p$. Now, of course, the sample information gives $x = 1$. Write out a development that shows that $\hat{p} = 1.0$ is the maximum likelihood estimator of the probability of a cure.

9.88 Consider the observation X from the negative binomial distribution given in Section 5.4. Find the maximum likelihood estimator for p , assuming k is known.

present in the soil and thus affect the amount available to deer. A large tract of land in the Fishburn Forest was selected for a prescribed burn. Soil samples were taken from 12 plots of equal area just prior to the burn and analyzed for calcium. Postburn calcium levels were analyzed from the same plots. These values, in kilograms per plot, are presented in the following table:

Plot	Calcium Level (kg/plot)	
	Preburn	Postburn
1	50	9
2	50	18
3	82	45
4	64	18
5	82	18
6	73	9
7	77	32
8	54	9
9	23	18
10	45	9
11	36	9
12	54	9

Construct a 95% confidence interval for the mean difference in calcium levels in the soil prior to and after the prescribed burn. Assume the distribution of differences in calcium levels to be approximately normal.

9.93 A health spa claims that a new exercise program will reduce a person's waist size by 2 centimeters on average over a 5-day period. The waist sizes, in centimeters, of 6 men who participated in this exercise program are recorded before and after the 5-day period in the following table:

Man	Waist Size Before	Waist Size After
1	90.4	91.7
2	95.5	93.9
3	98.7	97.4
4	115.9	112.8
5	104.0	101.3
6	85.6	84.0

By computing a 95% confidence interval for the mean reduction in waist size, determine whether the health spa's claim is valid. Assume the distribution of differences in waist sizes before and after the program to be approximately normal.

9.94 The Department of Civil Engineering at Virginia Tech compared a modified (M-5 hr) assay technique for recovering fecal coliforms in stormwater runoff from an urban area to a most probable number (MPN) technique. A total of 12 runoff samples were collected and analyzed by the two techniques. Fecal coliform counts per 100 milliliters are recorded in the following table.

Sample	MPN Count	M-5 hr Count
1	2300	2010
2	1200	930
3	450	400
4	210	436
5	270	4100
6	450	2090
7	154	219
8	179	169
9	192	194
10	230	174
11	340	274
12	194	183

Construct a 90% confidence interval for the difference in the mean fecal coliform counts between the M-5 hr and the MPN techniques. Assume that the count differences are approximately normally distributed.

9.95 An experiment was conducted to determine whether surface finish has an effect on the endurance limit of steel. There is a theory that polishing increases the average endurance limit (for reverse bending). From a practical point of view, polishing should not have any effect on the standard deviation of the endurance limit, which is known from numerous endurance limit experiments to be 4000 psi. An experiment was performed on 0.4% carbon steel using both unpolished and polished smooth-turned specimens. The data are as follows:

Endurance Limit (psi)	
Polished 0.4% Carbon	Unpolished 0.4% Carbon
85,500	82,600
91,900	82,400
89,400	81,700
84,000	79,500
89,900	79,400
78,700	69,800
87,500	79,900
83,100	83,400

Find a 95% confidence interval for the difference between the population means for the two methods, as-

suming that the populations are approximately normally distributed.

9.96 An anthropologist is interested in the proportion of individuals in two Indian tribes with double occipital hair whorls. Suppose that independent samples are taken from each of the two tribes, and it is found that 24 of 100 Indians from tribe *A* and 36 of 120 Indians from tribe *B* possess this characteristic. Construct a 95% confidence interval for the difference $p_B - p_A$ between the proportions of these two tribes with occipital hair whorls.

9.97 A manufacturer of electric irons produces these items in two plants. Both plants have the same suppliers of small parts. A saving can be made by purchasing thermostats for plant *B* from a local supplier. A single lot was purchased from the local supplier, and a test was conducted to see whether or not these new thermostats were as accurate as the old. The thermostats were tested on tile irons on the 550°F setting, and the actual temperatures were read to the nearest 0.1°F with a thermocouple. The data are as follows:

New Supplier (°F)					
530.3	559.3	549.4	544.0	551.7	566.3
549.9	556.9	536.7	558.8	538.8	543.3
559.1	555.0	538.6	551.1	565.4	554.9
550.0	554.9	554.7	536.1	569.1	
Old Supplier (°F)					
559.7	534.7	554.8	545.0	544.6	538.0
550.7	563.1	551.1	553.8	538.8	564.6
554.5	553.0	538.4	548.3	552.9	535.1
555.0	544.8	558.4	548.7	560.3	

Find 95% confidence intervals for σ_1^2/σ_2^2 and for σ_1/σ_2 , where σ_1^2 and σ_2^2 are the population variances of the thermostat readings for the new and old suppliers, respectively.

9.98 It is argued that the resistance of wire *A* is greater than the resistance of wire *B*. An experiment on the wires shows the following results (in ohms):

Wire A	Wire B
0.140	0.135
0.138	0.140
0.143	0.136
0.142	0.142
0.144	0.138
0.137	0.140

Assuming equal variances, what conclusions do you draw? Justify your answer.

9.99 An alternative form of estimation is accomplished through the method of moments. This method involves equating the population mean and variance to the corresponding sample mean $\bar{x}$ and sample variance

s^2 and solving for the parameters, the results being the **moment estimators**. In the case of a single parameter, only the means are used. Give an argument that in the case of the Poisson distribution the maximum likelihood estimator and moment estimators are the same.

9.100 Specify the moment estimators for μ and σ^2 for the normal distribution.

9.101 Specify the moment estimators for μ and σ^2 for the lognormal distribution.

9.102 Specify the moment estimators for α and β for the gamma distribution.

9.103 A survey was done with the hope of comparing salaries of chemical plant managers employed in two areas of the country, the northern and west central regions. An independent random sample of 300 plant managers was selected from each of the two regions. These managers were asked their annual salaries. The results are as follows

North	West Central
$\bar{x}_1 = \$102,300$	$\bar{x}_2 = \$98,500$
$s_1 = \$5700$	$s_2 = \$3800$

- Construct a 99% confidence interval for $\mu_1 - \mu_2$, the difference in the mean salaries.
- What assumption did you make in (a) about the distribution of annual salaries for the two regions? Is the assumption of normality necessary? Why or why not?
- What assumption did you make about the two variances? Is the assumption of equality of variances reasonable? Explain!

9.104 Consider Review Exercise 9.103. Let us assume that the data have not been collected yet and that previous statistics suggest that $\sigma_1 = \sigma_2 = \$4000$. Are the sample sizes in Review Exercise 9.103 sufficient to produce a 95% confidence interval on $\mu_1 - \mu_2$ having a width of only \$1000? Show all work.

9.105 A labor union is becoming defensive about gross absenteeism by its members. The union leaders had always claimed that, in a typical month, 95% of its members were absent less than 10 hours. The union decided to check this by monitoring a random sample of 300 of its members. The number of hours absent was recorded for each of the 300 members. The results were $\bar{x} = 6.5$ hours and $s = 2.5$ hours. Use the data to respond to this claim, using a one-sided tolerance limit and choosing the confidence level to be 99%. Be sure to interpret what you learn from the tolerance limit calculation.

9.106 A random sample of 30 firms dealing in wireless products was selected to determine the proportion of such firms that have implemented new software to improve productivity. It turned out that 8 of the 30 had implemented such software. Find a 95% confidence interval on p , the true proportion of such firms that have implemented new software.

9.107 Refer to Review Exercise 9.106. Suppose there is concern about whether the point estimate $\hat{p} = 8/30$ is accurate enough because the confidence interval around p is not sufficiently narrow. Using $\hat{p}$ as the estimate of p , how many companies would need to be sampled in order to have a 95% confidence interval with a width of only 0.05?

9.108 A manufacturer turns out a product item that is labeled either “defective” or “not defective.” In order to estimate the proportion defective, a random sample of 100 items is taken from production, and 10 are found to be defective. Following implementation of a quality improvement program, the experiment is conducted again. A new sample of 100 is taken, and this time only 6 are found to be defective.

- Give a 95% confidence interval on $p_1 - p_2$, where p_1 is the population proportion defective before improvement and p_2 is the proportion defective after improvement.
- Is there information in the confidence interval found in (a) that would suggest that $p_1 > p_2$? Explain.

9.109 A machine is used to fill boxes with product in an assembly line operation. Much concern centers around the variability in the number of ounces of product in a box. The standard deviation in weight of product is known to be 0.3 ounce. An improvement is implemented, after which a random sample of 20 boxes is selected and the sample variance is found to be 0.045 ounce². Find a 95% confidence interval on the variance in the weight of the product. Does it appear from the range of the confidence interval that the improvement of the process enhanced quality as far as variability is concerned? Assume normality on the distribution of weights of product.

9.110 A consumer group is interested in comparing operating costs for two different types of automobile engines. The group is able to find 15 owners whose cars have engine type A and 15 whose cars have engine type B . All 30 owners bought their cars at roughly the same time, and all have kept good records for a certain 12-month period. In addition, these owners drove roughly the same number of miles. The cost statistics are $\bar{y}_A = \$87.00/1000$ miles, $\bar{y}_B = \$75.00/1000$ miles, $s_A = \$5.99$, and $s_B = \$4.85$. Compute a 95% confidence interval to estimate $\mu_A - \mu_B$, the difference in

the mean operating costs. Assume normality and equal variances.

9.111 Consider the statistic S_p^2 , the pooled estimate of σ^2 discussed in Section 9.8. It is used when one is willing to assume that $\sigma_1^2 = \sigma_2^2 = \sigma^2$. Show that the estimator is unbiased for σ^2 [i.e., show that $E(S_p^2) = \sigma^2$]. You may make use of results from any theorem or example in this chapter.

9.112 A group of human factor researchers are concerned about reaction to a stimulus by airplane pilots in a certain cockpit arrangement. An experiment was conducted in a simulation laboratory, and 15 pilots were used with average reaction time of 3.2 seconds with a sample standard deviation of 0.6 second. It is of interest to characterize the extreme (i.e., worst case scenario). To that end, do the following:

- Give a particular important one-sided 99% confidence bound on the mean reaction time. What assumption, if any, must you make on the distribution of reaction times?
- Give a 99% one-sided prediction interval and give an interpretation of what it means. Must you make

an assumption about the distribution of reaction times to compute this bound?

- Compute a one-sided tolerance bound with 99% confidence that involves 95% of reaction times. Again, give an interpretation and assumptions about the distribution, if any. (Note: The one-sided tolerance limit values are also included in Table A.7.)

9.113 A certain supplier manufactures a type of rubber mat that is sold to automotive companies. The material used to produce the mats must have certain hardness characteristics. Defective mats are occasionally discovered and rejected. The supplier claims that the proportion defective is 0.05. A challenge was made by one of the clients who purchased the mats, so an experiment was conducted in which 400 mats are tested and 17 were found defective.

- Compute a 95% two-sided confidence interval on the proportion defective.
- Compute an appropriate 95% one-sided confidence interval on the proportion defective.
- Interpret both intervals from (a) and (b) and comment on the claim made by the supplier.

9.15 Potential Misconceptions and Hazards; Relationship to Material in Other Chapters

The concept of a *large-sample confidence interval* on a population is often confusing to the beginning student. It is based on the notion that even when σ is unknown and one is not convinced that the distribution being sampled is normal, a confidence interval on μ can be computed from

$$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}.$$

In practice, this formula is often used when the sample is too small. The genesis of this large sample interval is, of course, the Central Limit Theorem (CLT), under which normality is not necessary. Here the CLT requires a known σ , of which s is only an estimate. Thus, n must be at least as large as 30 and the underlying distribution must be close to symmetric, in which case the interval is still an approximation.

There are instances in which the appropriateness of the practical application of material in this chapter depends very much on the specific context. One very important illustration is the use of the t -distribution for the confidence interval on μ when σ is unknown. Strictly speaking, the use of the t -distribution requires that the distribution sampled from be normal. However, it is well known that any application of the t -distribution is reasonably insensitive (i.e., **robust**) to the normality assumption. This represents one of those fortunate situations which

occur often in the field of statistics in which a basic assumption does not hold and yet “everything turns out all right!” However, one population from which the sample is drawn cannot deviate substantially from normal. Thus, the normal probability plots discussed in Chapter 8 and the goodness-of-fit tests introduced in Chapter 10 often need be called upon to ascertain some sense of “nearness to normality.” This idea of “robustness to normality” will reappear in Chapter 10.

It is our experience that one of the most serious “misuses of statistics” in practice evolves from confusion about distinctions in the interpretation of the types of statistical intervals. Thus, the subsection in this chapter where differences among the three types of intervals are discussed is important. It is very likely that in practice the **confidence interval is heavily overused**. That is, it is used when there is really no interest in the mean; rather, the question is “Where is the next observation going to fall?” or often, more importantly, “Where is the large bulk of the distribution?” These are crucial questions that are not answered by computing an interval on the mean. The interpretation of a confidence interval is often misunderstood. It is tempting to conclude that the parameter falls inside the interval with probability 0.95. While this is a correct interpretation of a **Bayesian posterior interval** (readers are referred to Chapter 18 for more information on Bayesian inference), it is not the proper frequency interpretation.

A confidence interval merely suggests that if the experiment is conducted and data are observed again and again, about 95% of such intervals will contain the true parameter. Any beginning student of practical statistics should be very clear on the difference among these statistical intervals.

Another potential serious misuse of statistics centers around the use of the χ^2 -distribution for a confidence interval on a single variance. Again, normality of the distribution from which the sample is drawn is assumed. Unlike the use of the t -distribution, the use of the χ^2 test for this application is **not robust to the normality assumption** (i.e., the sampling distribution of $\frac{(n-1)S^2}{\sigma^2}$ deviates far from χ^2 if the underlying distribution is not normal). Thus, strict use of goodness-of-fit (Chapter 10) tests and/or normal probability plotting can be extremely important in such contexts. More information about this general issue will be given in future chapters.

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Chapter 10

One- and Two-Sample Tests of Hypotheses

10.1 Statistical Hypotheses: General Concepts

Often, the problem confronting the scientist or engineer is not so much the estimation of a population parameter, as discussed in Chapter 9, but rather the formation of a data-based decision procedure that can produce a conclusion about some scientific system. For example, a medical researcher may decide on the basis of experimental evidence whether coffee drinking increases the risk of cancer in humans; an engineer might have to decide on the basis of sample data whether there is a difference between the accuracy of two kinds of gauges; or a sociologist might wish to collect appropriate data to enable him or her to decide whether a person's blood type and eye color are independent variables. In each of these cases, the scientist or engineer *postulates* or *conjectures* something about a system. In addition, each must make use of experimental data and make a decision based on the data. In each case, the conjecture can be put in the form of a statistical hypothesis. Procedures that lead to the acceptance or rejection of statistical hypotheses such as these comprise a major area of statistical inference. First, let us define precisely what we mean by a **statistical hypothesis**.

Definition 10.1: A **statistical hypothesis** is an assertion or conjecture concerning one or more populations.

The truth or falsity of a statistical hypothesis is never known with absolute certainty unless we examine the entire population. This, of course, would be impractical in most situations. Instead, we take a random sample from the population of interest and use the data contained in this sample to provide evidence that either supports or does not support the hypothesis. Evidence from the sample that is inconsistent with the stated hypothesis leads to a rejection of the hypothesis.

The Role of Probability in Hypothesis Testing

It should be made clear to the reader that the decision procedure must include an awareness of the *probability of a wrong conclusion*. For example, suppose that the hypothesis postulated by the engineer is that the fraction defective p in a certain process is 0.10. The experiment is to observe a random sample of the product in question. Suppose that 100 items are tested and 12 items are found defective. It is reasonable to conclude that this evidence does not refute the condition that the binomial parameter $p = 0.10$, and thus it may lead one not to reject the hypothesis. However, it also does not refute $p = 0.12$ or perhaps even $p = 0.15$. As a result, the reader must be accustomed to understanding that **rejection of a hypothesis implies that the sample evidence refutes it**. Put another way, **rejection means that there is a small probability of obtaining the sample information observed when, in fact, the hypothesis is true**. For example, for our proportion-defective hypothesis, a sample of 100 revealing 20 defective items is certainly evidence for rejection. Why? If, indeed, $p = 0.10$, the probability of obtaining 20 or more defectives is approximately 0.002. With the resulting small risk of a wrong conclusion, it would seem safe to **reject the hypothesis** that $p = 0.10$. In other words, rejection of a hypothesis tends to all but “rule out” the hypothesis. On the other hand, it is very important to emphasize that acceptance or, rather, failure to reject does not rule out other possibilities. As a result, the *firm conclusion is established by the data analyst when a hypothesis is rejected*.

The formal statement of a hypothesis is often influenced by the structure of the probability of a wrong conclusion. If the scientist is interested in *strongly supporting* a contention, he or she hopes to arrive at the contention in the form of rejection of a hypothesis. If the medical researcher wishes to show strong evidence in favor of the contention that coffee drinking increases the risk of cancer, the hypothesis tested should be of the form “there is no increase in cancer risk produced by drinking coffee.” As a result, the contention is reached via a rejection. Similarly, to support the claim that one kind of gauge is more accurate than another, the engineer tests the hypothesis that there is no difference in the accuracy of the two kinds of gauges.

The foregoing implies that when the data analyst formalizes experimental evidence on the basis of hypothesis testing, the formal **statement of the hypothesis** is very important.

The Null and Alternative Hypotheses

The structure of hypothesis testing will be formulated with the use of the term **null hypothesis**, which refers to any hypothesis we wish to test and is denoted by H_0 . The rejection of H_0 leads to the acceptance of an **alternative hypothesis**, denoted by H_1 . An understanding of the different roles played by the null hypothesis (H_0) and the alternative hypothesis (H_1) is crucial to one's understanding of the rudiments of hypothesis testing. The alternative hypothesis H_1 usually represents the *question to be answered or the theory to be tested*, and thus its specification is crucial. The null hypothesis H_0 *nullifies or opposes* H_1 and is often the logical complement to H_1 . As the reader gains more understanding of hypothesis testing, he or she should note that the analyst arrives at one of the two following

conclusions:

reject H_0 in favor of H_1 because of sufficient evidence in the data or
fail to reject H_0 because of insufficient evidence in the data.

Note that the *conclusions do not involve a formal and literal “accept H_0 .”* The statement of H_0 often represents the “status quo” in opposition to the new idea, conjecture, and so on, stated in H_1 , while failure to reject H_0 represents the proper conclusion. In our binomial example, the practical issue may be a concern that the historical defective probability of 0.10 no longer is true. Indeed, the conjecture may be that p exceeds 0.10. We may then state

$$H_0: p = 0.10,$$

$$H_1: p > 0.10.$$

Now 12 defective items out of 100 does not refute $p = 0.10$, so the conclusion is “fail to reject H_0 .” However, if the data produce 20 out of 100 defective items, then the conclusion is “reject H_0 ” in favor of $H_1: p > 0.10$.

Though the applications of hypothesis testing are quite abundant in scientific and engineering work, perhaps the best illustration for a novice lies in the predicament encountered in a jury trial. The null and alternative hypotheses are

$$H_0: \text{defendant is innocent,}$$

$$H_1: \text{defendant is guilty.}$$

The indictment comes because of suspicion of guilt. The hypothesis H_0 (the status quo) stands in opposition to H_1 and is maintained unless H_1 is supported by evidence “beyond a reasonable doubt.” However, “failure to reject H_0 ” in this case does not imply innocence, but merely that the evidence was insufficient to convict. So the jury does not necessarily *accept H_0* but *fails to reject H_0* .

10.2 Testing a Statistical Hypothesis

To illustrate the concepts used in testing a statistical hypothesis about a population, we present the following example. A certain type of cold vaccine is known to be only 25% effective after a period of 2 years. To determine if a new and somewhat more expensive vaccine is superior in providing protection against the same virus for a longer period of time, suppose that 20 people are chosen at random and inoculated. (In an actual study of this type, the participants receiving the new vaccine might number several thousand. The number 20 is being used here only to demonstrate the basic steps in carrying out a statistical test.) If more than 8 of those receiving the new vaccine surpass the 2-year period without contracting the virus, the new vaccine will be considered superior to the one presently in use. The requirement that the number exceed 8 is somewhat arbitrary but appears reasonable in that it represents a modest gain over the 5 people who could be expected to receive protection if the 20 people had been inoculated with the vaccine already in use. We are essentially testing the null hypothesis that the new vaccine is equally effective after a period of 2 years as the one now commonly used. The alternative

hypothesis is that the new vaccine is in fact superior. This is equivalent to testing the hypothesis that the binomial parameter for the probability of a success on a given trial is $p = 1/4$ against the alternative that $p > 1/4$. This is usually written as follows:

$$\begin{aligned} H_0: & p = 0.25, \\ H_1: & p > 0.25. \end{aligned}$$

The Test Statistic

The **test statistic** on which we base our decision is X , the number of individuals in our test group who receive protection from the new vaccine for a period of at least 2 years. The possible values of X , from 0 to 20, are divided into two groups: those numbers less than or equal to 8 and those greater than 8. All possible scores greater than 8 constitute the **critical region**. The last number that we observe in passing into the critical region is called the **critical value**. In our illustration, the critical value is the number 8. Therefore, if $x > 8$, we reject H_0 in favor of the alternative hypothesis H_1 . If $x \leq 8$, we fail to reject H_0 . This decision criterion is illustrated in Figure 10.1.

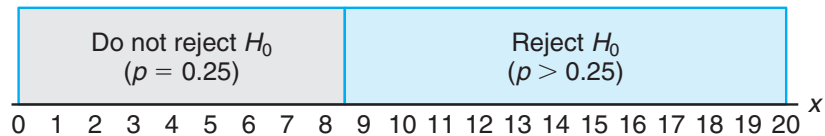


Figure 10.1: Decision criterion for testing $p = 0.25$ versus $p > 0.25$.

The Probability of a Type I Error

The decision procedure just described could lead to either of two wrong conclusions. For instance, the new vaccine may be no better than the one now in use (H_0 true) and yet, in this particular randomly selected group of individuals, more than 8 surpass the 2-year period without contracting the virus. We would be committing an error by rejecting H_0 in favor of H_1 when, in fact, H_0 is true. Such an error is called a **type I error**.

Definition 10.2: Rejection of the null hypothesis when it is true is called a **type I error**.

A second kind of error is committed if 8 or fewer of the group surpass the 2-year period successfully and we are unable to conclude that the vaccine is better when it actually is better (H_1 true). Thus, in this case, we fail to reject H_0 when in fact H_0 is false. This is called a **type II error**.

Definition 10.3: Nonrejection of the null hypothesis when it is false is called a **type II error**.

In testing any statistical hypothesis, there are four possible situations that determine whether our decision is correct or in error. These four situations are

summarized in Table 10.1.

Table 10.1: Possible Situations for Testing a Statistical Hypothesis

	H_0 is true	H_0 is false
Do not reject H_0	Correct decision	Type II error
Reject H_0	Type I error	Correct decision

The probability of committing a type I error, also called the **level of significance**, is denoted by the Greek letter α . In our illustration, a type I error will occur when more than 8 individuals inoculated with the new vaccine surpass the 2-year period without contracting the virus and researchers conclude that the new vaccine is better when it is actually equivalent to the one in use. Hence, if X is the number of individuals who remain free of the virus for at least 2 years,

$$\begin{aligned}\alpha &= P(\text{type I error}) = P\left(X > 8 \text{ when } p = \frac{1}{4}\right) = \sum_{x=9}^{20} b\left(x; 20, \frac{1}{4}\right) \\ &= 1 - \sum_{x=0}^8 b\left(x; 20, \frac{1}{4}\right) = 1 - 0.9591 = 0.0409.\end{aligned}$$

We say that the null hypothesis, $p = 1/4$, is being tested at the $\alpha = 0.0409$ level of significance. Sometimes the level of significance is called the **size of the test**. A critical region of size 0.0409 is very small, and therefore it is unlikely that a type I error will be committed. Consequently, it would be most unusual for more than 8 individuals to remain immune to a virus for a 2-year period using a new vaccine that is essentially equivalent to the one now on the market.

The Probability of a Type II Error

The probability of committing a type II error, denoted by β , is impossible to compute unless we have a specific alternative hypothesis. If we test the null hypothesis that $p = 1/4$ against the alternative hypothesis that $p = 1/2$, then we are able to compute the probability of not rejecting H_0 when it is false. We simply find the probability of obtaining 8 or fewer in the group that surpass the 2-year period when $p = 1/2$. In this case,

$$\begin{aligned}\beta &= P(\text{type II error}) = P\left(X \leq 8 \text{ when } p = \frac{1}{2}\right) \\ &= \sum_{x=0}^8 b\left(x; 20, \frac{1}{2}\right) = 0.2517.\end{aligned}$$

This is a rather high probability, indicating a test procedure in which it is quite likely that we shall reject the new vaccine when, in fact, it is superior to what is now in use. Ideally, we like to use a test procedure for which the type I and type II error probabilities are both small.

It is possible that the director of the testing program is willing to make a type II error if the more expensive vaccine is not significantly superior. In fact, the only

time he wishes to guard against the type II error is when the true value of p is at least 0.7. If $p = 0.7$, this test procedure gives

$$\begin{aligned}\beta &= P(\text{type II error}) = P(X \leq 8 \text{ when } p = 0.7) \\ &= \sum_{x=0}^8 b(x; 20, 0.7) = 0.0051.\end{aligned}$$

With such a small probability of committing a type II error, it is extremely unlikely that the new vaccine would be rejected when it was 70% effective after a period of 2 years. As the alternative hypothesis approaches unity, the value of β diminishes to zero.

The Role of α , β , and Sample Size

Let us assume that the director of the testing program is unwilling to commit a type II error when the alternative hypothesis $p = 1/2$ is true, even though we have found the probability of such an error to be $\beta = 0.2517$. It is always possible to reduce β by increasing the size of the critical region. For example, consider what happens to the values of α and β when we change our critical value to 7 so that all scores greater than 7 fall in the critical region and those less than or equal to 7 fall in the nonrejection region. Now, in testing $p = 1/4$ against the alternative hypothesis that $p = 1/2$, we find that

$$\alpha = \sum_{x=8}^{20} b\left(x; 20, \frac{1}{4}\right) = 1 - \sum_{x=0}^7 b\left(x; 20, \frac{1}{4}\right) = 1 - 0.8982 = 0.1018$$

and

$$\beta = \sum_{x=0}^7 b\left(x; 20, \frac{1}{2}\right) = 0.1316.$$

By adopting a new decision procedure, we have reduced the probability of committing a type II error at the expense of increasing the probability of committing a type I error. For a fixed sample size, a decrease in the probability of one error will usually result in an increase in the probability of the other error. Fortunately, **the probability of committing both types of error can be reduced by increasing the sample size**. Consider the same problem using a random sample of 100 individuals. If more than 36 of the group surpass the 2-year period, we reject the null hypothesis that $p = 1/4$ and accept the alternative hypothesis that $p > 1/4$. The critical value is now 36. All possible scores above 36 constitute the critical region, and all possible scores less than or equal to 36 fall in the acceptance region.

To determine the probability of committing a type I error, we shall use the normal curve approximation with

$$\mu = np = (100) \left(\frac{1}{4}\right) = 25 \quad \text{and} \quad \sigma = \sqrt{npq} = \sqrt{(100)(1/4)(3/4)} = 4.33.$$

Referring to Figure 10.2, we need the area under the normal curve to the right of $x = 36.5$. The corresponding z -value is

$$z = \frac{36.5 - 25}{4.33} = 2.66.$$

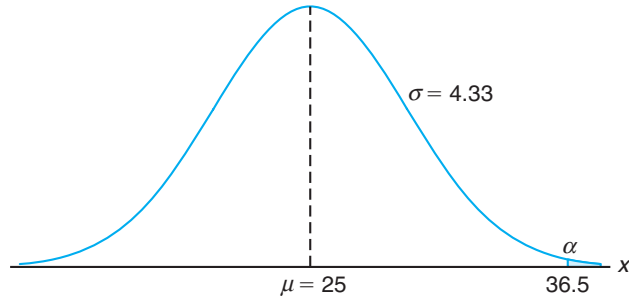


Figure 10.2: Probability of a type I error.

From Table A.3 we find that

$$\begin{aligned}\alpha = P(\text{type I error}) &= P\left(X > 36 \text{ when } p = \frac{1}{4}\right) \approx P(Z > 2.66) \\ &= 1 - P(Z < 2.66) = 1 - 0.9961 = 0.0039.\end{aligned}$$

If H_0 is false and the true value of H_1 is $p = 1/2$, we can determine the probability of a type II error using the normal curve approximation with

$$\mu = np = (100)(1/2) = 50 \quad \text{and} \quad \sigma = \sqrt{npq} = \sqrt{(100)(1/2)(1/2)} = 5.$$

The probability of a value falling in the nonrejection region when H_0 is true is given by the area of the shaded region to the left of $x = 36.5$ in Figure 10.3. The z -value corresponding to $x = 36.5$ is

$$z = \frac{36.5 - 50}{5} = -2.7.$$

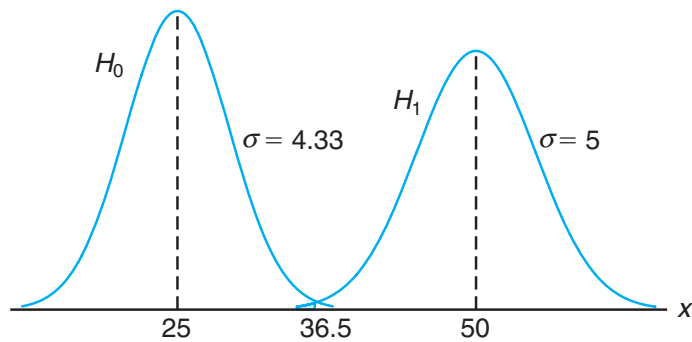


Figure 10.3: Probability of a type II error.

Therefore,

$$\beta = P(\text{type II error}) = P\left(X \leq 36 \text{ when } p = \frac{1}{2}\right) \approx P(Z < -2.7) = 0.0035.$$

Obviously, the type I and type II errors will rarely occur if the experiment consists of 100 individuals.

The illustration above underscores the strategy of the scientist in hypothesis testing. After the null and alternative hypotheses are stated, it is important to consider the sensitivity of the test procedure. By this we mean that there should be a determination, for a fixed α , of a reasonable value for the probability of wrongly accepting H_0 (i.e., the value of β) when the true situation represents some *important deviation from H_0* . A value for the sample size can usually be determined for which there is a reasonable balance between the values of α and β computed in this fashion. The vaccine problem provides an illustration.

Illustration with a Continuous Random Variable

The concepts discussed here for a discrete population can be applied equally well to continuous random variables. Consider the null hypothesis that the average weight of male students in a certain college is 68 kilograms against the alternative hypothesis that it is unequal to 68. That is, we wish to test

$$H_0: \mu = 68,$$

$$H_1: \mu \neq 68.$$

The alternative hypothesis allows for the possibility that $\mu < 68$ or $\mu > 68$.

A sample mean that falls close to the hypothesized value of 68 would be considered evidence in favor of H_0 . On the other hand, a sample mean that is considerably less than or more than 68 would be evidence inconsistent with H_0 and therefore favoring H_1 . The sample mean is the test statistic in this case. A critical region for the test statistic might arbitrarily be chosen to be the two intervals $\bar{x} < 67$ and $\bar{x} > 69$. The nonrejection region will then be the interval $67 \leq \bar{x} \leq 69$. This decision criterion is illustrated in Figure 10.4.

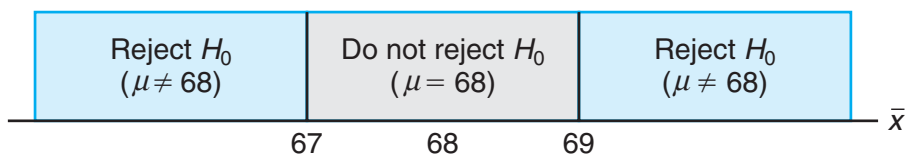


Figure 10.4: Critical region (in blue).

Let us now use the decision criterion of Figure 10.4 to calculate the probabilities of committing type I and type II errors when testing the null hypothesis that $\mu = 68$ kilograms against the alternative that $\mu \neq 68$ kilograms.

Assume the standard deviation of the population of weights to be $\sigma = 3.6$. For large samples, we may substitute s for σ if no other estimate of σ is available. Our decision statistic, based on a random sample of size $n = 36$, will be $\bar{X}$, the most efficient estimator of μ . From the Central Limit Theorem, we know that the sampling distribution of $\bar{X}$ is approximately normal with standard deviation $\sigma_{\bar{X}} = \sigma/\sqrt{n} = 3.6/6 = 0.6$.

The probability of committing a type I error, or the level of significance of our test, is equal to the sum of the areas that have been shaded in each tail of the distribution in Figure 10.5. Therefore,

$$\alpha = P(\bar{X} < 67 \text{ when } \mu = 68) + P(\bar{X} > 69 \text{ when } \mu = 68).$$

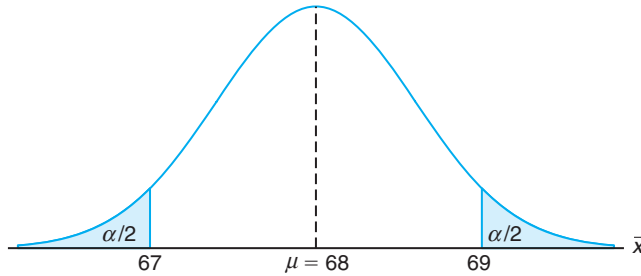


Figure 10.5: Critical region for testing $\mu = 68$ versus $\mu \neq 68$.

The z -values corresponding to $\bar{x}_1 = 67$ and $\bar{x}_2 = 69$ when H_0 is true are

$$z_1 = \frac{67 - 68}{0.6} = -1.67 \quad \text{and} \quad z_2 = \frac{69 - 68}{0.6} = 1.67.$$

Therefore,

$$\alpha = P(Z < -1.67) + P(Z > 1.67) = 2P(Z < -1.67) = 0.0950.$$

Thus, 9.5% of all samples of size 36 would lead us to reject $\mu = 68$ kilograms when, in fact, it is true. To reduce α , we have a choice of increasing the sample size or widening the fail-to-reject region. Suppose that we increase the sample size to $n = 64$. Then $\sigma_{\bar{X}} = 3.6/8 = 0.45$. Now

$$z_1 = \frac{67 - 68}{0.45} = -2.22 \quad \text{and} \quad z_2 = \frac{69 - 68}{0.45} = 2.22.$$

Hence,

$$\alpha = P(Z < -2.22) + P(Z > 2.22) = 2P(Z < -2.22) = 0.0264.$$

The reduction in α is not sufficient by itself to guarantee a good testing procedure. We must also evaluate β for various alternative hypotheses. If it is important to reject H_0 when the true mean is some value $\mu \geq 70$ or $\mu \leq 66$, then the probability of committing a type II error should be computed and examined for the alternatives $\mu = 66$ and $\mu = 70$. Because of symmetry, it is only necessary to consider the probability of not rejecting the null hypothesis that $\mu = 68$ when the alternative $\mu = 70$ is true. A type II error will result when the sample mean $\bar{x}$ falls between 67 and 69 when H_1 is true. Therefore, referring to Figure 10.6, we find that

$$\beta = P(67 \leq \bar{X} \leq 69 \text{ when } \mu = 70).$$

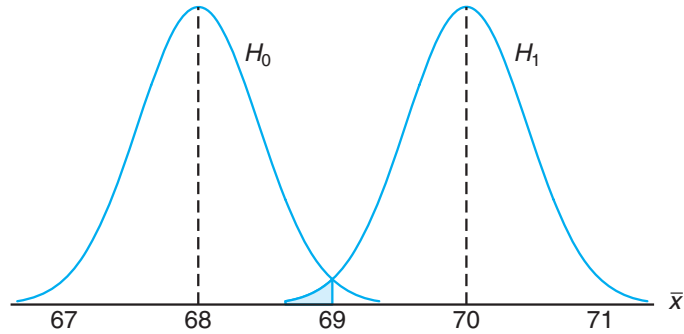


Figure 10.6: Probability of type II error for testing $\mu = 68$ versus $\mu = 70$.

The z -values corresponding to $\bar{x}_1 = 67$ and $\bar{x}_2 = 69$ when H_1 is true are

$$z_1 = \frac{67 - 70}{0.45} = -6.67 \quad \text{and} \quad z_2 = \frac{69 - 70}{0.45} = -2.22.$$

Therefore,

$$\begin{aligned} \beta &= P(-6.67 < Z < -2.22) = P(Z < -2.22) - P(Z < -6.67) \\ &= 0.0132 - 0.0000 = 0.0132. \end{aligned}$$

If the true value of μ is the alternative $\mu = 66$, the value of β will again be 0.0132. For all possible values of $\mu < 66$ or $\mu > 70$, the value of β will be even smaller when $n = 64$, and consequently there would be little chance of not rejecting H_0 when it is false.

The probability of committing a type II error increases rapidly when the true value of μ approaches, but is not equal to, the hypothesized value. Of course, this is usually the situation where we do not mind making a type II error. For example, if the alternative hypothesis $\mu = 68.5$ is true, we do not mind committing a type II error by concluding that the true answer is $\mu = 68$. The probability of making such an error will be high when $n = 64$. Referring to Figure 10.7, we have

$$\beta = P(67 \leq \bar{X} \leq 69 \text{ when } \mu = 68.5).$$

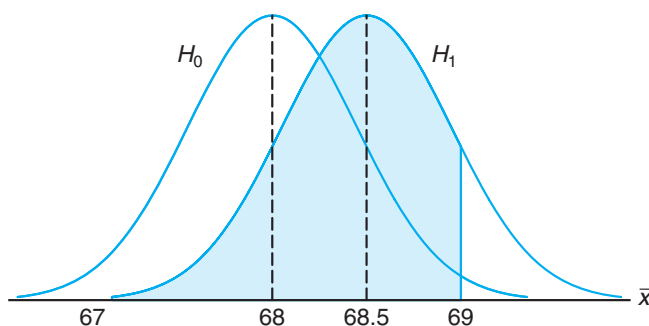
The z -values corresponding to $\bar{x}_1 = 67$ and $\bar{x}_2 = 69$ when $\mu = 68.5$ are

$$z_1 = \frac{67 - 68.5}{0.45} = -3.33 \quad \text{and} \quad z_2 = \frac{69 - 68.5}{0.45} = 1.11.$$

Therefore,

$$\begin{aligned} \beta &= P(-3.33 < Z < 1.11) = P(Z < 1.11) - P(Z < -3.33) \\ &= 0.8665 - 0.0004 = 0.8661. \end{aligned}$$

The preceding examples illustrate the following important properties:

Figure 10.7: Type II error for testing $\mu = 68$ versus $\mu = 68.5$.

Important
Properties of a
Test of
Hypothesis

1. The type I error and type II error are related. A decrease in the probability of one generally results in an increase in the probability of the other.
 2. The size of the critical region, and therefore the probability of committing a type I error, can always be reduced by adjusting the critical value(s).
 3. An increase in the sample size n will reduce α and β simultaneously.
 4. If the null hypothesis is false, β is a maximum when the true value of a parameter approaches the hypothesized value. The greater the distance between the true value and the hypothesized value, the smaller β will be.
-

One very important concept that relates to error probabilities is the notion of the **power** of a test.

Definition 10.4:

The **power** of a test is the probability of rejecting H_0 given that a specific alternative is true.

The power of a test can be computed as $1 - \beta$. Often **different types of tests are compared by contrasting power properties**. Consider the previous illustration, in which we were testing $H_0: \mu = 68$ and $H_1: \mu \neq 68$. As before, suppose we are interested in assessing the sensitivity of the test. The test is governed by the rule that we do not reject H_0 if $67 \leq \bar{x} \leq 69$. We seek the capability of the test to properly reject H_0 when indeed $\mu = 68.5$. We have seen that the probability of a type II error is given by $\beta = 0.8661$. Thus, the **power** of the test is $1 - 0.8661 = 0.1339$. In a sense, the power is a more succinct measure of how sensitive the test is for detecting differences between a mean of 68 and a mean of 68.5. In this case, if μ is truly 68.5, the test as described will *properly reject H_0 only 13.39% of the time*. As a result, the test would not be a good one if it was important that the analyst have a reasonable chance of truly distinguishing between a mean of 68.0 (specified by H_0) and a mean of 68.5. From the foregoing, it is clear that to produce a desirable power (say, greater than 0.8), one must either increase α or increase the sample size.

So far in this chapter, much of the discussion of hypothesis testing has focused on foundations and definitions. In the sections that follow, we get more specific

and put hypotheses in categories as well as discuss tests of hypotheses on various parameters of interest. We begin by drawing the distinction between a one-sided and a two-sided hypothesis.

One- and Two-Tailed Tests

A test of any statistical hypothesis where the alternative is **one sided**, such as

$$\begin{aligned}H_0: \theta &= \theta_0, \\H_1: \theta &> \theta_0\end{aligned}$$

or perhaps

$$\begin{aligned}H_0: \theta &= \theta_0, \\H_1: \theta &< \theta_0,\end{aligned}$$

is called a **one-tailed test**. Earlier in this section, we referred to the **test statistic** for a hypothesis. Generally, the critical region for the alternative hypothesis $\theta > \theta_0$ lies in the right tail of the distribution of the test statistic, while the critical region for the alternative hypothesis $\theta < \theta_0$ lies entirely in the left tail. (In a sense, the inequality symbol points in the direction of the critical region.) A one-tailed test was used in the vaccine experiment to test the hypothesis $p = 1/4$ against the one-sided alternative $p > 1/4$ for the binomial distribution. The one-tailed critical region is usually obvious; the reader should visualize the behavior of the test statistic and notice the obvious *signal* that would produce evidence supporting the alternative hypothesis.

A test of any statistical hypothesis where the alternative is **two sided**, such as

$$\begin{aligned}H_0: \theta &= \theta_0, \\H_1: \theta &\neq \theta_0,\end{aligned}$$

is called a **two-tailed test**, since the critical region is split into two parts, often having equal probabilities, in each tail of the distribution of the test statistic. The alternative hypothesis $\theta \neq \theta_0$ states that either $\theta < \theta_0$ or $\theta > \theta_0$. A two-tailed test was used to test the null hypothesis that $\mu = 68$ kilograms against the two-sided alternative $\mu \neq 68$ kilograms in the example of the continuous population of student weights.

How Are the Null and Alternative Hypotheses Chosen?

The null hypothesis H_0 will often be stated using the *equality sign*. With this approach, it is clear how the probability of type I error is controlled. However, there are situations in which “do not reject H_0 ” implies that the parameter θ might be any value defined by the natural complement to the alternative hypothesis. For example, in the vaccine example, where the alternative hypothesis is $H_1: p > 1/4$, it is quite possible that nonrejection of H_0 cannot rule out a value of p less than $1/4$. Clearly though, in the case of one-tailed tests, the statement of the alternative is the most important consideration.

Whether one sets up a one-tailed or a two-tailed test will depend on the conclusion to be drawn if H_0 is rejected. The location of the critical region can be determined only after H_1 has been stated. For example, in testing a new drug, one sets up the hypothesis that it is no better than similar drugs now on the market and tests this against the alternative hypothesis that the new drug is superior. Such an alternative hypothesis will result in a one-tailed test with the critical region in the right tail. However, if we wish to compare a new teaching technique with the conventional classroom procedure, the alternative hypothesis should allow for the new approach to be either inferior or superior to the conventional procedure. Hence, the test is two-tailed with the critical region divided equally so as to fall in the extreme left and right tails of the distribution of our statistic.

Example 10.1: A manufacturer of a certain brand of rice cereal claims that the average saturated fat content does not exceed 1.5 grams per serving. State the null and alternative hypotheses to be used in testing this claim and determine where the critical region is located.

Solution: The manufacturer's claim should be rejected only if μ is greater than 1.5 milligrams and should not be rejected if μ is less than or equal to 1.5 milligrams. We test

$$H_0: \mu = 1.5,$$

$$H_1: \mu > 1.5.$$

Nonrejection of H_0 does not rule out values less than 1.5 milligrams. Since we have a one-tailed test, the greater than symbol indicates that the critical region lies entirely in the right tail of the distribution of our test statistic $\bar{X}$. ■

Example 10.2: A real estate agent claims that 60% of all private residences being built today are 3-bedroom homes. To test this claim, a large sample of new residences is inspected; the proportion of these homes with 3 bedrooms is recorded and used as the test statistic. State the null and alternative hypotheses to be used in this test and determine the location of the critical region.

Solution: If the test statistic were substantially higher or lower than $p = 0.6$, we would reject the agent's claim. Hence, we should make the hypothesis

$$H_0: p = 0.6,$$

$$H_1: p \neq 0.6.$$

The alternative hypothesis implies a two-tailed test with the critical region divided equally in both tails of the distribution of $\hat{P}$, our test statistic. ■

10.3 The Use of P -Values for Decision Making in Testing Hypotheses

In testing hypotheses in which the test statistic is discrete, the critical region may be chosen arbitrarily and its size determined. If α is too large, it can be reduced by making an adjustment in the critical value. It may be necessary to increase the

sample size to offset the decrease that occurs automatically in the power of the test.

Over a number of generations of statistical analysis, it had become customary to choose an α of 0.05 or 0.01 and select the critical region accordingly. Then, of course, strict rejection or nonrejection of H_0 would depend on that critical region. For example, if the test is two tailed and α is set at the 0.05 level of significance and the test statistic involves, say, the standard normal distribution, then a z -value is observed from the data and the critical region is

$$z > 1.96 \quad \text{or} \quad z < -1.96,$$

where the value 1.96 is found as $z_{0.025}$ in Table A.3. A value of z in the critical region prompts the statement “The value of the test statistic is significant,” which we can then translate into the user’s language. For example, if the hypothesis is given by

$$H_0: \mu = 10,$$

$$H_1: \mu \neq 10,$$

one might say, “The mean differs significantly from the value 10.”

Preselection of a Significance Level

This preselection of a significance level α has its roots in the philosophy that the maximum risk of making a type I error should be controlled. However, this approach does not account for values of test statistics that are “close” to the critical region. Suppose, for example, in the illustration with $H_0: \mu = 10$ versus $H_1: \mu \neq 10$, a value of $z = 1.87$ is observed; strictly speaking, with $\alpha = 0.05$, the value is not significant. But the risk of committing a type I error if one rejects H_0 in this case could hardly be considered severe. In fact, in a two-tailed scenario, one can quantify this risk as

$$P = 2P(Z > 1.87 \text{ when } \mu = 10) = 2(0.0307) = 0.0614.$$

As a result, 0.0614 is the probability of obtaining a value of z as large as or larger (in magnitude) than 1.87 when in fact $\mu = 10$. Although this evidence against H_0 is not as strong as that which would result from rejection at an $\alpha = 0.05$ level, it is important information to the user. Indeed, continued use of $\alpha = 0.05$ or 0.01 is only a result of what standards have been passed down through the generations. **The P -value approach has been adopted extensively by users of applied statistics.** The approach is designed to give the user an alternative (in terms of a probability) to a mere “reject” or “do not reject” conclusion. The P -value computation also gives the user important information when the z -value falls well *into the ordinary critical region*. For example, if z is 2.73, it is informative for the user to observe that

$$P = 2(0.0032) = 0.0064,$$

and thus the z -value is significant at a level considerably less than 0.05. It is important to know that under the condition of H_0 , a value of $z = 2.73$ is an extremely rare event. That is, a value at least that large in magnitude would only occur 64 times in 10,000 experiments.

A Graphical Demonstration of a P -Value

One very simple way of explaining a P -value graphically is to consider two distinct samples. Suppose that two materials are being considered for coating a particular type of metal in order to inhibit corrosion. Specimens are obtained, and one collection is coated with material 1 and one collection coated with material 2. The sample sizes are $n_1 = n_2 = 10$, and corrosion is measured in percent of surface area affected. The hypothesis is that the samples came from common distributions with mean $\mu = 10$. Let us assume that the population variance is 1.0. Then we are testing

$$H_0: \mu_1 = \mu_2 = 10.$$

Let Figure 10.8 represent a point plot of the data; the data are placed on the distribution stated by the null hypothesis. Let us assume that the “ $\times$ ” data refer to material 1 and the “ $\circ$ ” data refer to material 2. Now it seems clear that the data do refute the null hypothesis. But how can this be summarized in one number? **The P -value can be viewed as simply the probability of obtaining these data given that both samples come from the same distribution.** Clearly, this probability is quite small, say 0.00000001! Thus, the small P -value clearly refutes H_0 , and the conclusion is that the population means are significantly different.

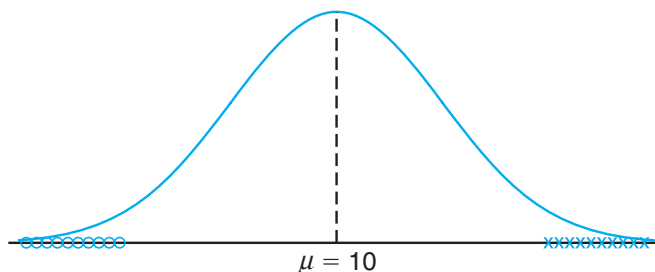


Figure 10.8: Data that are likely generated from populations having two different means.

Use of the P -value approach as an aid in decision-making is quite natural, and nearly all computer packages that provide hypothesis-testing computation print out P -values along with values of the appropriate test statistic. The following is a formal definition of a P -value.

Definition 10.5: A **P -value** is the lowest level (of significance) at which the observed value of the test statistic is significant.

How Does the Use of P -Values Differ from Classic Hypothesis Testing?

It is tempting at this point to summarize the procedures associated with testing, say, $H_0: \theta = \theta_0$. However, the student who is a novice in this area should understand that there are differences in approach and philosophy between the classic

fixed α approach that is climaxed with either a “reject H_0 ” or a “do not reject H_0 ” conclusion and the P -value approach. In the latter, no fixed α is determined and conclusions are drawn on the basis of the size of the P -value in harmony with the subjective judgment of the engineer or scientist. While modern computer software will output P -values, nevertheless it is important that readers understand both approaches in order to appreciate the totality of the concepts. Thus, we offer a brief list of procedural steps for both the classical and the P -value approach.

Approach to Hypothesis Testing with Fixed Probability of Type I Error	1. State the null and alternative hypotheses. 2. Choose a fixed significance level α. 3. Choose an appropriate test statistic and establish the critical region based on α. 4. Reject H_0 if the computed test statistic is in the critical region. Otherwise, do not reject. 5. Draw scientific or engineering conclusions.
Significance Testing (P -Value Approach)	1. State null and alternative hypotheses. 2. Choose an appropriate test statistic. 3. Compute the P-value based on the computed value of the test statistic. 4. Use judgment based on the P-value and knowledge of the scientific system.

In later sections of this chapter and chapters that follow, many examples and exercises emphasize the P -value approach to drawing scientific conclusions.

Exercises

10.1 Suppose that an allergist wishes to test the hypothesis that at least 30% of the public is allergic to some cheese products. Explain how the allergist could commit

- (a) a type I error;
- (b) a type II error.

10.2 A sociologist is concerned about the effectiveness of a training course designed to get more drivers to use seat belts in automobiles.

- (a) What hypothesis is she testing if she commits a type I error by erroneously concluding that the training course is ineffective?
- (b) What hypothesis is she testing if she commits a type II error by erroneously concluding that the training course is effective?

10.3 A large manufacturing firm is being charged with discrimination in its hiring practices.

- (a) What hypothesis is being tested if a jury commits a type I error by finding the firm guilty?
- (b) What hypothesis is being tested if a jury commits a type II error by finding the firm guilty?

10.4 A fabric manufacturer believes that the proportion of orders for raw material arriving late is $p = 0.6$. If a random sample of 10 orders shows that 3 or fewer arrived late, the hypothesis that $p = 0.6$ should be rejected in favor of the alternative $p < 0.6$. Use the binomial distribution.

- (a) Find the probability of committing a type I error if the true proportion is $p = 0.6$.
- (b) Find the probability of committing a type II error for the alternatives $p = 0.3$, $p = 0.4$, and $p = 0.5$.

10.5 Repeat Exercise 10.4 but assume that 50 orders are selected and the critical region is defined to be $x \leq 24$, where x is the number of orders in the sample that arrived late. Use the normal approximation.

10.6 The proportion of adults living in a small town who are college graduates is estimated to be $p = 0.6$. To test this hypothesis, a random sample of 15 adults is selected. If the number of college graduates in the sample is anywhere from 6 to 12, we shall not reject the null hypothesis that $p = 0.6$; otherwise, we shall conclude that $p \neq 0.6$.

- (a) Evaluate α assuming that $p = 0.6$. Use the binomial distribution.

- (b) Evaluate β for the alternatives $p = 0.5$ and $p = 0.7$.
 (c) Is this a good test procedure?

10.7 Repeat Exercise 10.6 but assume that 200 adults are selected and the fail-to-reject region is defined to be $110 \leq x \leq 130$, where x is the number of college graduates in our sample. Use the normal approximation.

10.8 In *Relief from Arthritis* published by Thorsons Publishers, Ltd., John E. Croft claims that over 40% of those who suffer from osteoarthritis receive measurable relief from an ingredient produced by a particular species of mussel found off the coast of New Zealand. To test this claim, the mussel extract is to be given to a group of 7 osteoarthritic patients. If 3 or more of the patients receive relief, we shall not reject the null hypothesis that $p = 0.4$; otherwise, we conclude that $p < 0.4$.

- (a) Evaluate α , assuming that $p = 0.4$.
 (b) Evaluate β for the alternative $p = 0.3$.

10.9 A dry cleaning establishment claims that a new spot remover will remove more than 70% of the spots to which it is applied. To check this claim, the spot remover will be used on 12 spots chosen at random. If fewer than 11 of the spots are removed, we shall not reject the null hypothesis that $p = 0.7$; otherwise, we conclude that $p > 0.7$.

- (a) Evaluate α , assuming that $p = 0.7$.
 (b) Evaluate β for the alternative $p = 0.9$.

10.10 Repeat Exercise 10.9 but assume that 100 spots are treated and the critical region is defined to be $x > 82$, where x is the number of spots removed.

10.11 Repeat Exercise 10.8 but assume that 70 patients are given the mussel extract and the critical region is defined to be $x < 24$, where x is the number of osteoarthritic patients who receive relief.

10.12 A random sample of 400 voters in a certain city are asked if they favor an additional 4% gasoline sales tax to provide badly needed revenues for street repairs. If more than 220 but fewer than 260 favor the sales tax, we shall conclude that 60% of the voters are for it.

- (a) Find the probability of committing a type I error if 60% of the voters favor the increased tax.
 (b) What is the probability of committing a type II error using this test procedure if actually only 48% of the voters are in favor of the additional gasoline tax?

10.13 Suppose, in Exercise 10.12, we conclude that 60% of the voters favor the gasoline sales tax if more than 214 but fewer than 266 voters in our sample favor it. Show that this new critical region results in a smaller value for α at the expense of increasing β .

10.14 A manufacturer has developed a new fishing line, which the company claims has a mean breaking strength of 15 kilograms with a standard deviation of 0.5 kilogram. To test the hypothesis that $\mu = 15$ kilograms against the alternative that $\mu < 15$ kilograms, a random sample of 50 lines will be tested. The critical region is defined to be $\bar{x} < 14.9$.

- (a) Find the probability of committing a type I error when H_0 is true.
 (b) Evaluate β for the alternatives $\mu = 14.8$ and $\mu = 14.9$ kilograms.

10.15 A soft-drink machine at a steak house is regulated so that the amount of drink dispensed is approximately normally distributed with a mean of 200 milliliters and a standard deviation of 15 milliliters. The machine is checked periodically by taking a sample of 9 drinks and computing the average content. If $\bar{x}$ falls in the interval $191 < \bar{x} < 209$, the machine is thought to be operating satisfactorily; otherwise, we conclude that $\mu \neq 200$ milliliters.

- (a) Find the probability of committing a type I error when $\mu = 200$ milliliters.
 (b) Find the probability of committing a type II error when $\mu = 215$ milliliters.

10.16 Repeat Exercise 10.15 for samples of size $n = 25$. Use the same critical region.

10.17 A new curing process developed for a certain type of cement results in a mean compressive strength of 5000 kilograms per square centimeter with a standard deviation of 120 kilograms. To test the hypothesis that $\mu = 5000$ against the alternative that $\mu < 5000$, a random sample of 50 pieces of cement is tested. The critical region is defined to be $\bar{x} < 4970$.

- (a) Find the probability of committing a type I error when H_0 is true.
 (b) Evaluate β for the alternatives $\mu = 4970$ and $\mu = 4960$.

10.18 If we plot the probabilities of failing to reject H_0 corresponding to various alternatives for μ (including the value specified by H_0) and connect all the points by a smooth curve, we obtain the **operating characteristic curve** of the test criterion, or simply the OC curve. Note that the probability of failing to reject H_0 when it is true is simply $1 - \alpha$. Operating characteristic curves are widely used in industrial applications to provide a visual display of the merits of the test criterion. With reference to Exercise 10.15, find the probabilities of failing to reject H_0 for the following 9 values of μ and plot the OC curve: 184, 188, 192, 196, 200, 204, 208, 212, and 216.

10.4 Single Sample: Tests Concerning a Single Mean

In this section, we formally consider tests of hypotheses on a single population mean. Many of the illustrations from previous sections involved tests on the mean, so the reader should already have insight into some of the details that are outlined here.

Tests on a Single Mean (Variance Known)

We should first describe the assumptions on which the experiment is based. The model for the underlying situation centers around an experiment with $X_1, X_2, \dots, X_n$ representing a random sample from a distribution with mean μ and variance $\sigma^2 > 0$. Consider first the hypothesis

$$H_0: \mu = \mu_0,$$

$$H_1: \mu \neq \mu_0.$$

The appropriate test statistic should be based on the random variable $\bar{X}$. In Chapter 8, the Central Limit Theorem was introduced, which essentially states that despite the distribution of X , the random variable $\bar{X}$ has approximately a normal distribution with mean μ and variance σ^2/n for reasonably large sample sizes. So, $\mu_{\bar{X}} = \mu$ and $\sigma_{\bar{X}}^2 = \sigma^2/n$. We can then determine a critical region based on the computed sample average, $\bar{x}$. It should be clear to the reader by now that there will be a two-tailed critical region for the test.

Standardization of $\bar{X}$

It is convenient to standardize $\bar{X}$ and formally involve the **standard normal** random variable Z , where

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

We know that *under* H_0 , that is, if $\mu = \mu_0$, $\sqrt{n}(\bar{X} - \mu_0)/\sigma$ follows an $n(x; 0, 1)$ distribution, and hence the expression

$$P\left(-z_{\alpha/2} < \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = 1 - \alpha$$

can be used to write an appropriate nonrejection region. The reader should keep in mind that, formally, the critical region is designed to control α , the probability of type I error. It should be obvious that a *two-tailed signal* of evidence is needed to support H_1 . Thus, given a computed value $\bar{x}$, the formal test involves rejecting H_0 if the computed *test statistic* z falls in the critical region described next.

Test Procedure
for a Single Mean
(Variance
Known)

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} > z_{\alpha/2} \quad \text{or} \quad z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} < -z_{\alpha/2}$$

If $-z_{\alpha/2} < z < z_{\alpha/2}$, do not reject H_0 . Rejection of H_0 , of course, implies acceptance of the alternative hypothesis $\mu \neq \mu_0$. With this definition of the critical region, it should be clear that there will be probability α of rejecting H_0 (falling into the critical region) when, indeed, $\mu = \mu_0$.

Although it is easier to understand the critical region written in terms of z , we can write the same critical region in terms of the computed average $\bar{x}$. The following can be written as an identical decision procedure:

$$\text{reject } H_0 \text{ if } \bar{x} < a \text{ or } \bar{x} > b,$$

where

$$a = \mu_0 - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \quad b = \mu_0 + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

Hence, for a significance level α , the critical values of the random variable z and $\bar{x}$ are both depicted in Figure 10.9.

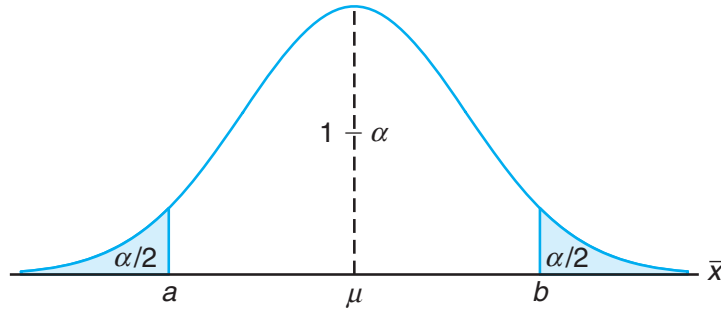


Figure 10.9: Critical region for the alternative hypothesis $\mu \neq \mu_0$.

Tests of one-sided hypotheses on the mean involve the same statistic described in the two-sided case. The difference, of course, is that the critical region is only in one tail of the standard normal distribution. For example, suppose that we seek to test

$$\begin{aligned} H_0: & \mu = \mu_0, \\ H_1: & \mu > \mu_0. \end{aligned}$$

The signal that favors H_1 comes from *large values* of z . Thus, rejection of H_0 results when the computed $z > z_\alpha$. Obviously, if the alternative is $H_1: \mu < \mu_0$, the critical region is entirely in the lower tail and thus rejection results from $z < -z_\alpha$. Although in a one-sided testing case the null hypothesis can be written as $H_0: \mu \leq \mu_0$ or $H_0: \mu \geq \mu_0$, it is usually written as $H_0: \mu = \mu_0$.

The following two examples illustrate tests on means for the case in which σ is known.

Example 10.3: A random sample of 100 recorded deaths in the United States during the past year showed an average life span of 71.8 years. Assuming a population standard deviation of 8.9 years, does this seem to indicate that the mean life span today is greater than 70 years? Use a 0.05 level of significance.

- Solution:**
1. $H_0: \mu = 70$ years.
 2. $H_1: \mu > 70$ years.
 3. $\alpha = 0.05$.
 4. Critical region: $z > 1.645$, where $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$.
 5. Computations: $\bar{x} = 71.8$ years, $\sigma = 8.9$ years, and hence $z = \frac{71.8 - 70}{8.9/\sqrt{100}} = 2.02$.
 6. Decision: Reject H_0 and conclude that the mean life span today is greater than 70 years.

The P -value corresponding to $z = 2.02$ is given by the area of the shaded region in Figure 10.10.

Using Table A.3, we have

$$P = P(Z > 2.02) = 0.0217.$$

As a result, the evidence in favor of H_1 is even stronger than that suggested by a 0.05 level of significance. 

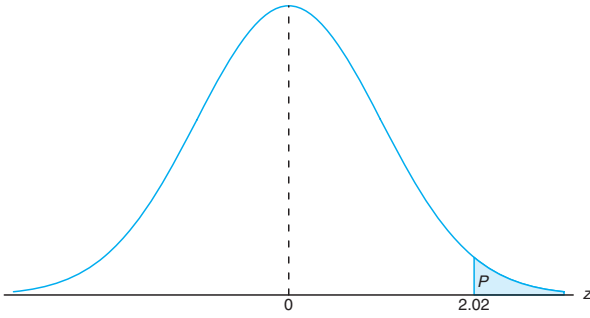
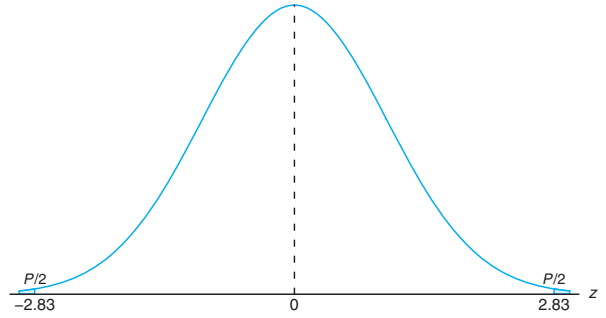
Example 10.4: A manufacturer of sports equipment has developed a new synthetic fishing line that the company claims has a mean breaking strength of 8 kilograms with a standard deviation of 0.5 kilogram. Test the hypothesis that $\mu = 8$ kilograms against the alternative that $\mu \neq 8$ kilograms if a random sample of 50 lines is tested and found to have a mean breaking strength of 7.8 kilograms. Use a 0.01 level of significance.

- Solution:**
1. $H_0: \mu = 8$ kilograms.
 2. $H_1: \mu \neq 8$ kilograms.
 3. $\alpha = 0.01$.
 4. Critical region: $z < -2.575$ and $z > 2.575$, where $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$.
 5. Computations: $\bar{x} = 7.8$ kilograms, $n = 50$, and hence $z = \frac{7.8 - 8}{0.5/\sqrt{50}} = -2.83$.
 6. Decision: Reject H_0 and conclude that the average breaking strength is not equal to 8 but is, in fact, less than 8 kilograms.

Since the test in this example is two tailed, the desired P -value is twice the area of the shaded region in Figure 10.11 to the left of $z = -2.83$. Therefore, using Table A.3, we have

$$P = P(|Z| > 2.83) = 2P(Z < -2.83) = 0.0046,$$

which allows us to reject the null hypothesis that $\mu = 8$ kilograms at a level of significance smaller than 0.01. 

Figure 10.10: P -value for Example 10.3.Figure 10.11: P -value for Example 10.4.

Relationship to Confidence Interval Estimation

The reader should realize by now that the hypothesis-testing approach to statistical inference in this chapter is very closely related to the confidence interval approach in Chapter 9. Confidence interval estimation involves computation of bounds within which it is “reasonable” for the parameter in question to lie. For the case of a single population mean μ with σ^2 known, the structure of both hypothesis testing and confidence interval estimation is based on the random variable

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

It turns out that the testing of $H_0: \mu = \mu_0$ against $H_1: \mu \neq \mu_0$ at a significance level α is equivalent to computing a $100(1 - \alpha)\%$ confidence interval on μ and rejecting H_0 if μ_0 is outside the confidence interval. If μ_0 is inside the confidence interval, the hypothesis is not rejected. The equivalence is very intuitive and quite simple to illustrate. Recall that with an observed value $\bar{x}$, failure to reject H_0 at significance level α implies that

$$-z_{\alpha/2} \leq \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \leq z_{\alpha/2},$$

which is equivalent to

$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu_0 \leq \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

The equivalence of confidence interval estimation to hypothesis testing extends to differences between two means, variances, ratios of variances, and so on. As a result, the student of statistics should not consider confidence interval estimation and hypothesis testing as separate forms of statistical inference. For example, consider Example 9.2 on page 271. The 95% confidence interval on the mean is given by the bounds (2.50, 2.70). Thus, with the same sample information, a two-sided hypothesis on μ involving any hypothesized value between 2.50 and 2.70 will not be rejected. As we turn to different areas of hypothesis testing, the equivalence to the confidence interval estimation will continue to be exploited.

Tests on a Single Sample (Variance Unknown)

One would certainly suspect that tests on a population mean μ with σ^2 unknown, like confidence interval estimation, should involve the use of Student t -distribution. Strictly speaking, the application of Student t for both confidence intervals and hypothesis testing is developed under the following assumptions. The random variables $X_1, X_2, \dots, X_n$ represent a random sample from a normal distribution with unknown μ and σ^2 . Then the random variable $\sqrt{n}(\bar{X} - \mu)/S$ has a Student t -distribution with $n - 1$ degrees of freedom. The structure of the test is identical to that for the case of σ known, with the exception that the value σ in the test statistic is replaced by the computed estimate S and the standard normal distribution is replaced by a t -distribution.

The t -Statistic
for a Test on a
Single Mean
(Variance
Unknown)

For the two-sided hypothesis

$$H_0: \mu = \mu_0,$$

$$H_1: \mu \neq \mu_0,$$

we reject H_0 at significance level α when the computed t -statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

exceeds $t_{\alpha/2, n-1}$ or is less than $-t_{\alpha/2, n-1}$.

The reader should recall from Chapters 8 and 9 that the t -distribution is symmetric around the value zero. Thus, this two-tailed critical region applies in a fashion similar to that for the case of known σ . For the two-sided hypothesis at significance level α , the two-tailed critical regions apply. For $H_1: \mu > \mu_0$, rejection results when $t > t_{\alpha, n-1}$. For $H_1: \mu < \mu_0$, the critical region is given by $t < -t_{\alpha, n-1}$.

Example 10.5: The Edison Electric Institute has published figures on the number of kilowatt hours used annually by various home appliances. It is claimed that a vacuum cleaner uses an average of 46 kilowatt hours per year. If a random sample of 12 homes included in a planned study indicates that vacuum cleaners use an average of 42 kilowatt hours per year with a standard deviation of 11.9 kilowatt hours, does this suggest at the 0.05 level of significance that vacuum cleaners use, on average, less than 46 kilowatt hours annually? Assume the population of kilowatt hours to be normal.

- Solution:**
1. $H_0: \mu = 46$ kilowatt hours.
 2. $H_1: \mu < 46$ kilowatt hours.
 3. $\alpha = 0.05$.
 4. Critical region: $t < -1.796$, where $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ with 11 degrees of freedom.
 5. Computations: $\bar{x} = 42$ kilowatt hours, $s = 11.9$ kilowatt hours, and $n = 12$. Hence,

$$t = \frac{42 - 46}{11.9/\sqrt{12}} = -1.16, \quad P = P(T < -1.16) \approx 0.135.$$

6. Decision: Do not reject H_0 and conclude that the average number of kilowatt hours used annually by home vacuum cleaners is not significantly less than 46. J

Comment on the Single-Sample t -Test

The reader has probably noticed that the equivalence of the two-tailed t -test for a single mean and the computation of a confidence interval on μ with σ replaced by s is maintained. For example, consider Example 9.5 on page 275. Essentially, we can view that computation as one in which we have found all values of μ_0 , the hypothesized mean volume of containers of sulfuric acid, for which the hypothesis $H_0: \mu = \mu_0$ will not be rejected at $\alpha = 0.05$. Again, this is consistent with the statement “Based on the sample information, values of the population mean volume between 9.74 and 10.26 liters are not unreasonable.”

Comments regarding the normality assumption are worth emphasizing at this point. We have indicated that when σ is known, the Central Limit Theorem allows for the use of a test statistic or a confidence interval which is based on Z , the standard normal random variable. Strictly speaking, of course, the Central Limit Theorem, and thus the use of the standard normal distribution, does not apply unless σ is known. In Chapter 8, the development of the t -distribution was given. There we pointed out that normality on $X_1, X_2, \dots, X_n$ was an underlying assumption. Thus, *strictly speaking*, the Student’s t -tables of percentage points for tests or confidence intervals should not be used unless it is known that the sample comes from a normal population. In practice, σ can rarely be assumed to be known. However, a very good estimate may be available from previous experiments. Many statistics textbooks suggest that one can safely replace σ by s in the test statistic

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

when $n \geq 30$ with a bell-shaped population and still use the Z -tables for the appropriate critical region. The implication here is that the Central Limit Theorem is indeed being invoked and one is relying on the fact that $s \approx \sigma$. Obviously, when this is done, the results must be viewed as approximate. Thus, a computed P -value (from the Z -distribution) of 0.15 may be 0.12 or perhaps 0.17, or a computed confidence interval may be a 93% confidence interval rather than a 95% interval as desired. Now what about situations where $n \leq 30$? The user cannot rely on s being close to σ , and in order to take into account the inaccuracy of the estimate, the confidence interval should be wider or the critical value larger in magnitude. The t -distribution percentage points accomplish this but are correct only when the sample is from a normal distribution. Of course, normal probability plots can be used to ascertain some sense of the deviation of normality in a data set.

For small samples, it is often difficult to detect deviations from a normal distribution. (Goodness-of-fit tests are discussed in a later section of this chapter.) For bell-shaped distributions of the random variables $X_1, X_2, \dots, X_n$, the use of the t -distribution for tests or confidence intervals is likely to produce quite good results. When in doubt, the user should resort to nonparametric procedures, which are presented in Chapter 16.

Annotated Computer Printout for Single-Sample t -Test

It should be of interest for the reader to see an annotated computer printout showing the result of a single-sample t -test. Suppose that an engineer is interested in testing the bias in a pH meter. Data are collected on a neutral substance (pH = 7.0). A sample of the measurements were taken with the data as follows:

7.07 7.00 7.10 6.97 7.00 7.03 7.01 7.01 6.98 7.08

It is, then, of interest to test

$$H_0: \mu = 7.0,$$

$$H_1: \mu \neq 7.0.$$

In this illustration, we use the computer package *MINITAB* to illustrate the analysis of the data set above. Notice the key components of the printout shown in Figure 10.12. Of course, the mean $\bar{y}$ is 7.0250, StDev is simply the sample standard deviation $s = 0.044$, and SE Mean is the estimated standard error of the mean and is computed as $s/\sqrt{n} = 0.0139$. The t -value is the ratio

$$(7.0250 - 7)/0.0139 = 1.80.$$

pH-meter									
7.07	7.00	7.10	6.97	7.00	7.03	7.01	7.01	6.98	7.08
MTB > Onet 'pH-meter'; SUBC> Test 7.									
One-Sample T: pH-meter Test of mu = 7 vs not = 7									
Variable	N	Mean	StDev	SE Mean	95% CI		T	P	
pH-meter	10	7.02500	0.04403	0.01392	(6.99350, 7.05650)		1.80	0.106	

Figure 10.12: *MINITAB* printout for one sample t -test for pH meter.

The P -value of 0.106 suggests results that are inconclusive. There is no evidence suggesting a strong rejection of H_0 (based on an α of 0.05 or 0.10), **yet one certainly cannot truly conclude that the pH meter is unbiased**. Notice that the sample size of 10 is rather small. An increase in sample size (perhaps another experiment) may sort things out. A discussion regarding appropriate sample size appears in Section 10.6.

10.5 Two Samples: Tests on Two Means

The reader should now understand the relationship between tests and confidence intervals, and can only heavily rely on details supplied by the confidence interval material in Chapter 9. Tests concerning two means represent a set of very important analytical tools for the scientist or engineer. The experimental setting is very much like that described in Section 9.8. Two independent random samples of sizes

n_1 and n_2 , respectively, are drawn from two populations with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2 . We know that the random variable

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

has a standard normal distribution. Here we are assuming that n_1 and n_2 are sufficiently large that the Central Limit Theorem applies. Of course, if the two populations are normal, the statistic above has a standard normal distribution even for small n_1 and n_2 . Obviously, if we can assume that $\sigma_1 = \sigma_2 = \sigma$, the statistic above reduces to

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma\sqrt{1/n_1 + 1/n_2}}.$$

The two statistics above serve as a basis for the development of the test procedures involving two means. The equivalence between tests and confidence intervals, along with the technical detail involving tests on one mean, allow a simple transition to tests on two means.

The two-sided hypothesis on two means can be written generally as

$$H_0: \mu_1 - \mu_2 = d_0.$$

Obviously, the alternative can be two sided or one sided. Again, the distribution used is the distribution of the test statistic under H_0 . Values $\bar{x}_1$ and $\bar{x}_2$ are computed and, for σ_1 and σ_2 known, the test statistic is given by

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}},$$

with a two-tailed critical region in the case of a two-sided alternative. That is, reject H_0 in favor of H_1 : $\mu_1 - \mu_2 \neq d_0$ if $z > z_{\alpha/2}$ or $z < -z_{\alpha/2}$. One-tailed critical regions are used in the case of the one-sided alternatives. The reader should, as before, study the test statistic and be satisfied that for, say, H_1 : $\mu_1 - \mu_2 > d_0$, the signal favoring H_1 comes from large values of z . Thus, the upper-tailed critical region applies.

Unknown But Equal Variances

The more prevalent situations involving tests on two means are those in which variances are unknown. If the scientist involved is willing to assume that both distributions are normal and that $\sigma_1 = \sigma_2 = \sigma$, the *pooled t-test* (often called the two-sample *t-test*) may be used. The test statistic (see Section 9.8) is given by the following test procedure.

Two-Sample
Pooled t -Test

For the two-sided hypothesis

$$H_0: \mu_1 = \mu_2,$$

$$H_1: \mu_1 \neq \mu_2,$$

we reject H_0 at significance level α when the computed t -statistic

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{1/n_1 + 1/n_2}},$$

where

$$s_p^2 = \frac{s_1^2(n_1 - 1) + s_2^2(n_2 - 1)}{n_1 + n_2 - 2}$$

exceeds $t_{\alpha/2, n_1+n_2-2}$ or is less than $-t_{\alpha/2, n_1+n_2-2}$.

Recall from Chapter 9 that the degrees of freedom for the t -distribution are a result of pooling of information from the two samples to estimate σ^2 . One-sided alternatives suggest one-sided critical regions, as one might expect. For example, for $H_1: \mu_1 - \mu_2 > d_0$, reject $H_0: \mu_1 - \mu_2 = d_0$ when $t > t_{\alpha, n_1+n_2-2}$.

Example 10.6: An experiment was performed to compare the abrasive wear of two different laminated materials. Twelve pieces of material 1 were tested by exposing each piece to a machine measuring wear. Ten pieces of material 2 were similarly tested. In each case, the depth of wear was observed. The samples of material 1 gave an average (coded) wear of 85 units with a sample standard deviation of 4, while the samples of material 2 gave an average of 81 with a sample standard deviation of 5. Can we conclude at the 0.05 level of significance that the abrasive wear of material 1 exceeds that of material 2 by more than 2 units? Assume the populations to be approximately normal with equal variances.

Solution: Let μ_1 and μ_2 represent the population means of the abrasive wear for material 1 and material 2, respectively.

1. $H_0: \mu_1 - \mu_2 = 2$.
2. $H_1: \mu_1 - \mu_2 > 2$.
3. $\alpha = 0.05$.
4. Critical region: $t > 1.725$, where $t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{1/n_1 + 1/n_2}}$ with $v = 20$ degrees of freedom.
5. Computations:

$$\begin{array}{lll} \bar{x}_1 = 85, & s_1 = 4, & n_1 = 12, \\ \bar{x}_2 = 81, & s_2 = 5, & n_2 = 10. \end{array}$$

Hence

$$\begin{aligned}
 s_p &= \sqrt{\frac{(11)(16) + (9)(25)}{12 + 10 - 2}} = 4.478, \\
 t &= \frac{(85 - 81) - 2}{4.478\sqrt{1/12 + 1/10}} = 1.04, \\
 P &= P(T > 1.04) \approx 0.16. \quad (\text{See Table A.4.})
 \end{aligned}$$

6. Decision: Do not reject H_0 . We are unable to conclude that the abrasive wear of material 1 exceeds that of material 2 by more than 2 units. 

Unknown But Unequal Variances

There are situations where the analyst is **not** able to assume that $\sigma_1 = \sigma_2$. Recall from Section 9.8 that, if the populations are normal, the statistic

$$T' = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

has an approximate t -distribution with approximate degrees of freedom

$$v = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{(s_1^2/n_1)^2/(n_1 - 1) + (s_2^2/n_2)^2/(n_2 - 1)}.$$

As a result, the test procedure is to *not reject* H_0 when

$$-t_{\alpha/2, v} < t' < t_{\alpha/2, v},$$

with v given as above. Again, as in the case of the pooled t -test, one-sided alternatives suggest one-sided critical regions.

Paired Observations

A study of the two-sample t -test or confidence interval on the difference between means should suggest the need for experimental design. Recall the discussion of experimental units in Chapter 9, where it was suggested that the conditions of the two populations (often referred to as the two treatments) should be assigned randomly to the experimental units. This is done to avoid biased results due to systematic differences between experimental units. In other words, in hypothesis-testing jargon, it is important that any significant difference found between means be due to the different conditions of the populations and not due to the experimental units in the study. For example, consider Exercise 9.40 in Section 9.9. The 20 seedlings play the role of the experimental units. Ten of them are to be treated with nitrogen and 10 with no nitrogen. It may be very important that this assignment to the “nitrogen” and “no-nitrogen” treatments be random to ensure that systematic differences between the seedlings do not interfere with a valid comparison between the means.

In Example 10.6, time of measurement is the most likely choice for the experimental unit. The 22 pieces of material should be measured in random order. We

need to guard against the possibility that wear measurements made close together in time might tend to give similar results. **Systematic (nonrandom) differences in experimental units are not expected.** However, random assignments guard against the problem.

References to planning of experiments, randomization, choice of sample size, and so on, will continue to influence much of the development in Chapters 13, 14, and 15. Any scientist or engineer whose interest lies in analysis of real data should study this material. The pooled t -test is extended in Chapter 13 to cover more than two means.

Testing of two means can be accomplished when data are in the form of paired observations, as discussed in Chapter 9. In this pairing structure, the conditions of the two populations (treatments) are assigned randomly within homogeneous units. Computation of the confidence interval for $\mu_1 - \mu_2$ in the situation with paired observations is based on the random variable

$$T = \frac{\bar{D} - \mu_D}{S_d/\sqrt{n}},$$

where $\bar{D}$ and S_d are random variables representing the sample mean and standard deviation of the differences of the observations in the experimental units. As in the case of the *pooled t-test*, the assumption is that the observations from each population are normal. This two-sample problem is essentially reduced to a one-sample problem by using the computed differences $d_1, d_2, \dots, d_n$. Thus, the hypothesis reduces to

$$H_0: \mu_D = d_0.$$

The computed test statistic is then given by

$$t = \frac{\bar{d} - d_0}{s_d/\sqrt{n}}.$$

Critical regions are constructed using the t -distribution with $n - 1$ degrees of freedom.

Problem of Interaction in a Paired t -Test

Not only will the case study that follows illustrate the use of the paired t -test but the discussion will shed considerable light on the difficulties that arise when there is an interaction between the treatments and the experimental units in the paired t structure. Recall that interaction between factors was introduced in Section 1.7 in a discussion of general types of statistical studies. The concept of interaction will be an important issue from Chapter 13 through Chapter 15.

There are some types of statistical tests in which the existence of interaction results in difficulty. The paired t -test is one such example. In Section 9.9, the paired structure was used in the computation of a confidence interval on the difference between two means, and the advantage in pairing was revealed for situations in which the experimental units are homogeneous. The pairing results in a reduction in σ_D , the standard deviation of a difference $D_i = X_{1i} - X_{2i}$, as discussed in

Section 9.9. If interaction exists between treatments and experimental units, the advantage gained in pairing may be substantially reduced. Thus, in Example 9.13 on page 293, the no interaction assumption allowed the difference in mean TCDD levels (plasma vs. fat tissue) to be the same across veterans. A quick glance at the data would suggest that there is no significant violation of the assumption of no interaction.

In order to demonstrate how interaction influences $\text{Var}(D)$ and hence the quality of the paired t -test, it is instructive to revisit the i th difference given by $D_i = X_{1i} - X_{2i} = (\mu_1 - \mu_2) + (\epsilon_1 - \epsilon_2)$, where X_{1i} and X_{2i} are taken on the i th experimental unit. If the pairing unit is homogeneous, the errors in X_{1i} and in X_{2i} should be similar and not independent. We noted in Chapter 9 that the positive covariance between the errors results in a reduced $\text{Var}(D)$. Thus, the size of the difference in the treatments and the relationship between the errors in X_{1i} and X_{2i} contributed by the experimental unit will tend to allow a significant difference to be detected.

What Conditions Result in Interaction?

Let us consider a situation in which the experimental units are not homogeneous. Rather, consider the i th experimental unit with random variables X_{1i} and X_{2i} that are not similar. Let ϵ_{1i} and ϵ_{2i} be random variables representing the errors in the values X_{1i} and X_{2i} , respectively, at the i th unit. Thus, we may write

$$X_{1i} = \mu_1 + \epsilon_{1i} \text{ and } X_{2i} = \mu_2 + \epsilon_{2i}.$$

The errors with expectation zero may tend to cause the response values X_{1i} and X_{2i} to move in opposite directions, resulting in a negative value for $\text{Cov}(\epsilon_{1i}, \epsilon_{2i})$ and hence negative $\text{Cov}(X_{1i}, X_{2i})$. In fact, the model may be complicated even more by the fact that $\sigma_1^2 = \text{Var}(\epsilon_{1i}) \neq \sigma_2^2 = \text{Var}(\epsilon_{2i})$. The variance and covariance parameters may vary among the n experimental units. Thus, unlike in the homogeneous case, D_i will tend to be quite different across experimental units due to the heterogeneous nature of the difference in $\epsilon_1 - \epsilon_2$ among the units. This produces the interaction between treatments and units. In addition, for a specific experimental unit (see Theorem 4.9),

$$\sigma_D^2 = \text{Var}(D) = \text{Var}(\epsilon_1) + \text{Var}(\epsilon_2) - 2 \text{Cov}(\epsilon_1, \epsilon_2)$$

is inflated by the negative covariance term, and thus the advantage gained in pairing in the homogeneous unit case is lost in the case described here. While the inflation in $\text{Var}(D)$ will vary from case to case, there is a danger in some cases that the increase in variance may neutralize any difference that exists between μ_1 and μ_2 . Of course, a large value of $\bar{d}$ in the t -statistic may reflect a treatment difference that overcomes the inflated variance estimate, s_d^2 .

Case Study 10.1: Blood Sample Data: In a study conducted in the Forestry and Wildlife Department at Virginia Tech, J. A. Wesson examined the influence of the drug succinylcholine on the circulation levels of androgens in the blood. Blood samples were taken from wild, free-ranging deer immediately after they had received an intramuscular injection of succinylcholine administered using darts and a capture gun. A second blood sample was obtained from each deer 30 minutes after the

first sample, after which the deer was released. The levels of androgens at time of capture and 30 minutes later, measured in nanograms per milliliter (ng/mL), for 15 deer are given in Table 10.2.

Assuming that the populations of androgen levels at time of injection and 30 minutes later are normally distributed, test at the 0.05 level of significance whether the androgen concentrations are altered after 30 minutes.

Table 10.2: Data for Case Study 10.1

Deer	Androgen (ng/mL)		d_i
	At Time of Injection	30 Minutes after Injection	
1	2.76	7.02	4.26
2	5.18	3.10	-2.08
3	2.68	5.44	2.76
4	3.05	3.99	0.94
5	4.10	5.21	1.11
6	7.05	10.26	3.21
7	6.60	13.91	7.31
8	4.79	18.53	13.74
9	7.39	7.91	0.52
10	7.30	4.85	-2.45
11	11.78	11.10	-0.68
12	3.90	3.74	-0.16
13	26.00	94.03	68.03
14	67.48	94.03	26.55
15	17.04	41.70	24.66

Solution: Let μ_1 and μ_2 be the average androgen concentration at the time of injection and 30 minutes later, respectively. We proceed as follows:

1. H_0 : $\mu_1 = \mu_2$ or $\mu_D = \mu_1 - \mu_2 = 0$.
2. H_1 : $\mu_1 \neq \mu_2$ or $\mu_D = \mu_1 - \mu_2 \neq 0$.
3. $\alpha = 0.05$.
4. Critical region: $t < -2.145$ and $t > 2.145$, where $t = \frac{\bar{d} - d_0}{s_D/\sqrt{n}}$ with $v = 14$ degrees of freedom.
5. Computations: The sample mean and standard deviation for the d_i are

$$\bar{d} = 9.848 \quad \text{and} \quad s_d = 18.474.$$

Therefore,

$$t = \frac{9.848 - 0}{18.474/\sqrt{15}} = 2.06.$$

6. Though the t -statistic is not significant at the 0.05 level, from Table A.4,

$$P = P(|T| > 2.06) \approx 0.06.$$

As a result, there is some evidence that there is a difference in mean circulating levels of androgen. J

The assumption of no interaction would imply that the effect on androgen levels of the deer is roughly the same in the data for both treatments, i.e., at the time of injection of succinylcholine and 30 minutes following injection. This can be expressed with the two factors switching roles; for example, the difference in treatments is roughly the same across the units (i.e., the deer). There certainly are some deer/treatment combinations for which the no interaction assumption seems to hold, but there is hardly any strong evidence that the experimental units are homogeneous. However, the nature of the interaction and the resulting increase in $\text{Var}(\bar{D})$ appear to be dominated by a substantial difference in the treatments. This is further demonstrated by the fact that 11 of the 15 deer exhibited positive signs for the computed d_i and the negative d_i (for deer 2, 10, 11, and 12) are small in magnitude compared to the 12 positive ones. Thus, it appears that the mean level of androgen is significantly higher 30 minutes following injection than at injection, and the conclusions may be stronger than $p = 0.06$ would suggest.

Annotated Computer Printout for Paired t -Test

Figure 10.13 displays a *SAS* computer printout for a paired t -test using the data of Case Study 10.1. Notice that the printout looks like that for a single sample t -test and, of course, that is exactly what is accomplished, since the test seeks to determine if $\bar{d}$ is significantly different from zero.

Analysis Variable : Diff				
N	Mean	Std Error	t Value	Pr > t
15	9.8480000	4.7698699	2.06	0.0580

Figure 10.13: *SAS* printout of paired t -test for data of Case Study 10.1.

Summary of Test Procedures

As we complete the formal development of tests on population means, we offer Table 10.3, which summarizes the test procedure for the cases of a single mean and two means. Notice the approximate procedure when distributions are normal and variances are unknown but not assumed to be equal. This statistic was introduced in Chapter 9.

10.6 Choice of Sample Size for Testing Means

In Section 10.2, we demonstrated how the analyst can exploit relationships among the sample size, the significance level α , and the power of the test to achieve a certain standard of quality. In most practical circumstances, the experiment should be planned, with a choice of sample size made prior to the data-taking process if possible. The sample size is usually determined to achieve good power for a fixed α and fixed specific alternative. This fixed alternative may be in the

Table 10.3: Tests Concerning Means

H_0	Value of Test Statistic	H_1	Critical Region
$\mu = \mu_0$	$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}; \sigma \text{ known}$	$\mu < \mu_0$ $\mu > \mu_0$ $\mu \neq \mu_0$	$z < -z_\alpha$ $z > z_\alpha$ $z < -z_{\alpha/2} \text{ or } z > z_{\alpha/2}$
$\mu = \mu_0$	$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}; v = n - 1,$ $\sigma \text{ unknown}$	$\mu < \mu_0$ $\mu > \mu_0$ $\mu \neq \mu_0$	$t < -t_\alpha$ $t > t_\alpha$ $t < -t_{\alpha/2} \text{ or } t > t_{\alpha/2}$
$\mu_1 - \mu_2 = d_0$	$z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}};$ $\sigma_1 \text{ and } \sigma_2 \text{ known}$	$\mu_1 - \mu_2 < d_0$ $\mu_1 - \mu_2 > d_0$ $\mu_1 - \mu_2 \neq d_0$	$z < -z_\alpha$ $z > z_\alpha$ $z < -z_{\alpha/2} \text{ or } z > z_{\alpha/2}$
$\mu_1 - \mu_2 = d_0$	$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{1/n_1 + 1/n_2}};$ $v = n_1 + n_2 - 2,$ $\sigma_1 = \sigma_2 \text{ but unknown,}$ $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$	$\mu_1 - \mu_2 < d_0$ $\mu_1 - \mu_2 > d_0$ $\mu_1 - \mu_2 \neq d_0$	$t < -t_\alpha$ $t > t_\alpha$ $t < -t_{\alpha/2} \text{ or } t > t_{\alpha/2}$
$\mu_1 - \mu_2 = d_0$	$t' = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{s_1^2/n_1 + s_2^2/n_2}{(s_1^2/n_1 + s_2^2/n_2)^2}}};$ $v = \frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1},$ $\sigma_1 \neq \sigma_2 \text{ and unknown}$	$\mu_1 - \mu_2 < d_0$ $\mu_1 - \mu_2 > d_0$ $\mu_1 - \mu_2 \neq d_0$	$t' < -t_\alpha$ $t' > t_\alpha$ $t' < -t_{\alpha/2} \text{ or } t' > t_{\alpha/2}$
$\mu_D = d_0$ paired observations	$t = \frac{\bar{d} - d_0}{s_d/\sqrt{n}};$ $v = n - 1$	$\mu_D < d_0$ $\mu_D > d_0$ $\mu_D \neq d_0$	$t < -t_\alpha$ $t > t_\alpha$ $t < -t_{\alpha/2} \text{ or } t > t_{\alpha/2}$

form of $\mu - \mu_0$ in the case of a hypothesis involving a single mean or $\mu_1 - \mu_2$ in the case of a problem involving two means. Specific cases will provide illustrations.

Suppose that we wish to test the hypothesis

$$H_0: \mu = \mu_0,$$

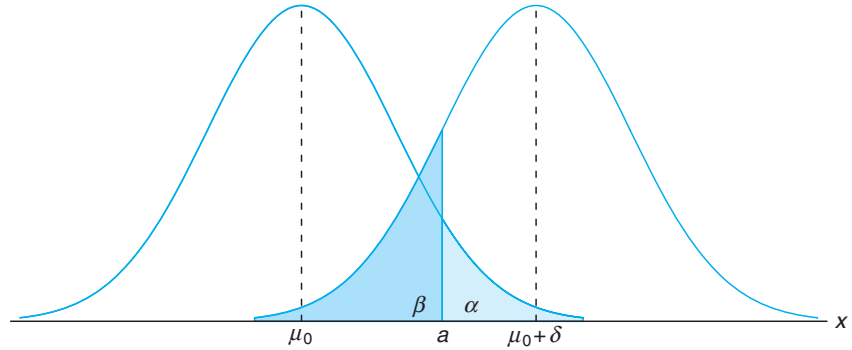
$$H_1: \mu > \mu_0,$$

with a significance level α , when the variance σ^2 is known. For a specific alternative, say $\mu = \mu_0 + \delta$, the power of our test is shown in Figure 10.14 to be

$$1 - \beta = P(\bar{X} > a \text{ when } \mu = \mu_0 + \delta).$$

Therefore,

$$\begin{aligned} \beta &= P(\bar{X} < a \text{ when } \mu = \mu_0 + \delta) \\ &= P\left[\frac{\bar{X} - (\mu_0 + \delta)}{\sigma/\sqrt{n}} < \frac{a - (\mu_0 + \delta)}{\sigma/\sqrt{n}} \text{ when } \mu = \mu_0 + \delta\right]. \end{aligned}$$

Figure 10.14: Testing $\mu = \mu_0$ versus $\mu = \mu_0 + \delta$.

Under the alternative hypothesis $\mu = \mu_0 + \delta$, the statistic

$$\frac{\bar{X} - (\mu_0 + \delta)}{\sigma/\sqrt{n}}$$

is the standard normal variable Z . So

$$\beta = P\left(Z < \frac{a - \mu_0}{\sigma/\sqrt{n}} - \frac{\delta}{\sigma/\sqrt{n}}\right) = P\left(Z < z_\alpha - \frac{\delta}{\sigma/\sqrt{n}}\right),$$

from which we conclude that

$$-z_\beta = z_\alpha - \frac{\delta\sqrt{n}}{\sigma},$$

and hence

$$\text{Choice of sample size: } n = \frac{(z_\alpha + z_\beta)^2 \sigma^2}{\delta^2},$$

a result that is also true when the alternative hypothesis is $\mu < \mu_0$.

In the case of a two-tailed test, we obtain the power $1 - \beta$ for a specified alternative when

$$n \approx \frac{(z_{\alpha/2} + z_\beta)^2 \sigma^2}{\delta^2}.$$

Example 10.7: Suppose that we wish to test the hypothesis

$$H_0: \mu = 68 \text{ kilograms,}$$

$$H_1: \mu > 68 \text{ kilograms}$$

for the weights of male students at a certain college, using an $\alpha = 0.05$ level of significance, when it is known that $\sigma = 5$. Find the sample size required if the power of our test is to be 0.95 when the true mean is 69 kilograms.

Solution: Since $\alpha = \beta = 0.05$, we have $z_\alpha = z_\beta = 1.645$. For the alternative $\beta = 69$, we take $\delta = 1$ and then

$$n = \frac{(1.645 + 1.645)^2(25)}{1} = 270.6.$$

Therefore, 271 observations are required if the test is to reject the null hypothesis 95% of the time when, in fact, μ is as large as 69 kilograms. ■

Two-Sample Case

A similar procedure can be used to determine the sample size $n = n_1 = n_2$ required for a specific power of the test in which two population means are being compared. For example, suppose that we wish to test the hypothesis

$$H_0: \mu_1 - \mu_2 = d_0,$$

$$H_1: \mu_1 - \mu_2 \neq d_0,$$

when σ_1 and σ_2 are known. For a specific alternative, say $\mu_1 - \mu_2 = d_0 + \delta$, the power of our test is shown in Figure 10.15 to be

$$1 - \beta = P(|\bar{X}_1 - \bar{X}_2| > a \text{ when } \mu_1 - \mu_2 = d_0 + \delta).$$

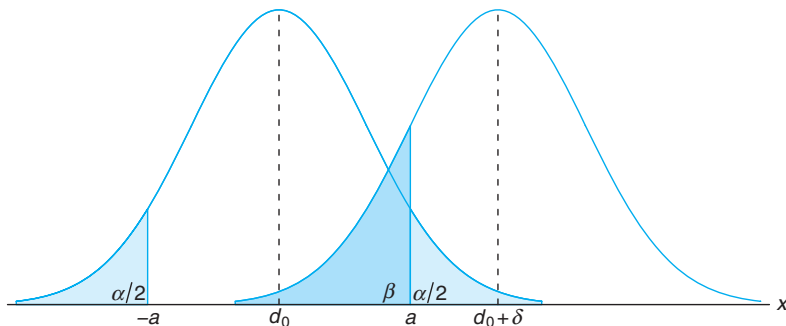


Figure 10.15: Testing $\mu_1 - \mu_2 = d_0$ versus $\mu_1 - \mu_2 = d_0 + \delta$.

Therefore,

$$\begin{aligned} \beta &= P(-a < \bar{X}_1 - \bar{X}_2 < a \text{ when } \mu_1 - \mu_2 = d_0 + \delta) \\ &= P\left[\frac{-a - (d_0 + \delta)}{\sqrt{(\sigma_1^2 + \sigma_2^2)/n}} < \frac{(\bar{X}_1 - \bar{X}_2) - (d_0 + \delta)}{\sqrt{(\sigma_1^2 + \sigma_2^2)/n}} \right. \\ &\quad \left. < \frac{a - (d_0 + \delta)}{\sqrt{(\sigma_1^2 + \sigma_2^2)/n}} \text{ when } \mu_1 - \mu_2 = d_0 + \delta\right]. \end{aligned}$$

Under the alternative hypothesis $\mu_1 - \mu_2 = d_0 + \delta$, the statistic

$$\frac{\bar{X}_1 - \bar{X}_2 - (d_0 + \delta)}{\sqrt{(\sigma_1^2 + \sigma_2^2)/n}}$$

is the standard normal variable Z . Now, writing

$$-z_{\alpha/2} = \frac{-a - d_0}{\sqrt{(\sigma_1^2 + \sigma_2^2)/n}} \quad \text{and} \quad z_{\alpha/2} = \frac{a - d_0}{\sqrt{(\sigma_1^2 + \sigma_2^2)/n}},$$

we have

$$\beta = P \left[-z_{\alpha/2} - \frac{\delta}{\sqrt{(\sigma_1^2 + \sigma_2^2)/n}} < Z < z_{\alpha/2} - \frac{\delta}{\sqrt{(\sigma_1^2 + \sigma_2^2)/n}} \right],$$

from which we conclude that

$$-z_{\beta} \approx z_{\alpha/2} - \frac{\delta}{\sqrt{(\sigma_1^2 + \sigma_2^2)/n}},$$

and hence

$$n \approx \frac{(z_{\alpha/2} + z_{\beta})^2(\sigma_1^2 + \sigma_2^2)}{\delta^2}.$$

For the one-tailed test, the expression for the required sample size when $n = n_1 = n_2$ is

$$\text{Choice of sample size: } n = \frac{(z_{\alpha} + z_{\beta})^2(\sigma_1^2 + \sigma_2^2)}{\delta^2}.$$

When the population variance (or variances, in the two-sample situation) is unknown, the choice of sample size is not straightforward. In testing the hypothesis $\mu = \mu_0$ when the true value is $\mu = \mu_0 + \delta$, the statistic

$$\frac{\bar{X} - (\mu_0 + \delta)}{S/\sqrt{n}}$$

does not follow the t -distribution, as one might expect, but instead follows the **noncentral t -distribution**. However, tables or charts based on the noncentral t -distribution do exist for determining the appropriate sample size if some estimate of σ is available or if δ is a multiple of σ . Table A.8 gives the sample sizes needed to control the values of α and β for various values of

$$\Delta = \frac{|\delta|}{\sigma} = \frac{|\mu - \mu_0|}{\sigma}$$

for both one- and two-tailed tests. In the case of the two-sample t -test in which the variances are unknown but assumed equal, we obtain the sample sizes $n = n_1 = n_2$ needed to control the values of α and β for various values of

$$\Delta = \frac{|\delta|}{\sigma} = \frac{|\mu_1 - \mu_2 - d_0|}{\sigma}$$

from Table A.9.

Example 10.8: In comparing the performance of two catalysts on the effect of a reaction yield, a two-sample t -test is to be conducted with $\alpha = 0.05$. The variances in the yields

are considered to be the same for the two catalysts. How large a sample for each catalyst is needed to test the hypothesis

$$H_0: \mu_1 = \mu_2,$$

$$H_1: \mu_1 \neq \mu_2$$

if it is essential to detect a difference of 0.8σ between the catalysts with probability 0.9?

Solution: From Table A.9, with $\alpha = 0.05$ for a two-tailed test, $\beta = 0.1$, and

$$\Delta = \frac{|0.8\sigma|}{\sigma} = 0.8,$$

we find the required sample size to be $n = 34$. ■

In practical situations, it might be difficult to force a scientist or engineer to make a commitment on information from which a value of Δ can be found. The reader is reminded that the Δ -value quantifies the kind of difference between the means that the scientist considers important, that is, a difference considered *significant* from a scientific, not a statistical, point of view. Example 10.8 illustrates how this choice is often made, namely, by selecting a fraction of σ . Obviously, if the sample size is based on a choice of $|\delta|$ that is a small fraction of σ , the resulting sample size may be quite large compared to what the study allows.

10.7 Graphical Methods for Comparing Means

In Chapter 1, considerable attention was directed to displaying data in graphical form, such as stem-and-leaf plots and box-and-whisker plots. In Section 8.8, quantile plots and quantile-quantile normal plots were used to provide a “picture” to summarize a set of experimental data. Many computer software packages produce graphical displays. As we proceed to other forms of data analysis (e.g., regression analysis and analysis of variance), graphical methods become even more informative.

Graphical aids cannot be used as a replacement for the test procedure itself. Certainly, the value of the test statistic indicates the proper type of evidence in support of H_0 or H_1 . However, a pictorial display provides a good illustration and is often a better communicator of evidence to the beneficiary of the analysis. Also, a picture will often clarify why a significant difference was found. Failure of an important assumption may be exposed by a summary type of graphical tool.

For the comparison of means, side-by-side box-and-whisker plots provide a telling display. The reader should recall that these plots display the 25th percentile, 75th percentile, and the median in a data set. In addition, the whiskers display the extremes in a data set. Consider Exercise 10.40 at the end of this section. Plasma ascorbic acid levels were measured in two groups of pregnant women, smokers and nonsmokers. Figure 10.16 shows the box-and-whisker plots for both groups of women. Two things are very apparent. Taking into account variability, there appears to be a negligible difference in the sample means. In addition, the variability in the two groups appears to be somewhat different. Of course, the analyst must keep in mind the rather sizable differences between the sample sizes in this case.

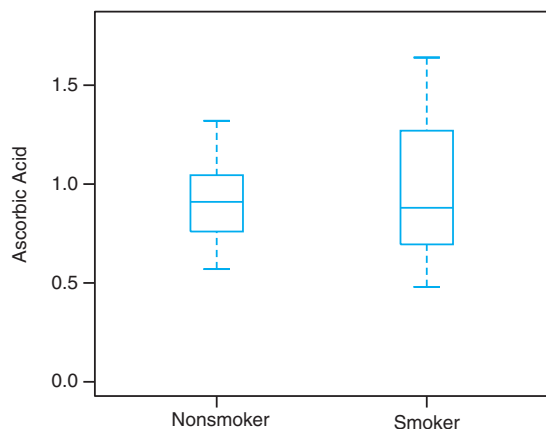


Figure 10.16: Two box-and-whisker plots of plasma ascorbic acid in smokers and nonsmokers.

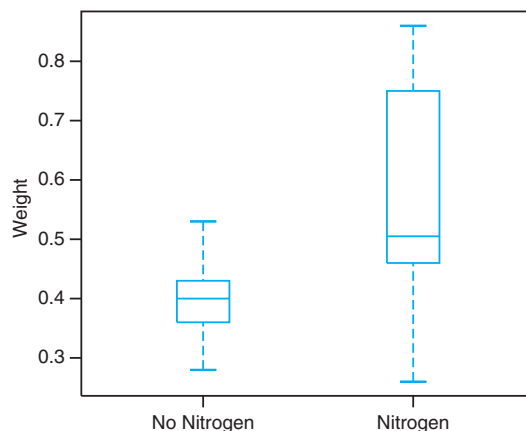


Figure 10.17: Two box-and-whisker plots of seedling data.

Consider Exercise 9.40 in Section 9.9. Figure 10.17 shows the multiple box-and-whisker plot for the data on 10 seedlings, half given nitrogen and half given no nitrogen. The display reveals a smaller variability for the group containing no nitrogen. In addition, the lack of overlap of the box plots suggests a significant difference between the mean stem weights for the two groups. It would appear that the presence of nitrogen increases the stem weights and perhaps increases the variability in the weights.

There are no certain rules of thumb regarding when two box-and-whisker plots give evidence of significant difference between the means. However, a rough guideline is that if the 25th percentile line for one sample exceeds the median line for the other sample, there is strong evidence of a difference between means.

More emphasis is placed on graphical methods in a real-life case study presented later in this chapter.

Annotated Computer Printout for Two-Sample t -Test

Consider once again Exercise 9.40 on page 294, where seedling data under conditions of nitrogen and no nitrogen were collected. Test

$$H_0: \mu_{\text{NIT}} = \mu_{\text{NON}},$$

$$H_1: \mu_{\text{NIT}} > \mu_{\text{NON}},$$

where the population means indicate mean weights. Figure 10.18 is an annotated computer printout generated using the *SAS* package. Notice that sample standard deviation and standard error are shown for both samples. The t -statistics under the assumption of equal variance and unequal variance are both given. From the box-and-whisker plot of Figure 10.17 it would certainly appear that the equal variance assumption is violated. A P -value of 0.0229 suggests a conclusion of unequal means. This concurs with the diagnostic information given in Figure 10.18. Incidentally, notice that t and t' are equal in this case, since $n_1 = n_2$.

TTEST Procedure					
Variable Weight					
Mineral	N	Mean	Std Dev	Std Err	
No nitrogen	10	0.3990	0.0728	0.0230	
Nitrogen	10	0.5650	0.1867	0.0591	
Variances					
	DF	t Value	Pr > t		
Equal	18	2.62	0.0174		
Unequal	11.7	2.62	0.0229		
Test the Equality of Variances					
Variable	Num DF	Den DF	F Value	Pr > F	
Weight	9	9	6.58	0.0098	

Figure 10.18: SAS printout for two-sample t -test.

Exercises

10.19 In a research report, Richard H. Weindruch of the UCLA Medical School claims that mice with an average life span of 32 months will live to be about 40 months old when 40% of the calories in their diet are replaced by vitamins and protein. Is there any reason to believe that $\mu < 40$ if 64 mice that are placed on this diet have an average life of 38 months with a standard deviation of 5.8 months? Use a P -value in your conclusion.

10.20 A random sample of 64 bags of white cheddar popcorn weighed, on average, 5.23 ounces with a standard deviation of 0.24 ounce. Test the hypothesis that $\mu = 5.5$ ounces against the alternative hypothesis, $\mu < 5.5$ ounces, at the 0.05 level of significance.

10.21 An electrical firm manufactures light bulbs that have a lifetime that is approximately normally distributed with a mean of 800 hours and a standard deviation of 40 hours. Test the hypothesis that $\mu = 800$ hours against the alternative, $\mu \neq 800$ hours, if a random sample of 30 bulbs has an average life of 788 hours. Use a P -value in your answer.

10.22 In the American Heart Association journal *Hypertension*, researchers report that individuals who practice Transcendental Meditation (TM) lower their blood pressure significantly. If a random sample of 225 male TM practitioners meditate for 8.5 hours per week with a standard deviation of 2.25 hours, does that suggest that, on average, men who use TM meditate more than 8 hours per week? Quote a P -value in your conclusion.

10.23 Test the hypothesis that the average content of containers of a particular lubricant is 10 liters if the

contents of a random sample of 10 containers are 10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, and 9.8 liters. Use a 0.01 level of significance and assume that the distribution of contents is normal.

10.24 The average height of females in the freshman class of a certain college has historically been 162.5 centimeters with a standard deviation of 6.9 centimeters. Is there reason to believe that there has been a change in the average height if a random sample of 50 females in the present freshman class has an average height of 165.2 centimeters? Use a P -value in your conclusion. Assume the standard deviation remains the same.

10.25 It is claimed that automobiles are driven on average more than 20,000 kilometers per year. To test this claim, 100 randomly selected automobile owners are asked to keep a record of the kilometers they travel. Would you agree with this claim if the random sample showed an average of 23,500 kilometers and a standard deviation of 3900 kilometers? Use a P -value in your conclusion.

10.26 According to a dietary study, high sodium intake may be related to ulcers, stomach cancer, and migraine headaches. The human requirement for salt is only 220 milligrams per day, which is surpassed in most single servings of ready-to-eat cereals. If a random sample of 20 similar servings of a certain cereal has a mean sodium content of 244 milligrams and a standard deviation of 24.5 milligrams, does this suggest at the 0.05 level of significance that the average sodium content for a single serving of such cereal is greater than 220 milligrams? Assume the distribution of sodium contents to be normal.

10.27 A study at the University of Colorado at Boulder shows that running increases the percent resting metabolic rate (RMR) in older women. The average RMR of 30 elderly women runners was 34.0% higher than the average RMR of 30 sedentary elderly women, and the standard deviations were reported to be 10.5 and 10.2%, respectively. Was there a significant increase in RMR of the women runners over the sedentary women? Assume the populations to be approximately normally distributed with equal variances. Use a P -value in your conclusions.

10.28 According to *Chemical Engineering*, an important property of fiber is its water absorbency. The average percent absorbency of 25 randomly selected pieces of cotton fiber was found to be 20 with a standard deviation of 1.5. A random sample of 25 pieces of acetate yielded an average percent of 12 with a standard deviation of 1.25. Is there strong evidence that the population mean percent absorbency is significantly higher for cotton fiber than for acetate? Assume that the percent absorbency is approximately normally distributed and that the population variances in percent absorbency for the two fibers are the same. Use a significance level of 0.05.

10.29 Past experience indicates that the time required for high school seniors to complete a standardized test is a normal random variable with a mean of 35 minutes. If a random sample of 20 high school seniors took an average of 33.1 minutes to complete this test with a standard deviation of 4.3 minutes, test the hypothesis, at the 0.05 level of significance, that $\mu = 35$ minutes against the alternative that $\mu < 35$ minutes.

10.30 A random sample of size $n_1 = 25$, taken from a normal population with a standard deviation $\sigma_1 = 5.2$, has a mean $\bar{x}_1 = 81$. A second random sample of size $n_2 = 36$, taken from a different normal population with a standard deviation $\sigma_2 = 3.4$, has a mean $\bar{x}_2 = 76$. Test the hypothesis that $\mu_1 = \mu_2$ against the alternative, $\mu_1 \neq \mu_2$. Quote a P -value in your conclusion.

10.31 A manufacturer claims that the average tensile strength of thread A exceeds the average tensile strength of thread B by at least 12 kilograms. To test this claim, 50 pieces of each type of thread were tested under similar conditions. Type A thread had an average tensile strength of 86.7 kilograms with a standard deviation of 6.28 kilograms, while type B thread had an average tensile strength of 77.8 kilograms with a standard deviation of 5.61 kilograms. Test the manufacturer's claim using a 0.05 level of significance.

10.32 *Amstat News* (December 2004) lists median salaries for associate professors of statistics at research institutions and at liberal arts and other institutions in the United States. Assume that a sample of 200

associate professors from research institutions has an average salary of \$70,750 per year with a standard deviation of \$6000. Assume also that a sample of 200 associate professors from other types of institutions has an average salary of \$65,200 with a standard deviation of \$5000. Test the hypothesis that the mean salary for associate professors in research institutions is \$2000 higher than for those in other institutions. Use a 0.01 level of significance.

10.33 A study was conducted to see if increasing the substrate concentration has an appreciable effect on the velocity of a chemical reaction. With a substrate concentration of 1.5 moles per liter, the reaction was run 15 times, with an average velocity of 7.5 micromoles per 30 minutes and a standard deviation of 1.5. With a substrate concentration of 2.0 moles per liter, 12 runs were made, yielding an average velocity of 8.8 micromoles per 30 minutes and a sample standard deviation of 1.2. Is there any reason to believe that this increase in substrate concentration causes an increase in the mean velocity of the reaction of more than 0.5 micromole per 30 minutes? Use a 0.01 level of significance and assume the populations to be approximately normally distributed with equal variances.

10.34 A study was made to determine if the subject matter in a physics course is better understood when a lab constitutes part of the course. Students were randomly selected to participate in either a 3-semester-hour course without labs or a 4-semester-hour course with labs. In the section with labs, 11 students made an average grade of 85 with a standard deviation of 4.7, and in the section without labs, 17 students made an average grade of 79 with a standard deviation of 6.1. Would you say that the laboratory course increases the average grade by as much as 8 points? Use a P -value in your conclusion and assume the populations to be approximately normally distributed with equal variances.

10.35 To find out whether a new serum will arrest leukemia, 9 mice, all with an advanced stage of the disease, are selected. Five mice receive the treatment and 4 do not. Survival times, in years, from the time the experiment commenced are as follows:

Treatment	2.1	5.3	1.4	4.6	0.9
No Treatment	1.9	0.5	2.8	3.1	

At the 0.05 level of significance, can the serum be said to be effective? Assume the two populations to be normally distributed with equal variances.

10.36 Engineers at a large automobile manufacturing company are trying to decide whether to purchase brand A or brand B tires for the company's new models. To help them arrive at a decision, an experiment is conducted using 12 of each brand. The tires are run

until they wear out. The results are as follows:

- Brand *A* : $\bar{x}_1 = 37,900$ kilometers,
 $s_1 = 5100$ kilometers.
- Brand *B* : $\bar{x}_1 = 39,800$ kilometers,
 $s_2 = 5900$ kilometers.

Test the hypothesis that there is no difference in the average wear of the two brands of tires. Assume the populations to be approximately normally distributed with equal variances. Use a *P*-value.

10.37 In Exercise 9.42 on page 295, test the hypothesis that the fuel economy of Volkswagen mini-trucks, on average, exceeds that of similarly equipped Toyota mini-trucks by 4 kilometers per liter. Use a 0.10 level of significance.

10.38 A UCLA researcher claims that the average life span of mice can be extended by as much as 8 months when the calories in their diet are reduced by approximately 40% from the time they are weaned. The restricted diets are enriched to normal levels by vitamins and protein. Suppose that a random sample of 10 mice is fed a normal diet and has an average life span of 32.1 months with a standard deviation of 3.2 months, while a random sample of 15 mice is fed the restricted diet and has an average life span of 37.6 months with a standard deviation of 2.8 months. Test the hypothesis, at the 0.05 level of significance, that the average life span of mice on this restricted diet is increased by 8 months against the alternative that the increase is less than 8 months. Assume the distributions of life spans for the regular and restricted diets are approximately normal with equal variances.

10.39 The following data represent the running times of films produced by two motion-picture companies:

Company	Time (minutes)						
1	102	86	98	109	92		
2	81	165	97	134	92	87	114

Test the hypothesis that the average running time of films produced by company 2 exceeds the average running time of films produced by company 1 by 10 minutes against the one-sided alternative that the difference is less than 10 minutes. Use a 0.1 level of significance and assume the distributions of times to be approximately normal with unequal variances.

10.40 In a study conducted at Virginia Tech, the plasma ascorbic acid levels of pregnant women were compared for smokers versus nonsmokers. Thirty-two women in the last three months of pregnancy, free of major health disorders and ranging in age from 15 to 32 years, were selected for the study. Prior to the collection of 20 ml of blood, the participants were told to avoid breakfast, forgo their vitamin supplements, and avoid foods high in ascorbic acid content. From the

blood samples, the following plasma ascorbic acid values were determined, in milligrams per 100 milliliters:

Plasma Ascorbic Acid Values		
Nonsmokers		Smokers
0.97	1.16	0.48
0.72	0.86	0.71
1.00	0.85	0.98
0.81	0.58	0.68
0.62	0.57	1.18
1.32	0.64	1.36
1.24	0.98	0.78
0.99	1.09	1.64
0.90	0.92	
0.74	0.78	
0.88	1.24	
0.94	1.18	

Is there sufficient evidence to conclude that there is a difference between plasma ascorbic acid levels of smokers and nonsmokers? Assume that the two sets of data came from normal populations with unequal variances. Use a *P*-value.

10.41 A study was conducted by the Department of Zoology at Virginia Tech to determine if there is a significant difference in the density of organisms at two different stations located on Cedar Run, a secondary stream in the Roanoke River drainage basin. Sewage from a sewage treatment plant and overflow from the Federal Mogul Corporation settling pond enter the stream near its headwaters. The following data give the density measurements, in number of organisms per square meter, at the two collecting stations:

Number of Organisms per Square Meter			
Station 1		Station 2	
5030	4980	2800	2810
13,700	11,910	4670	1330
10,730	8130	6890	3320
11,400	26,850	7720	1230
860	17,660	7030	2130
2200	22,800	7330	2190
4250	1130		
15,040	1690		

Can we conclude, at the 0.05 level of significance, that the average densities at the two stations are equal? Assume that the observations come from normal populations with different variances.

10.42 Five samples of a ferrous-type substance were used to determine if there is a difference between a laboratory chemical analysis and an X-ray fluorescence analysis of the iron content. Each sample was split into two subsamples and the two types of analysis were applied. Following are the coded data showing the iron content analysis:

Analysis	Sample				
	1	2	3	4	5
X-ray	2.0	2.0	2.3	2.1	2.4
Chemical	2.2	1.9	2.5	2.3	2.4

Assuming that the populations are normal, test at the 0.05 level of significance whether the two methods of analysis give, on the average, the same result.

10.43 According to published reports, practice under fatigued conditions distorts mechanisms that govern performance. An experiment was conducted using 15 college males, who were trained to make a continuous horizontal right-to-left arm movement from a microswitch to a barrier, knocking over the barrier coincident with the arrival of a clock sweep hand to the 6 o'clock position. The absolute value of the difference between the time, in milliseconds, that it took to knock over the barrier and the time for the sweep hand to reach the 6 o'clock position (500 msec) was recorded. Each participant performed the task five times under prefatigue and postfatigue conditions, and the sums of the absolute differences for the five performances were recorded.

Subject	Absolute Time Differences	
	Prefatigue	Postfatigue
1	158	91
2	92	59
3	65	215
4	98	226
5	33	223
6	89	91
7	148	92
8	58	177
9	142	134
10	117	116
11	74	153
12	66	219
13	109	143
14	57	164
15	85	100

An increase in the mean absolute time difference when the task is performed under postfatigue conditions would support the claim that practice under fatigued conditions distorts mechanisms that govern performance. Assuming the populations to be normally distributed, test this claim.

10.44 In a study conducted by the Department of Human Nutrition and Foods at Virginia Tech, the following data were recorded on sorbic acid residuals, in parts per million, in ham immediately after dipping in a sorbate solution and after 60 days of storage:

Slice	Sorbic Acid Residuals in Ham	
	Before Storage	After Storage
1	224	116
2	270	96
3	400	239
4	444	329
5	590	437
6	660	597
7	1400	689
8	680	576

Assuming the populations to be normally distributed, is there sufficient evidence, at the 0.05 level of significance, to say that the length of storage influences sorbic acid residual concentrations?

10.45 A taxi company manager is trying to decide whether the use of radial tires instead of regular belted tires improves fuel economy. Twelve cars were equipped with radial tires and driven over a prescribed test course. Without changing drivers, the same cars were then equipped with regular belted tires and driven once again over the test course. The gasoline consumption, in kilometers per liter, was recorded as follows:

Car	Kilometers per Liter	
	Radial Tires	Belted Tires
1	4.2	4.1
2	4.7	4.9
3	6.6	6.2
4	7.0	6.9
5	6.7	6.8
6	4.5	4.4
7	5.7	5.7
8	6.0	5.8
9	7.4	6.9
10	4.9	4.7
11	6.1	6.0
12	5.2	4.9

Can we conclude that cars equipped with radial tires give better fuel economy than those equipped with belted tires? Assume the populations to be normally distributed. Use a P -value in your conclusion.

10.46 In Review Exercise 9.91 on page 313, use the t -distribution to test the hypothesis that the diet reduces a woman's weight by 4.5 kilograms on average against the alternative hypothesis that the mean difference in weight is less than 4.5 kilograms. Use a P -value.

10.47 How large a sample is required in Exercise 10.20 if the power of the test is to be 0.90 when the true mean is 5.20? Assume that $\sigma = 0.24$.

10.48 If the distribution of life spans in Exercise 10.19 is approximately normal, how large a sample is required in order that the probability of committing a type II error be 0.1 when the true mean is 35.9 months? Assume that $\sigma = 5.8$ months.

10.49 How large a sample is required in Exercise 10.24 if the power of the test is to be 0.95 when the true average height differs from 162.5 by 3.1 centimeters? Use $\alpha = 0.02$.

10.50 How large should the samples be in Exercise 10.31 if the power of the test is to be 0.95 when the true difference between thread types A and B is 8 kilograms?

10.51 How large a sample is required in Exercise 10.22 if the power of the test is to be 0.8 when the true mean meditation time exceeds the hypothesized value by 1.2σ ? Use $\alpha = 0.05$.

10.52 For testing

$$H_0: \mu = 14,$$

$$H_1: \mu \neq 14,$$

an $\alpha = 0.05$ level t -test is being considered. What sample size is necessary in order for the probability to be 0.1 of falsely failing to reject H_0 when the true population mean differs from 14 by 0.5? From a preliminary sample we estimate σ to be 1.25.

10.53 A study was conducted at the Department of Veterinary Medicine at Virginia Tech to determine if the “strength” of a wound from surgical incision is affected by the temperature of the knife. Eight dogs were used in the experiment. “Hot” and “cold” incisions were made on the abdomen of each dog, and the strength was measured. The resulting data appear below.

Dog	Knife	Strength
1	Hot	5120
1	Cold	8200
2	Hot	10,000
2	Cold	8600
3	Hot	10,000
3	Cold	9200
4	Hot	10,000
4	Cold	6200

Dog	Knife	Strength
5	Hot	10,000
5	Cold	10,000
6	Hot	7900
6	Cold	5200
7	Hot	510
7	Cold	885
8	Hot	1020
8	Cold	460

(a) Write an appropriate hypothesis to determine if there is a significant difference in strength between the hot and cold incisions.

(b) Test the hypothesis using a paired t -test. Use a P -value in your conclusion.

10.54 Nine subjects were used in an experiment to determine if exposure to carbon monoxide has an impact on breathing capability. The data were collected by personnel in the Health and Physical Education Department at Virginia Tech and were analyzed in the Statistics Consulting Center at Hokie Land. The subjects were exposed to breathing chambers, one of which contained a high concentration of CO. Breathing frequency measures were made for each subject for each chamber. The subjects were exposed to the breathing chambers in random sequence. The data give the breathing frequency, in number of breaths taken per minute. Make a one-sided test of the hypothesis that mean breathing frequency is the same for the two environments. Use $\alpha = 0.05$. Assume that breathing frequency is approximately normal.

Subject	With CO	Without CO
1	30	30
2	45	40
3	26	25
4	25	23
5	34	30
6	51	49
7	46	41
8	32	35
9	30	28

10.8 One Sample: Test on a Single Proportion

Tests of hypotheses concerning proportions are required in many areas. Politicians are certainly interested in knowing what fraction of the voters will favor them in the next election. All manufacturing firms are concerned about the proportion of defective items when a shipment is made. Gamblers depend on a knowledge of the proportion of outcomes that they consider favorable.

We shall consider the problem of testing the hypothesis that the proportion of successes in a binomial experiment equals some specified value. That is, we are testing the null hypothesis H_0 that $p = p_0$, where p is the parameter of the binomial distribution. The alternative hypothesis may be one of the usual one-sided

or two-sided alternatives:

$$p < p_0, \quad p > p_0, \quad \text{or} \quad p \neq p_0.$$

The appropriate random variable on which we base our decision criterion is the binomial random variable X , although we could just as well use the statistic $\hat{p} = X/n$. Values of X that are far from the mean $\mu = np_0$ will lead to the rejection of the null hypothesis. Because X is a discrete binomial variable, it is unlikely that a critical region can be established whose size is *exactly* equal to a prespecified value of α . For this reason it is preferable, in dealing with small samples, to base our decisions on P -values. To test the hypothesis

$$H_0: p = p_0,$$

$$H_1: p < p_0,$$

we use the binomial distribution to compute the P -value

$$P = P(X \leq x \text{ when } p = p_0).$$

The value x is the number of successes in our sample of size n . If this P -value is less than or equal to α , our test is significant at the α level and we reject H_0 in favor of H_1 . Similarly, to test the hypothesis

$$H_0: p = p_0,$$

$$H_1: p > p_0,$$

at the α -level of significance, we compute

$$P = P(X \geq x \text{ when } p = p_0)$$

and reject H_0 in favor of H_1 if this P -value is less than or equal to α . Finally, to test the hypothesis

$$H_0: p = p_0,$$

$$H_1: p \neq p_0,$$

at the α -level of significance, we compute

$$P = 2P(X \leq x \text{ when } p = p_0) \quad \text{if } x < np_0$$

or

$$P = 2P(X \geq x \text{ when } p = p_0) \quad \text{if } x > np_0$$

and reject H_0 in favor of H_1 if the computed P -value is less than or equal to α .

The steps for testing a null hypothesis about a proportion against various alternatives using the binomial probabilities of Table A.1 are as follows:

Testing a Proportion (Small Samples)	1. $H_0: p = p_0$.
	2. One of the alternatives $H_1: p < p_0, p > p_0, \text{ or } p \neq p_0$.
	3. Choose a level of significance equal to α .
	4. Test statistic: Binomial variable X with $p = p_0$.
	5. Computations: Find x , the number of successes, and compute the appropriate P -value.
	6. Decision: Draw appropriate conclusions based on the P -value.

Example 10.9: A builder claims that heat pumps are installed in 70% of all homes being constructed today in the city of Richmond, Virginia. Would you agree with this claim if a random survey of new homes in this city showed that 8 out of 15 had heat pumps installed? Use a 0.10 level of significance.

- Solution:**
1. $H_0: p = 0.7$.
 2. $H_1: p \neq 0.7$.
 3. $\alpha = 0.10$.
 4. Test statistic: Binomial variable X with $p = 0.7$ and $n = 15$.
 5. Computations: $x = 8$ and $np_0 = (15)(0.7) = 10.5$. Therefore, from Table A.1, the computed P -value is

$$P = 2P(X \leq 8 \text{ when } p = 0.7) = 2 \sum_{x=0}^8 b(x; 15, 0.7) = 0.2622 > 0.10.$$

6. Decision: Do not reject H_0 . Conclude that there is insufficient reason to doubt the builder's claim. ■

In Section 5.2, we saw that binomial probabilities can be obtained from the actual binomial formula or from Table A.1 when n is small. For large n , approximation procedures are required. When the hypothesized value p_0 is very close to 0 or 1, the Poisson distribution, with parameter $\mu = np_0$, may be used. However, the normal curve approximation, with parameters $\mu = np_0$ and $\sigma^2 = np_0q_0$, is usually preferred for large n and is very accurate as long as p_0 is not extremely close to 0 or to 1. If we use the normal approximation, the **z -value for testing $p = p_0$** is given by

$$z = \frac{x - np_0}{\sqrt{np_0q_0}} = \frac{\hat{p} - p_0}{\sqrt{p_0q_0/n}},$$

which is a value of the standard normal variable Z . Hence, for a two-tailed test at the α -level of significance, the critical region is $z < -z_{\alpha/2}$ or $z > z_{\alpha/2}$. For the one-sided alternative $p < p_0$, the critical region is $z < -z_\alpha$, and for the alternative $p > p_0$, the critical region is $z > z_\alpha$.

Example 10.10: A commonly prescribed drug for relieving nervous tension is believed to be only 60% effective. Experimental results with a new drug administered to a random sample of 100 adults who were suffering from nervous tension show that 70 received relief. Is this sufficient evidence to conclude that the new drug is superior to the one commonly prescribed? Use a 0.05 level of significance.

- Solution:**
1. $H_0: p = 0.6$.
 2. $H_1: p > 0.6$.
 3. $\alpha = 0.05$.
 4. Critical region: $z > 1.645$.

5. Computations: $x = 70$, $n = 100$, $\hat{p} = 70/100 = 0.7$, and

$$z = \frac{0.7 - 0.6}{\sqrt{(0.6)(0.4)/100}} = 2.04, \quad P = P(Z > 2.04) < 0.0207.$$

6. Decision: Reject H_0 and conclude that the new drug is superior. ■

10.9 Two Samples: Tests on Two Proportions

Situations often arise where we wish to test the hypothesis that two proportions are equal. For example, we might want to show evidence that the proportion of doctors who are pediatricians in one state is equal to the proportion in another state. A person may decide to give up smoking only if he or she is convinced that the proportion of smokers with lung cancer exceeds the proportion of nonsmokers with lung cancer.

In general, we wish to test the null hypothesis that two proportions, or binomial parameters, are equal. That is, we are testing $p_1 = p_2$ against one of the alternatives $p_1 < p_2$, $p_1 > p_2$, or $p_1 \neq p_2$. Of course, this is equivalent to testing the null hypothesis that $p_1 - p_2 = 0$ against one of the alternatives $p_1 - p_2 < 0$, $p_1 - p_2 > 0$, or $p_1 - p_2 \neq 0$. The statistic on which we base our decision is the random variable $\hat{P}_1 - \hat{P}_2$. Independent samples of sizes n_1 and n_2 are selected at random from two binomial populations and the proportions of successes $\hat{P}_1$ and $\hat{P}_2$ for the two samples are computed.

In our construction of confidence intervals for p_1 and p_2 we noted, for n_1 and n_2 sufficiently large, that the point estimator $\hat{P}_1$ minus $\hat{P}_2$ was approximately normally distributed with mean

$$\mu_{\hat{P}_1 - \hat{P}_2} = p_1 - p_2$$

and variance

$$\sigma_{\hat{P}_1 - \hat{P}_2}^2 = \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}.$$

Therefore, our critical region(s) can be established by using the standard normal variable

$$Z = \frac{(\hat{P}_1 - \hat{P}_2) - (p_1 - p_2)}{\sqrt{p_1 q_1 / n_1 + p_2 q_2 / n_2}}.$$

When H_0 is true, we can substitute $p_1 = p_2 = p$ and $q_1 = q_2 = q$ (where p and q are the common values) in the preceding formula for Z to give the form

$$Z = \frac{\hat{P}_1 - \hat{P}_2}{\sqrt{pq(1/n_1 + 1/n_2)}}.$$

To compute a value of Z , however, we must estimate the parameters p and q that appear in the radical. Upon pooling the data from both samples, the **pooled estimate of the proportion p** is

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2},$$

where x_1 and x_2 are the numbers of successes in each of the two samples. Substituting $\hat{p}$ for p and $\hat{q} = 1 - \hat{p}$ for q , the **z-value for testing $p_1 = p_2$** is determined from the formula

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\hat{q}(1/n_1 + 1/n_2)}}.$$

The critical regions for the appropriate alternative hypotheses are set up as before, using critical points of the standard normal curve. Hence, for the alternative $p_1 \neq p_2$ at the α -level of significance, the critical region is $z < -z_{\alpha/2}$ or $z > z_{\alpha/2}$. For a test where the alternative is $p_1 < p_2$, the critical region is $z < -z_{\alpha}$, and when the alternative is $p_1 > p_2$, the critical region is $z > z_{\alpha}$.

Example 10.11: A vote is to be taken among the residents of a town and the surrounding county to determine whether a proposed chemical plant should be constructed. The construction site is within the town limits, and for this reason many voters in the county believe that the proposal will pass because of the large proportion of town voters who favor the construction. To determine if there is a significant difference in the proportions of town voters and county voters favoring the proposal, a poll is taken. If 120 of 200 town voters favor the proposal and 240 of 500 county residents favor it, would you agree that the proportion of town voters favoring the proposal is higher than the proportion of county voters? Use an $\alpha = 0.05$ level of significance.

Solution: Let p_1 and p_2 be the true proportions of voters in the town and county, respectively, favoring the proposal.

1. $H_0: p_1 = p_2$.
2. $H_1: p_1 > p_2$.
3. $\alpha = 0.05$.
4. Critical region: $z > 1.645$.
5. Computations:

$$\begin{aligned}\hat{p}_1 &= \frac{x_1}{n_1} = \frac{120}{200} = 0.60, & \hat{p}_2 &= \frac{x_2}{n_2} = \frac{240}{500} = 0.48, & \text{and} \\ \hat{p} &= \frac{x_1 + x_2}{n_1 + n_2} = \frac{120 + 240}{200 + 500} = 0.51.\end{aligned}$$

Therefore,

$$z = \frac{0.60 - 0.48}{\sqrt{(0.51)(0.49)(1/200 + 1/500)}} = 2.9,$$

$$P = P(Z > 2.9) = 0.0019.$$

6. Decision: Reject H_0 and agree that the proportion of town voters favoring the proposal is higher than the proportion of county voters. ■

Exercises

10.55 A marketing expert for a pasta-making company believes that 40% of pasta lovers prefer lasagna. If 9 out of 20 pasta lovers choose lasagna over other pastas, what can be concluded about the expert's claim? Use a 0.05 level of significance.

10.56 Suppose that, in the past, 40% of all adults favored capital punishment. Do we have reason to believe that the proportion of adults favoring capital punishment has increased if, in a random sample of 15 adults, 8 favor capital punishment? Use a 0.05 level of significance.

10.57 A new radar device is being considered for a certain missile defense system. The system is checked by experimenting with aircraft in which a kill or a no kill is simulated. If, in 300 trials, 250 kills occur, accept or reject, at the 0.04 level of significance, the claim that the probability of a kill with the new system does not exceed the 0.8 probability of the existing device.

10.58 It is believed that at least 60% of the residents in a certain area favor an annexation suit by a neighboring city. What conclusion would you draw if only 110 in a sample of 200 voters favored the suit? Use a 0.05 level of significance.

10.59 A fuel oil company claims that one-fifth of the homes in a certain city are heated by oil. Do we have reason to believe that fewer than one-fifth are heated by oil if, in a random sample of 1000 homes in this city, 136 are heated by oil? Use a P -value in your conclusion.

10.60 At a certain college, it is estimated that at most 25% of the students ride bicycles to class. Does this seem to be a valid estimate if, in a random sample of 90 college students, 28 are found to ride bicycles to class? Use a 0.05 level of significance.

10.61 In a winter of an epidemic flu, the parents of 2000 babies were surveyed by researchers at a well-known pharmaceutical company to determine if the company's new medicine was effective after two days. Among 120 babies who had the flu and were given the medicine, 29 were cured within two days. Among 280 babies who had the flu but were not given the medicine, 56 recovered within two days. Is there any significant indication that supports the company's claim of the effectiveness of the medicine?

10.62 In a controlled laboratory experiment, scientists at the University of Minnesota discovered that 25% of a certain strain of rats subjected to a 20% coffee bean diet and then force-fed a powerful cancer-causing chemical later developed cancerous tumors. Would we have reason to believe that the proportion of rats developing tumors when subjected to this diet has increased if the experiment were repeated and 16 of 48 rats developed tumors? Use a 0.05 level of significance.

10.63 In a study to estimate the proportion of residents in a certain city and its suburbs who favor the construction of a nuclear power plant, it is found that 63 of 100 urban residents favor the construction while only 59 of 125 suburban residents are in favor. Is there a significant difference between the proportions of urban and suburban residents who favor construction of the nuclear plant? Make use of a P -value.

10.64 In a study on the fertility of married women conducted by Martin O'Connell and Carolyn C. Rogers for the Census Bureau in 1979, two groups of childless wives aged 25 to 29 were selected at random, and each was asked if she eventually planned to have a child. One group was selected from among wives married less than two years and the other from among wives married five years. Suppose that 240 of the 300 wives married less than two years planned to have children some day compared to 288 of the 400 wives married five years. Can we conclude that the proportion of wives married less than two years who planned to have children is significantly higher than the proportion of wives married five years? Make use of a P -value.

10.65 An urban community would like to show that the incidence of breast cancer is higher in their area than in a nearby rural area. (PCB levels were found to be higher in the soil of the urban community.) If it is found that 20 of 200 adult women in the urban community have breast cancer and 10 of 150 adult women in the rural community have breast cancer, can we conclude at the 0.05 level of significance that breast cancer is more prevalent in the urban community?

10.66 Group Project: The class should be divided into pairs of students for this project. Suppose it is conjectured that at least 25% of students at your university exercise for more than two hours a week. Collect data from a random sample of 50 students. Ask each student if he or she works out for at least two hours per week. Then do the computations that allow either rejection or nonrejection of the above conjecture. Show all work and quote a P -value in your conclusion.

10.10 One- and Two-Sample Tests Concerning Variances

In this section, we are concerned with testing hypotheses concerning population variances or standard deviations. Applications of one- and two-sample tests on variances are certainly not difficult to motivate. Engineers and scientists are confronted with studies in which they are required to demonstrate that measurements involving products or processes adhere to specifications set by consumers. The specifications are often met if the process variance is sufficiently small. Attention is also focused on comparative experiments between methods or processes, where inherent reproducibility or variability must formally be compared. In addition, to determine if the equal variance assumption is violated, a test comparing two variances is often applied prior to conducting a t -test on two means.

Let us first consider the problem of testing the null hypothesis H_0 that the population variance σ^2 equals a specified value σ_0^2 against one of the usual alternatives $\sigma^2 < \sigma_0^2$, $\sigma^2 > \sigma_0^2$, or $\sigma^2 \neq \sigma_0^2$. The appropriate statistic on which to base our decision is the chi-squared statistic of Theorem 8.4, which was used in Chapter 9 to construct a confidence interval for σ^2 . Therefore, if we assume that the distribution of the population being sampled is normal, the chi-squared value for testing $\sigma^2 = \sigma_0^2$ is given by

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2},$$

where n is the sample size, s^2 is the sample variance, and σ_0^2 is the value of σ^2 given by the null hypothesis. If H_0 is true, χ^2 is a value of the chi-squared distribution with $v = n - 1$ degrees of freedom. Hence, for a two-tailed test at the α -level of significance, the critical region is $\chi^2 < \chi_{1-\alpha/2}^2$ or $\chi^2 > \chi_{\alpha/2}^2$. For the one-sided alternative $\sigma^2 < \sigma_0^2$, the critical region is $\chi^2 < \chi_{1-\alpha}^2$, and for the one-sided alternative $\sigma^2 > \sigma_0^2$, the critical region is $\chi^2 > \chi_{\alpha}^2$.

Robustness of χ^2 -Test to Assumption of Normality

The reader may have discerned that various tests depend, at least theoretically, on the assumption of normality. In general, many procedures in applied statistics have theoretical underpinnings that depend on the normal distribution. These procedures vary in the degree of their dependency on the assumption of normality. A procedure that is reasonably insensitive to the assumption is called a **robust procedure** (i.e., robust to normality). The χ^2 -test on a single variance is very nonrobust to normality (i.e., the practical success of the procedure depends on normality). As a result, the P -value computed may be appreciably different from the actual P -value if the population sampled is not normal. Indeed, it is quite feasible that a statistically significant P -value may not truly signal $H_1: \sigma \neq \sigma_0$; rather, a significant value may be a result of the violation of the normality assumptions. Therefore, the analyst should approach the use of this particular χ^2 -test with caution.

Example 10.12: A manufacturer of car batteries claims that the life of the company's batteries is approximately normally distributed with a standard deviation equal to 0.9 year.

If a random sample of 10 of these batteries has a standard deviation of 1.2 years, do you think that $\sigma > 0.9$ year? Use a 0.05 level of significance.

Solution: 1. $H_0: \sigma^2 = 0.81$.

2. $H_1: \sigma^2 > 0.81$.

3. $\alpha = 0.05$.

4. Critical region: From Figure 10.19 we see that the null hypothesis is rejected when $\chi^2 > 16.919$, where $\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$, with $v = 9$ degrees of freedom.

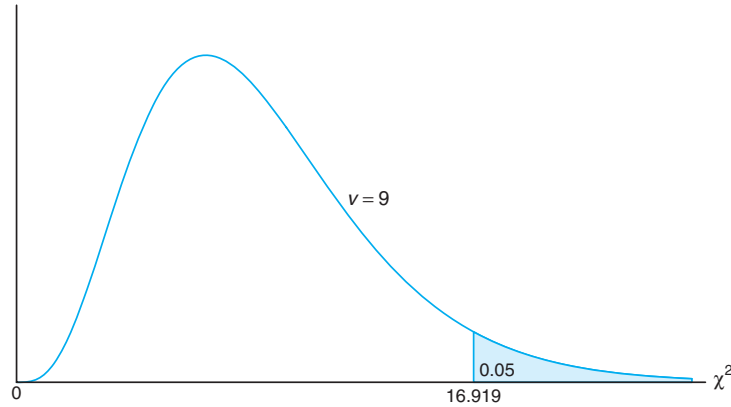


Figure 10.19: Critical region for the alternative hypothesis $\sigma > 0.9$.

5. Computations: $s^2 = 1.44$, $n = 10$, and

$$\chi^2 = \frac{(9)(1.44)}{0.81} = 16.0, \quad P \approx 0.07.$$

6. Decision: The χ^2 -statistic is not significant at the 0.05 level. However, based on the P -value 0.07, there is evidence that $\sigma > 0.9$. ■

Now let us consider the problem of testing the equality of the variances σ_1^2 and σ_2^2 of two populations. That is, we shall test the null hypothesis H_0 that $\sigma_1^2 = \sigma_2^2$ against one of the usual alternatives

$$\sigma_1^2 < \sigma_2^2, \quad \sigma_1^2 > \sigma_2^2, \quad \text{or} \quad \sigma_1^2 \neq \sigma_2^2.$$

For independent random samples of sizes n_1 and n_2 , respectively, from the two populations, the **f -value for testing $\sigma_1^2 = \sigma_2^2$** is the ratio

$$f = \frac{s_1^2}{s_2^2},$$

where s_1^2 and s_2^2 are the variances computed from the two samples. If the two populations are approximately normally distributed and the null hypothesis is true, according to Theorem 8.8 the ratio $f = s_1^2/s_2^2$ is a value of the F -distribution with $v_1 = n_1 - 1$ and $v_2 = n_2 - 1$ degrees of freedom. Therefore, the critical regions

of size α corresponding to the one-sided alternatives $\sigma_1^2 < \sigma_2^2$ and $\sigma_1^2 > \sigma_2^2$ are, respectively, $f < f_{1-\alpha}(v_1, v_2)$ and $f > f_{\alpha}(v_1, v_2)$. For the two-sided alternative $\sigma_1^2 \neq \sigma_2^2$, the critical region is $f < f_{1-\alpha/2}(v_1, v_2)$ or $f > f_{\alpha/2}(v_1, v_2)$.

Example 10.13: In testing for the difference in the abrasive wear of the two materials in Example 10.6, we assumed that the two unknown population variances were equal. Were we justified in making this assumption? Use a 0.10 level of significance.

Solution: Let σ_1^2 and σ_2^2 be the population variances for the abrasive wear of material 1 and material 2, respectively.

1. $H_0: \sigma_1^2 = \sigma_2^2$.
2. $H_1: \sigma_1^2 \neq \sigma_2^2$.
3. $\alpha = 0.10$.
4. Critical region: From Figure 10.20, we see that $f_{0.05}(11, 9) = 3.11$, and, by using Theorem 8.7, we find

$$f_{0.95}(11, 9) = \frac{1}{f_{0.05}(9, 11)} = 0.34.$$

Therefore, the null hypothesis is rejected when $f < 0.34$ or $f > 3.11$, where $f = s_1^2/s_2^2$ with $v_1 = 11$ and $v_2 = 9$ degrees of freedom.

5. Computations: $s_1^2 = 16$, $s_2^2 = 25$, and hence $f = \frac{16}{25} = 0.64$.
6. Decision: Do not reject H_0 . Conclude that there is insufficient evidence that the variances differ. └

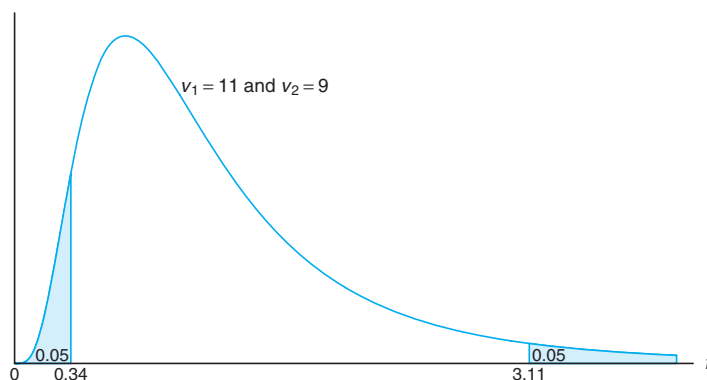


Figure 10.20: Critical region for the alternative hypothesis $\sigma_1^2 \neq \sigma_2^2$.

F-Test for Testing Variances in SAS

Figure 10.18 on page 356 displays the printout of a two-sample t -test where two means from the seedling data in Exercise 9.40 were compared. Box-and-whisker plots in Figure 10.17 on page 355 suggest that variances are not homogeneous, and thus the t' -statistic and its corresponding P -value are relevant. Note also that

the printout displays the F -statistic for $H_0: \sigma_1 = \sigma_2$ with a P -value of 0.0098, additional evidence that more variability is to be expected when nitrogen is used than under the no-nitrogen condition.

Exercises

10.67 The content of containers of a particular lubricant is known to be normally distributed with a variance of 0.03 liter. Test the hypothesis that $\sigma^2 = 0.03$ against the alternative that $\sigma^2 \neq 0.03$ for the random sample of 10 containers in Exercise 10.23 on page 356. Use a P -value in your conclusion.

10.68 Past experience indicates that the time required for high school seniors to complete a standardized test is a normal random variable with a standard deviation of 6 minutes. Test the hypothesis that $\sigma = 6$ against the alternative that $\sigma < 6$ if a random sample of the test times of 20 high school seniors has a standard deviation $s = 4.51$. Use a 0.05 level of significance.

10.69 Aflatoxins produced by mold on peanut crops in Virginia must be monitored. A sample of 64 batches of peanuts reveals levels of 24.17 ppm, on average, with a variance of 4.25 ppm. Test the hypothesis that $\sigma^2 = 4.2$ ppm against the alternative that $\sigma^2 \neq 4.2$ ppm. Use a P -value in your conclusion.

10.70 Past data indicate that the amount of money contributed by the working residents of a large city to a volunteer rescue squad is a normal random variable with a standard deviation of \$1.40. It has been suggested that the contributions to the rescue squad from just the employees of the sanitation department are much more variable. If the contributions of a random sample of 12 employees from the sanitation department have a standard deviation of \$1.75, can we conclude at the 0.01 level of significance that the standard deviation of the contributions of all sanitation workers is greater than that of all workers living in the city?

10.71 A soft-drink dispensing machine is said to be out of control if the variance of the contents exceeds 1.15 deciliters. If a random sample of 25 drinks from this machine has a variance of 2.03 deciliters, does this indicate at the 0.05 level of significance that the machine is out of control? Assume that the contents are approximately normally distributed.

10.72 Large-Sample Test of $\sigma^2 = \sigma_0^2$: When $n \geq 30$, we can test the null hypothesis that $\sigma^2 = \sigma_0^2$, or $\sigma = \sigma_0$, by computing

$$z = \frac{s - \sigma_0}{\sigma_0 / \sqrt{2n}},$$

which is a value of a random variable whose sampling distribution is approximately the standard normal distribution.

- (a) With reference to Example 10.4, test at the 0.05 level of significance whether $\sigma = 10.0$ years against the alternative that $\sigma \neq 10.0$ years.
- (b) It is suspected that the variance of the distribution of distances in kilometers traveled on 5 liters of fuel by a new automobile model equipped with a diesel engine is less than the variance of the distribution of distances traveled by the same model equipped with a six-cylinder gasoline engine, which is known to be $\sigma^2 = 6.25$. If 72 test runs of the diesel model have a variance of 4.41, can we conclude at the 0.05 level of significance that the variance of the distances traveled by the diesel model is less than that of the gasoline model?

10.73 A study is conducted to compare the lengths of time required by men and women to assemble a certain product. Past experience indicates that the distribution of times for both men and women is approximately normal but the variance of the times for women is less than that for men. A random sample of times for 11 men and 14 women produced the following data:

Men	Women
$n_1 = 11$	$n_2 = 14$
$s_1 = 6.1$	$s_2 = 5.3$

Test the hypothesis that $\sigma_1^2 = \sigma_2^2$ against the alternative that $\sigma_1^2 > \sigma_2^2$. Use a P -value in your conclusion.

10.74 For Exercise 10.41 on page 358, test the hypothesis at the 0.05 level of significance that $\sigma_1^2 = \sigma_2^2$ against the alternative that $\sigma_1^2 \neq \sigma_2^2$, where σ_1^2 and σ_2^2 are the variances of the number of organisms per square meter of water at the two different locations on Cedar Run.

10.75 With reference to Exercise 10.39 on page 358, test the hypothesis that $\sigma_1^2 = \sigma_2^2$ against the alternative that $\sigma_1^2 \neq \sigma_2^2$, where σ_1^2 and σ_2^2 are the variances for the running times of films produced by company 1 and company 2, respectively. Use a P -value.

10.76 Two types of instruments for measuring the amount of sulfur monoxide in the atmosphere are being compared in an air-pollution experiment. Researchers